

Chapter 2

Dynamics of Inviscid, Incompressible Flow

2.1. Introduction

Dynamics is the field of science that deals with substances in motion. To quantitatively describe the motion of a substance we must first model the physical phenomena with a consistent set of mathematical equations. Engineering analysis of problems involving a substance in motion requires the application of at least some of the fundamental laws of physics, which include but are not limited to *conservation of mass*, *Newton's second law*, and the *first law of thermodynamics*. Obviously, not all of the laws of physics are always required to solve any one particular problem. On the other hand, most problems involving a substance in motion cannot be solved using the laws of physics alone. It is usually necessary to include constitutive equations or *equations of state*, which describe the behavior of the particular substance under consideration.

The first step in the process of modeling any physical phenomena with a consistent set of mathematical equations is choosing the physical system that we are attempting to analyze. There are two types of systems commonly used for the mathematical modeling of physical phenomena in engineering mechanics. These are the *Lagrangian system* and the *Eulerian control volume*.

A **Lagrangian system** is defined to be an identifiable quantity of matter having fixed mass. Everything external to the system is typically referred to as the *surroundings* and the system is separated from the surroundings by a closed surface commonly called the *system boundary*. This boundary may be either fixed or movable. In general, both heat and work can cross the boundary of a Lagrangian system. However, because the system is defined to always contain the same fixed quantity of matter, mass never crosses the boundary of any Lagrangian system. For example, Fig. 2.1.1 shows a Lagrangian system consisting of a fixed quantity of gas confined by a cylinder and piston. If a constant force is applied to the piston while the cylinder is heated from some external source, the temperature and volume of the gas will increase. As the gas expands, the system boundary moves, transferring energy to the piston in the form of work.

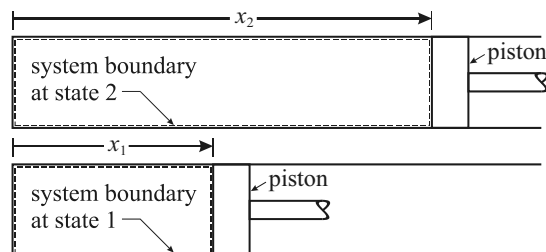


Figure 2.1.1. Example of a Lagrangian system.

An **Eulerian control volume** is defined to be an arbitrary volume in space through which matter flows. The boundary of the control volume is a closed surface commonly called the *control surface*. This boundary may be either a real surface like the walls of a solid channel or an imaginary surface separating one region of a fluid from another. The control volume may be either at rest or in motion. Mass as well as heat transfer and work may cross the boundary of an Eulerian control volume. Thus, the identity and amount of matter contained within the control volume may change with time. For example, Fig. 2.1.2 shows an Eulerian control volume that encloses a turbofan engine. Air enters the control volume through the inlet section i and passes through the fan. Part of this air exits the control volume through the bypass section b and the remainder passes through the compressor and enters the combustion chamber. Fuel also enters this control volume through a fuel line (not shown) and is mixed with the air in the combustion chamber. The fuel-air mixture is ignited in the combustion chamber, and the products of combustion pass through the turbine and nozzle to exit the control volume through the nozzle exit section e .

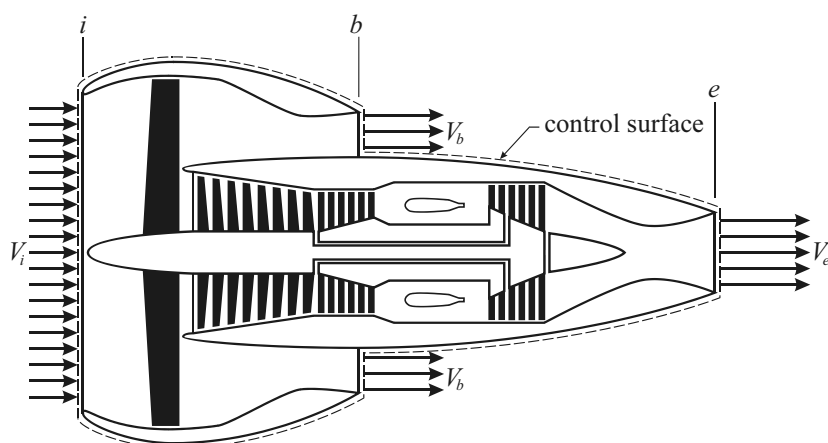


Figure 2.1.2. Example of an Eulerian control volume.

In the Lagrangian description of the Newtonian laws of physics, one follows the history of individual quantities of matter as they move and deform. In the Eulerian description, one concentrates on what happens at individual locations in space as moving and deforming matter passes through. The Lagrangian description is ideal for the study of solid mechanics, where it is easy to follow a fixed quantity of matter as it moves and deforms. For example, in the study of astrodynamics it is relatively easy to follow a solid object as it moves through space under the influence of gravitational forces. Likewise, in the study of structural mechanics we can easily follow the matter in a solid beam as it deflects under a load. The Lagrangian description is also useful for the study of fluid statics, because there is no motion in a static fluid and a fixed volume always contains the same fixed quantity of matter. However, in the study of fluid dynamics we are typically interested in the flow of fluids through devices and around objects. For such problems it

is often very difficult to distinguish and follow the same fixed quantity of fluid as it deforms and moves through a device or around an object. Furthermore, we are often more interested in the effect that the fluid motion has on the device or object than we are in the motion of the fluid itself. For example, in the aerodynamics of flight, our primary purpose for studying the motion of the fluid in the flow about an aircraft is to predict the lift and drag forces exerted on the aircraft by the fluid. Hence, it is typically more convenient to model the physical phenomena associated with fluid dynamics using the Eulerian control volume.

In this chapter we will use the Eulerian control volume to develop a mathematical model for predicting the velocity and pressure fields associated with the dynamic flow of inviscid, incompressible fluids. As we shall see, this model accurately predicts the fluid velocity and pressure distributions outside the boundary layer for flow about streamlined objects at low angles of attack, high Reynolds numbers, and low Mach numbers, which are conditions that occur commonly for aircraft in flight. Because the boundary layer is thin, this model accurately predicts the pressure distribution on the surface of an aircraft under these conditions. Furthermore, because viscous forces contribute little to the lift under such conditions, the inviscid, incompressible flow model developed in this chapter provides a relatively simple and accurate tool for studying the production of aerodynamic lift on an aircraft in flight.

2.2. Conservation of Mass: The Continuity Equation

By definition, a Lagrangian system always contains the same fixed quantity of matter. Thus, when a Lagrangian system is used for the development of a mathematical model for some physical phenomenon, by our choice and definition of the Lagrangian system, mass is conserved automatically. On the other hand, when a mathematical model for a physical phenomenon is developed using an Eulerian control volume, we must include in the formulation a separate equation to ensure the conservation of mass.

From their introductory class in thermodynamics, most students should be familiar with applying the principle of conservation of mass to a control volume. *Within the laws of Newtonian physics, mass can be neither created nor destroyed; so conservation of mass simply requires that the time rate of change of the mass contained within any control volume plus the net instantaneous mass flow rate exiting the control volume must sum to zero.* For example, consider the control volume that encloses the turbofan engine shown in Fig. 2.1.2. Let m_{cv} denote the instantaneous total mass contained within the control volume; and let $\dot{m}_i$, $\dot{m}_b$, and $\dot{m}_e$ denote the mass flow rates crossing sections i , b , and e , respectively. The mass flow rate of the fuel entering the control volume through the fuel line (not shown) is designated as $\dot{m}_f$. Because $\dot{m}_i$ and $\dot{m}_f$ are entering the control volume while $\dot{m}_b$ and $\dot{m}_e$ are exiting the control volume, application of the conservation of mass to this particular control volume requires

$$\frac{dm_{cv}}{dt} + \dot{m}_b + \dot{m}_e - \dot{m}_i - \dot{m}_f = 0 \quad (2.2.1)$$

In general, the concentration of mass within the control volume as well as the mass flow rate through the control surface can vary with both time and position.

Consider an arbitrary Eulerian control volume $\mathcal{V}$ enclosed by a control surface $\mathcal{S}$ in a fluid flow field as shown in Fig. 2.2.1. The same fluid fills and surrounds this control volume and the instantaneous local fluid density is denoted as ρ . In the limit as the differential volume $d\mathcal{V}$ approaches zero, the mass contained within $d\mathcal{V}$ is simply the product $\rho d\mathcal{V}$, and **the total mass contained within the control volume at any instant in time can be expressed as the volume integral**

$$m_{cv} = \iiint_{\mathcal{V}} \rho d\mathcal{V} \quad (2.2.2)$$

which may be a function of time t .

Let $\mathbf{V}_s$ denote the instantaneous local velocity of the control surface and let $\mathbf{V}$ denote the instantaneous local fluid velocity. As shown in Fig. 2.2.2, in the limit as both $d\mathcal{S}$ and dt approach zero, the volume of fluid passing through the differential surface area $d\mathcal{S}$ in the differential time increment dt is given by $|\mathbf{V} - \mathbf{V}_s| dt d\mathcal{S} \cos\theta$, where $|\mathbf{V} - \mathbf{V}_s|$ is the magnitude of the vector $\mathbf{V} - \mathbf{V}_s$ and θ is the angle from $\mathbf{V} - \mathbf{V}_s$ to the unit outward normal to the control surface $\mathbf{n}$. From Eq. (1.4.11), the dot product of the vector $\mathbf{V} - \mathbf{V}_s$ and the unit vector $\mathbf{n}$ can be written as $(\mathbf{V} - \mathbf{V}_s) \cdot \mathbf{n} = |\mathbf{V} - \mathbf{V}_s| \cos\theta$. Dividing the differential volume $|\mathbf{V} - \mathbf{V}_s| dt d\mathcal{S} \cos\theta = (\mathbf{V} - \mathbf{V}_s) \cdot \mathbf{n} dt d\mathcal{S}$ shown in Fig. 2.2.2 by the differential time increment yields the volume flow rate $(\mathbf{V} - \mathbf{V}_s) \cdot \mathbf{n} d\mathcal{S}$, which is exiting the control volume through $d\mathcal{S}$. Multiplying by the instantaneous local fluid density ρ and integrating over the entire control surface $\mathcal{S}$, **the total net mass flow rate exiting a control volume at any instant in time can be expressed as the surface integral**

$$\dot{m}_e = \iint_{\mathcal{S}} \rho (\mathbf{V} - \mathbf{V}_s) \cdot \mathbf{n} d\mathcal{S} \quad (2.2.3)$$

This net mass flow rate could be either positive or negative and may vary with time.

To reiterate, within the Newtonian approximation, mass can be neither created nor destroyed. Thus, conservation of mass requires that the time rate of change of the total mass contained within a control volume, combined with the net mass flow rate exiting the

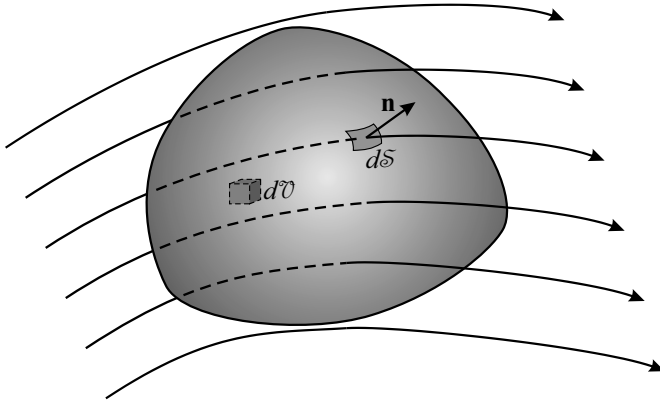


Figure 2.2.1. Eulerian control volume in a fluid flow field.

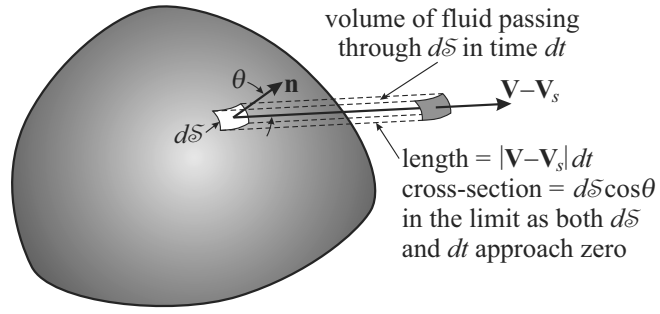


Figure 2.2.2. Volume of fluid passing through a differential area on a control surface.

control volume through the control surface, must sum to zero. Hence, from Eqs. (2.2.2) and (2.2.3), **the conservation of mass for an arbitrary control volume requires**

$$\frac{d}{dt} \iiint_V \rho d\mathcal{V} + \iint_S \rho (\mathbf{V} - \mathbf{V}_s) \cdot \mathbf{n} d\mathcal{S} = 0 \quad (2.2.4)$$

Equation (2.2.4) is the integral form of what is commonly called the **continuity equation**. Of course, Newtonian physics neglects the effects of *general relativity*, so Eq. (2.2.4) assumes that all velocities are small compared to the speed of light.

If the flow field is steady and the control volume is *fixed* (i.e., $\mathbf{V}_s = 0$), the density at any fixed position within the control volume must be invariant with time. Because we are considering a fixed control volume that does not change position, size, or shape with time, the first term in Eq. (2.2.4) is zero for steady flow. Thus, **the continuity equation for steady flow through a fixed Eulerian control volume becomes**

$$\iint_S \rho \mathbf{V} \cdot \mathbf{n} d\mathcal{S} = 0 \quad (2.2.5)$$

If the flow field is incompressible and the control volume is fixed, the fluid density is constant at all points within the control volume and for all time, even if the velocity field is unsteady. Because we are considering a fixed control volume, the first term in Eq. (2.2.4) vanishes for incompressible flow, and the constant fluid density can be taken outside the surface integral in the second term. Hence, **the continuity equation for any incompressible flow through a fixed Eulerian control volume reduces to**

$$\iint_S \mathbf{V} \cdot \mathbf{n} d\mathcal{S} = 0 \quad (2.2.6)$$

The integral forms of the continuity equation given by Eqs. (2.2.4) through (2.2.6) apply to control volumes of any size and shape. However, before the volume and surface integrals shown symbolically in these equations can be evaluated, the density ρ , velocity $\mathbf{V}$, outward normal $\mathbf{n}$, differential volume $d\mathcal{V}$, differential area $d\mathcal{S}$, and the boundaries of $\mathcal{V}$ and $\mathcal{S}$ must be expressed in terms of a convenient coordinate system.

EXAMPLE 2.2.1. Consider steady flow through the turbofan engine shown in Fig. 2.1.2. Air enters the engine through inlet section i with a mass flow rate $\dot{m}_i$. Part of this air, $\dot{m}_b$, exits the engine through bypass section b and the remainder, $\dot{m}_i - \dot{m}_b$, passes through the compressor and enters the combustion chamber. Fuel also enters this engine at the mass flow rate $\dot{m}_f$ through a fuel line (not shown) and is mixed with the air in the combustion chamber. The products of combustion exit the engine through the nozzle exit section e . Flow through a turbofan engine is typically characterized in terms of what are called the *bypass ratio*, $\dot{m}_b/(\dot{m}_i - \dot{m}_b)$, and the *fuel-air ratio*, $\dot{m}_f/(\dot{m}_i - \dot{m}_b)$. The engine is operating at 35,000 ft with the following parameters:

$$\begin{aligned} \text{bypass ratio} &\equiv R_b = 8.40, \quad \text{fuel-air ratio} \equiv R_f = 0.0060, \\ \text{inlet area} &\equiv \mathcal{S}_i = 164.2 \text{ ft}, \quad T_i = 394 \text{ R}, \quad p_i = 499 \text{ lbf/ft}^2, \quad M_i = 0.80, \\ \text{bypass exit area} &\equiv \mathcal{S}_b = 113.6 \text{ ft}, \quad T_b = 400 \text{ R}, \quad p_b = 525 \text{ lbf/ft}^2, \\ \text{nozzle exit area} &\equiv \mathcal{S}_e = 18.4 \text{ ft}, \quad T_e = 547 \text{ R}, \quad p_e = 532 \text{ lbf/ft}^2 \end{aligned}$$

As shown in Fig. 2.1.2, the fluid velocity vectors at sections i , b , and e are assumed constant and normal to the control surface. If the gas constants for the products of combustion are the same as for air, $R = 1716.561 \text{ lbf}\cdot\text{ft}/\text{slug}\cdot\text{R} = 1716.561 \text{ ft}^2/\text{s}^2\cdot\text{R}$ and $\gamma = 1.4$, find the velocity and Mach number at sections b and e .

Solution. For the fixed control volume shown in Fig. 2.1.2, $\mathbf{V}_s = 0$ and $\mathbf{V}\cdot\mathbf{n}$ is zero everywhere on the control surface except in the fuel line and on sections i , b , and e . Thus, the steady flow continuity equation given by Eq. (2.2.5) reduces to

$$\iint_{\mathcal{S}_f} \rho_f (-V_f) d\mathcal{S} + \iint_{\mathcal{S}_i} \rho_i (-V_i) d\mathcal{S} + \iint_{\mathcal{S}_b} \rho_b (V_b) d\mathcal{S} + \iint_{\mathcal{S}_e} \rho_e (V_e) d\mathcal{S} = 0$$

Because the density and velocity are assumed to be constant over each of these four sections, the integration yields

$$-\rho_f V_f \mathcal{S}_f - \rho_i V_i \mathcal{S}_i + \rho_b V_b \mathcal{S}_b + \rho_e V_e \mathcal{S}_e = 0 \quad (2.2.7)$$

From the definition of the bypass ratio

$$R_b \equiv \rho_b V_b \mathcal{S}_b / (\rho_i V_i \mathcal{S}_i - \rho_b V_b \mathcal{S}_b)$$

or after solving for the bypass flow rate and velocity

$$\rho_b V_b \mathcal{S}_b = \rho_i V_i \mathcal{S}_i R_b / (1 + R_b) \quad (2.2.8)$$

$$\begin{aligned} V_b &= V_i \frac{\rho_i}{\rho_b} \frac{\mathcal{S}_i}{\mathcal{S}_b} \frac{R_b}{1 + R_b} = M_i \sqrt{\gamma R T_i} \frac{p_i / R T_i}{p_b / R T_b} \frac{\mathcal{S}_i}{\mathcal{S}_b} \frac{R_b}{1 + R_b} \\ &= 0.80 \sqrt{1.4 (1716.561 \text{ ft}^2/\text{s}^2\cdot\text{R}) (394 \text{ R})} \frac{499/394}{525/400} \frac{164.2}{113.6} \frac{8.40}{1 + 8.40} \\ &= \underline{970 \text{ ft/s}} \end{aligned}$$

From the definition of the Mach number

$$M_b = V_b / \sqrt{\gamma R T_b} = 970 \text{ ft/s} / \sqrt{1.4(1716.561 \text{ ft}^2/\text{s}^2 \cdot \text{R})(400 \text{ R})} = \underline{0.99}$$

From the definition of the fuel-air ratio

$$R_f \equiv \rho_f V_f \mathcal{S}_f / (\rho_i V_i \mathcal{S}_i - \rho_b V_b \mathcal{S}_b)$$

or after solving for the fuel flow rate and using Eq. (2.2.8) to eliminate the bypass flow rate

$$\rho_f V_f \mathcal{S}_f = \rho_i V_i \mathcal{S}_i R_f / (1 + R_b) \quad (2.2.9)$$

Using Eqs. (2.2.8) and (2.2.9) to eliminate the bypass flow rate and fuel flow rate from Eq. (2.2.7) and solving for the nozzle exit velocity yields

$$\begin{aligned} V_e &= V_i \frac{\rho_i}{\rho_e} \frac{\mathcal{S}_i}{\mathcal{S}_e} \frac{1 + R_f}{1 + R_b} = M_i \sqrt{\gamma R T_i} \frac{p_i / R T_i}{p_e / R T_e} \frac{\mathcal{S}_i}{\mathcal{S}_e} \frac{1 + R_f}{1 + R_b} \\ &= 0.80 \sqrt{1.4(1716.561 \text{ ft}^2/\text{s}^2 \cdot \text{R})(394 \text{ R})} \frac{499/394}{532/547} \frac{164.2}{18.4} \frac{1 + 0.0060}{1 + 8.40} \\ &= \underline{968 \text{ ft/s}} \end{aligned}$$

and the Mach number at the nozzle exit is

$$M_e = V_e / \sqrt{\gamma R T_e} = 968 \text{ ft/s} / \sqrt{1.4(1716.561 \text{ ft}^2/\text{s}^2 \cdot \text{R})(547 \text{ R})} = \underline{0.84}$$

Continuity at a Point

The conservation of mass is expressed in Eq. (2.2.4) as an integral over some region in space. This form of the continuity equation is convenient for obtaining approximate solutions to fluid flow problems by assuming that the fluid density and velocity vector are constant over certain portions of the control surface, as was done in Example 1.2.1. This fluid flow model is commonly called the *uniform-flow approximation*. Strictly speaking, the flow across the inlet and exit sections of a turbofan engine may not be uniform. Nevertheless, the agreement between experimental data and results obtained from such one-dimensional analysis is adequate to make the uniform-flow approximation of great practical value for some applications.

Although Eq. (2.2.4) is generally valid for fluid flow problems, its use is limited. Often there is insufficient knowledge regarding the details of the flow to afford accurate approximations for the integrals in Eq. (2.2.4). Approximations like the uniform-flow approximation often lead to significant error. If more detailed information is required regarding spatial variations in the flow, it can be acquired in one of two ways. One can obtain information on spatial variations in a fluid flow field experimentally. However, obtaining such experimental data is often very time consuming and expensive. The other option is an analytical method that allows us to predict the spatial variations in a fluid flow field. This method leads to differential equations rather than integral equations.

Consider flow through the fixed differential Cartesian element shown in Fig. 2.2.3. The element is centered at an arbitrary fixed point with coordinates (x, y, z) where the fluid velocity vector is $\mathbf{V}$ and the density is ρ . The element's lengths in the three coordinate directions are dx , dy , and dz , respectively. As shown in Fig. 2.2.3, in the limit as dx , dy , and dz approach zero, $\rho \mathbf{V} \cdot \mathbf{n} d\mathcal{S}$ on the left-hand face of this differential element is

$$\begin{aligned} (\rho \mathbf{V} \cdot \mathbf{n} d\mathcal{S})_{\text{left}} &= \left(\rho - \frac{\partial \rho}{\partial x} \frac{dx}{2} \right) \left[- \left(V_x - \frac{\partial V_x}{\partial x} \frac{dx}{2} \right) \right] dy dz \\ &= \left(-\rho V_x + V_x \frac{\partial \rho}{\partial x} \frac{dx}{2} + \rho \frac{\partial V_x}{\partial x} \frac{dx}{2} - \frac{\partial \rho}{\partial x} \frac{\partial V_x}{\partial x} \frac{dx^2}{4} \right) dy dz \end{aligned}$$

Similarly, $\rho \mathbf{V} \cdot \mathbf{n} d\mathcal{S}$ on the right-hand face of this differential element is

$$\begin{aligned} (\rho \mathbf{V} \cdot \mathbf{n} d\mathcal{S})_{\text{right}} &= \left(\rho + \frac{\partial \rho}{\partial x} \frac{dx}{2} \right) \left(V_x + \frac{\partial V_x}{\partial x} \frac{dx}{2} \right) dy dz \\ &= \left(\rho V_x + V_x \frac{\partial \rho}{\partial x} \frac{dx}{2} + \rho \frac{\partial V_x}{\partial x} \frac{dx}{2} + \frac{\partial \rho}{\partial x} \frac{\partial V_x}{\partial x} \frac{dx^2}{4} \right) dy dz \end{aligned}$$

Thus, the sum of $\rho \mathbf{V} \cdot \mathbf{n} d\mathcal{S}$ on the left- and right-hand faces of this differential element is

$$(\rho \mathbf{V} \cdot \mathbf{n} d\mathcal{S})_{\text{left}} + (\rho \mathbf{V} \cdot \mathbf{n} d\mathcal{S})_{\text{right}} = \left(V_x \frac{\partial \rho}{\partial x} + \rho \frac{\partial V_x}{\partial x} \right) dx dy dz \quad (2.2.10)$$

Also shown in Fig. 2.2.3, in the limit as dx , dy , and dz approach zero, the sum of $\rho \mathbf{V} \cdot \mathbf{n} d\mathcal{S}$ on the bottom and top faces of this differential element can be obtained from

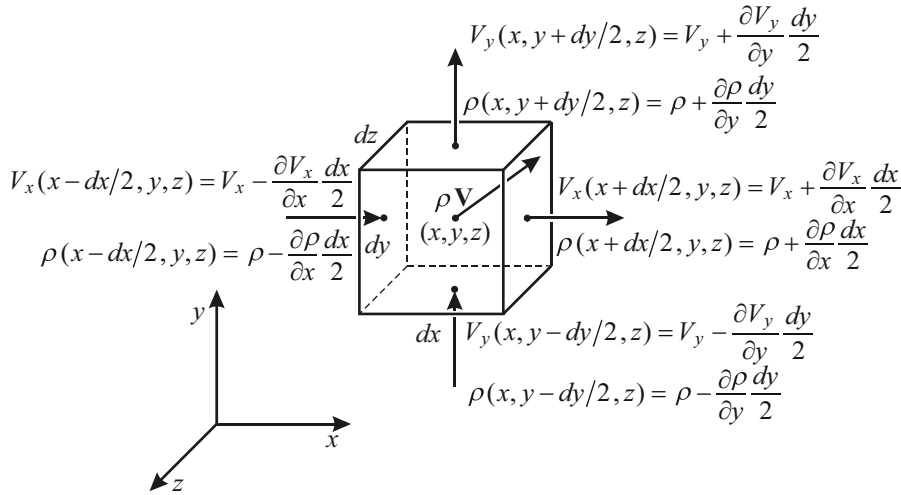


Figure 2.2.3. Density and normal velocity components at the surface of a fixed differential fluid element in Cartesian coordinates.

$$\begin{aligned}
 (\rho \mathbf{V} \cdot \mathbf{n} d\mathcal{S})_{\text{bottom}} &= \left(\rho - \frac{\partial \rho}{\partial y} \frac{dy}{2} \right) \left[- \left(V_y - \frac{\partial V_y}{\partial y} \frac{dy}{2} \right) \right] dx dz \\
 &= \left(-\rho V_y + V_y \frac{\partial \rho}{\partial y} \frac{dy}{2} + \rho \frac{\partial V_y}{\partial y} \frac{dy}{2} - \frac{\partial \rho}{\partial y} \frac{\partial V_y}{\partial y} \frac{dy^2}{4} \right) dx dz \\
 (\rho \mathbf{V} \cdot \mathbf{n} d\mathcal{S})_{\text{top}} &= \left(\rho + \frac{\partial \rho}{\partial y} \frac{dy}{2} \right) \left(V_y + \frac{\partial V_y}{\partial y} \frac{dy}{2} \right) dx dz \\
 &= \left(\rho V_y + V_y \frac{\partial \rho}{\partial y} \frac{dy}{2} + \rho \frac{\partial V_y}{\partial y} \frac{dy}{2} + \frac{\partial \rho}{\partial y} \frac{\partial V_y}{\partial y} \frac{dy^2}{4} \right) dx dz \\
 (\rho \mathbf{V} \cdot \mathbf{n} d\mathcal{S})_{\text{bottom}} + (\rho \mathbf{V} \cdot \mathbf{n} d\mathcal{S})_{\text{top}} &= \left(V_y \frac{\partial \rho}{\partial y} + \rho \frac{\partial V_y}{\partial y} \right) dx dy dz \quad (2.2.11)
 \end{aligned}$$

Similarly, the sum of $\rho \mathbf{V} \cdot \mathbf{n} d\mathcal{S}$ on the back and front faces of this differential element, which are not shown in Fig. 2.2.3, is found to be

$$(\rho \mathbf{V} \cdot \mathbf{n} d\mathcal{S})_{\text{back}} + (\rho \mathbf{V} \cdot \mathbf{n} d\mathcal{S})_{\text{front}} = \left(V_z \frac{\partial \rho}{\partial z} + \rho \frac{\partial V_z}{\partial z} \right) dx dy dz \quad (2.2.12)$$

The sum of Eqs. (2.2.10), (2.2.11), and (2.2.12) gives **the net mass flow rate exiting this fixed differential control volume** in the limit as dx , dy , and dz approach zero,

$$\begin{aligned}
 \iint_{\mathcal{S}} \rho \mathbf{V} \cdot \mathbf{n} d\mathcal{S} &= \left(V_x \frac{\partial \rho}{\partial x} + V_y \frac{\partial \rho}{\partial y} + V_z \frac{\partial \rho}{\partial z} \right) dx dy dz \\
 &\quad + \rho \left(\frac{\partial V_x}{\partial x} + \frac{\partial V_y}{\partial y} + \frac{\partial V_z}{\partial z} \right) dx dy dz \quad (2.2.13)
 \end{aligned}$$

Because the product of dx , dy , and dz is the volume of the differential element shown in Fig. 2.2.3 and ρ is the mass per unit volume at the element center, in the limit as dx , dy , and dz approach zero, **the mass contained within this fixed differential element is**

$$\iiint_V \rho d\mathcal{V} = \rho dx dy dz \quad (2.2.14)$$

Because this is a fixed control volume, $\mathbf{V}_s = 0$ and $\rho(\mathbf{V} - \mathbf{V}_s) \cdot \mathbf{n} d\mathcal{S} = \rho \mathbf{V} \cdot \mathbf{n} d\mathcal{S}$. Thus, using Eqs. (2.2.13) and (2.2.14) in Eq. (2.2.4) and dividing by dx , dy , and dz yields **the continuity equation at a fixed point within a fluid flow field in Cartesian coordinates**

$$\frac{\partial \rho}{\partial t} + V_x \frac{\partial \rho}{\partial x} + V_y \frac{\partial \rho}{\partial y} + V_z \frac{\partial \rho}{\partial z} + \rho \left(\frac{\partial V_x}{\partial x} + \frac{\partial V_y}{\partial y} + \frac{\partial V_z}{\partial z} \right) = 0 \quad (2.2.15)$$

The differential continuity equation given by Eq. (2.2.15) is valid for any fluid flow field; steady or unsteady, compressible or incompressible, viscous or inviscid, Newtonian or non-Newtonian.

The sum of the first four terms in Eq. (2.2.15) is commonly called the *substantial derivative* of ρ and is denoted as

$$\frac{D\rho}{Dt} \equiv \frac{\partial\rho}{\partial t} + V_x \frac{\partial\rho}{\partial x} + V_y \frac{\partial\rho}{\partial y} + V_z \frac{\partial\rho}{\partial z} \equiv \frac{\partial\rho}{\partial t} + \mathbf{V} \cdot \nabla \rho \quad (2.2.16)$$

where $\nabla \rho$ is the gradient of ρ as defined in Eq. (1.4.22). The physical significance of the substantial derivative is that it represents the time rate of change in the fluid density at a point that moves with the fluid. It should also be noted that the sum of the three terms in the brackets in Eq. (2.2.15) is the divergence of $\mathbf{V}$ as defined in Eq. (1.4.23),

$$\nabla \cdot \mathbf{V} \equiv \frac{\partial V_x}{\partial x} + \frac{\partial V_y}{\partial y} + \frac{\partial V_z}{\partial z} \quad (2.2.17)$$

Thus, using the definitions from Eqs. (2.2.16) and (2.2.17) in Eq. (2.2.15), **the continuity equation at a fixed point within a fluid flow field can be written in vector notation,**

$$\frac{\partial\rho}{\partial t} + \mathbf{V} \cdot \nabla \rho + \rho \nabla \cdot \mathbf{V} = 0 \quad (2.2.18)$$

Although Eq. (2.2.18) was developed here in Cartesian coordinates, it is valid for any coordinate system. This can be shown from an alternate development, which is purely mathematical. Although Eq. (2.2.4) applies to any control volume, if the control volume is fixed relative to the coordinate system, the integration limits are independent of time, and the time derivative of the integral in the first term in Eq. (2.2.4) can be written as the integral of the time derivative. Furthermore, since $\mathbf{V}_s$ is zero for a fixed control volume, $(\mathbf{V} - \mathbf{V}_s) \cdot \mathbf{n} = \mathbf{V} \cdot \mathbf{n}$. Hence, **for a fixed control volume, Eq. (2.2.4) can be written**

$$\iiint_{\mathcal{V}} \frac{\partial\rho}{\partial t} d\mathcal{V} + \iint_{\mathcal{S}} \rho \mathbf{V} \cdot \mathbf{n} d\mathcal{S} = 0 \quad (2.2.19)$$

A mathematical identity known as the **divergence theorem** requires that, for any arbitrary vector field $\mathbf{Q}$, the integral of the divergence of $\mathbf{Q}$ over an arbitrary volume $\mathcal{V}$ enclosed by the surface $\mathcal{S}$ must be equal to the dot product of $\mathbf{Q}$ with the unit outward surface normal $\mathbf{n}$ integrated over the enclosing surface, i.e.,

$$\iiint_{\mathcal{V}} \nabla \cdot \mathbf{Q} d\mathcal{V} = \iint_{\mathcal{S}} \mathbf{Q} \cdot \mathbf{n} d\mathcal{S} \quad (2.2.20)$$

If we let $\mathbf{Q} = \rho \mathbf{V}$, then the divergence theorem requires

$$\iiint_{\mathcal{V}} \nabla \cdot (\rho \mathbf{V}) d\mathcal{V} = \iint_{\mathcal{S}} \rho \mathbf{V} \cdot \mathbf{n} d\mathcal{S} \quad (2.2.21)$$

From the product rule for differentiation, it is readily shown that the divergence of the product of a scalar field and a vector field obeys the associative law,

$$\nabla \cdot (\rho \mathbf{V}) = \mathbf{V} \cdot \nabla \rho + \rho \nabla \cdot \mathbf{V} \quad (2.2.22)$$

Thus, using the mathematical identities given by Eqs. (2.2.21) and (2.2.22), the surface integral in the second term in Eq. (2.2.19) can be written as a volume integral,

$$\iiint_V \frac{\partial \rho}{\partial t} d\mathcal{V} + \iiint_V (\mathbf{V} \cdot \nabla \rho + \rho \nabla \cdot \mathbf{V}) d\mathcal{V} = 0$$

After combining the two volume integrals, **the continuity equation for flow through an arbitrary fixed control volume can be written as**

$$\iiint_V \left(\frac{\partial \rho}{\partial t} + \mathbf{V} \cdot \nabla \rho + \rho \nabla \cdot \mathbf{V} \right) d\mathcal{V} = 0 \quad (2.2.23)$$

It is important to recognize that Eqs. (2.2.4) and (2.2.23) are equivalent statements of the conservation of mass when applied to a fixed control volume. Since only mathematical identities and the assumption of a fixed control volume were used to develop Eq. (2.2.23) from Eq. (2.2.4), these equations are mathematically identical for a fixed control volume.

Further examination of Eq. (2.2.23) reveals that, if the integrand were finite over any portion of the control volume, the integral over one part of the control volume must be equal in magnitude and opposite in sign to the integral over some other part of the control volume. However, because the fixed volume $\mathcal{V}$ is completely arbitrary, the integral must be zero over any and all portions of the control volume. This is possible only if the integrand is zero at every point within the control volume. Hence, Eq. (2.2.23) can be satisfied for an arbitrary fixed volume $\mathcal{V}$ only if

$$\frac{\partial \rho}{\partial t} + \mathbf{V} \cdot \nabla \rho + \rho \nabla \cdot \mathbf{V} = 0$$

at all points within the flow field. This is exactly the result previously developed for Cartesian coordinates and presented in Eq. (2.2.18). Because the mathematical identities used to develop Eq. (2.2.23) from Eq. (2.2.4) are independent of the coordinate system, **Eq. (2.2.18) is not restricted to Cartesian coordinates.**

Using Eqs. (1.4.27) and (1.4.28) in Eq. (2.2.18) yields **the continuity equation at a fixed point within a fluid flow field in cylindrical coordinates**

$$\frac{\partial \rho}{\partial t} + V_r \frac{\partial \rho}{\partial r} + \frac{V_\theta}{r} \frac{\partial \rho}{\partial \theta} + V_z \frac{\partial \rho}{\partial z} + \rho \left(\frac{\partial V_r}{\partial r} + \frac{1}{r} \frac{\partial V_\theta}{\partial \theta} + \frac{\partial V_z}{\partial z} + \frac{V_r}{r} \right) = 0 \quad (2.2.24)$$

Similarly, using Eqs. (1.4.33) and (1.4.34) in Eq. (2.2.18) yields **the continuity equation at a fixed point within a fluid flow field in spherical coordinates**

$$\begin{aligned} \frac{\partial \rho}{\partial t} + V_r \frac{\partial \rho}{\partial r} + \frac{V_\theta}{r} \frac{\partial \rho}{\partial \theta} + \frac{V_\phi}{r \sin \theta} \frac{\partial \rho}{\partial \phi} \\ + \rho \left(\frac{\partial V_r}{\partial r} + \frac{1}{r} \frac{\partial V_\theta}{\partial \theta} + \frac{1}{r \sin \theta} \frac{\partial V_\phi}{\partial \phi} + \frac{2V_r}{r} + \frac{V_\theta \cos \theta}{r \sin \theta} \right) = 0 \end{aligned} \quad (2.2.25)$$

Equations (2.2.15), (2.2.18), (2.2.24), and (2.2.25) are valid for any fluid flow field.

If the flow is incompressible, the fluid density is constant for all time and at all points within the flow field, even if the velocity is unsteady. Because density is constant in an incompressible flow, the time derivative and the gradient of ρ are identically zero. Thus, from Eq. (2.2.18), **the continuity equation at a fixed point in an incompressible flow field reduces to**

$$\nabla \cdot \mathbf{V} = 0 \quad (2.2.26)$$

In **Cartesian coordinates** this yields

$$\frac{\partial V_x}{\partial x} + \frac{\partial V_y}{\partial y} + \frac{\partial V_z}{\partial z} = 0 \quad (2.2.27)$$

in **cylindrical coordinates** Eq. (2.2.26) becomes

$$\frac{\partial V_r}{\partial r} + \frac{1}{r} \frac{\partial V_\theta}{\partial \theta} + \frac{\partial V_z}{\partial z} + \frac{V_r}{r} = 0 \quad (2.2.28)$$

and in **spherical coordinates** the incompressible continuity equation requires

$$\frac{\partial V_r}{\partial r} + \frac{1}{r} \frac{\partial V_\theta}{\partial \theta} + \frac{1}{r \sin \theta} \frac{\partial V_\phi}{\partial \phi} + \frac{2V_r}{r} + \frac{V_\theta \cos \theta}{r \sin \theta} = 0 \quad (2.2.29)$$

EXAMPLE 2.2.2. Consider a compressible fluid flow field that has the following velocity components in Cartesian coordinates:

$$V_x = V_0 x \cos(\omega t)/(c - y), \quad V_y = 0, \quad V_z = 0$$

where V_0 , ω , and c are constants. The density ρ is a function of y and t only. At time $t=0$, the density is $\rho(y,0)=\rho_0$, where ρ_0 is constant. Find the density $\rho(y,t)$.

Solution. For an unsteady, compressible flow, the continuity equation in Cartesian coordinates is given by Eq. (2.2.15)

$$\frac{\partial \rho}{\partial t} + V_x \frac{\partial \rho}{\partial x} + V_y \frac{\partial \rho}{\partial y} + V_z \frac{\partial \rho}{\partial z} + \rho \left(\frac{\partial V_x}{\partial x} + \frac{\partial V_y}{\partial y} + \frac{\partial V_z}{\partial z} \right) = 0$$

From the given velocity components, this yields

$$\frac{\partial \rho}{\partial t} + V_0 \frac{x \cos(\omega t)}{c - y} \frac{\partial \rho}{\partial x} + \rho V_0 \frac{\cos(\omega t)}{c - y} = 0$$

Because ρ is a function of y and t only, the change in ρ with respect to x is zero and the continuity equation can be further reduced to

$$\frac{\partial \rho}{\partial t} + \rho V_0 \frac{\cos(\omega t)}{c - y} = 0$$

This partial differential equation can be solved by separation of variables

$$\int \frac{d\rho}{\rho} = -\frac{V_0}{c-y} \int \cos(\omega t) dt$$

which yields

$$\ln \rho = -\frac{V_0}{\omega(c-y)} \sin(\omega t) + f(y)$$

This general solution contains the arbitrary function $f(y)$, which can be evaluated from the initial condition, i.e., $\rho(y,0)=\rho_0$. Applying this initial condition yields $\ln \rho_0 = f(y)$ and the complete solution is

$$\rho(y,t) = \rho_0 \exp\left[-\frac{V_0}{\omega(c-y)} \sin(\omega t)\right]$$

EXAMPLE 2.2.3. Consider an incompressible fluid flow field with the following velocity components in cylindrical coordinates:

$$V_\theta = -V_\infty(1+R^2/r^2) \sin \theta, \quad V_z = 0$$

where V_∞ and R are constants. The r -velocity component is a function of r and θ only. At the radial coordinate $r=R$, the r -velocity component is $V_r(R,\theta)=0$. Find the relation for the r -velocity component at an arbitrary point, $V_r(r,\theta)$.

Solution. For an incompressible flow field, the continuity equation in cylindrical coordinates is given by Eq. (2.2.28)

$$\frac{\partial V_r}{\partial r} + \frac{1}{r} \frac{\partial V_\theta}{\partial \theta} + \frac{\partial V_z}{\partial z} + \frac{V_r}{r} = 0$$

From the known velocity components, this reduces to

$$\frac{\partial V_r}{\partial r} - V_\infty \left(\frac{1}{r} + \frac{R^2}{r^3} \right) \cos \theta + \frac{V_r}{r} = 0$$

or after rearranging

$$r \frac{\partial V_r}{\partial r} + V_r = V_\infty \left(1 + \frac{R^2}{r^2} \right) \cos \theta$$

$$\frac{\partial(rV_r)}{\partial r} = V_\infty \left(1 + \frac{R^2}{r^2} \right) \cos \theta$$

This partial differential equation is readily solved by separating the variables and integrating, which yields

$$rV_r = V_\infty \left(r - \frac{R^2}{r} \right) \cos \theta + f(\theta)$$

This general solution contains the arbitrary function $f(\theta)$, which can be evaluated from the boundary condition. Applying the boundary condition $V_r(R, \theta) = 0$ yields

$$0 = V_\infty \left(R - \frac{R^2}{R} \right) \cos \theta + f(\theta) \quad \text{or} \quad f(\theta) = 0$$

and

$$V_r = V_\infty \left(1 - \frac{R^2}{r^2} \right) \cos \theta$$

2.3. Newton's Second Law: The Euler Equations

Students should be familiar with Newton's laws of motion from an introductory course in dynamics. These fundamental laws of physics relate the motion of a body to the forces acting on the body. **Newton's first law** states that, *when the net force acting on a body is zero, a body at rest remains at rest, and a body in motion remains in motion along a straight line at constant velocity.* **Newton's second law** states that, *the acceleration of a body is proportional to the total net force acting on the body and inversely proportional to the mass of the body.* **Newton's third law** states that, *when one body exerts a force on a second body, the second body exerts an equal and opposite force on the first body.*

Students should be very familiar with the statement of Newton's second law in the commonly used form, *force equals mass times acceleration*, which is typically written as

$$\mathbf{F} = m\mathbf{a} \quad (2.3.1)$$

where $\mathbf{F}$ is the total net force vector acting on a body, m is the mass of the body, and $\mathbf{a}$ is the acceleration vector of the body's center of mass. Students should also remember that the statement of Newton's second law given by Eq. (2.3.1) applies to only a Lagrangian system, which is defined to be an identifiable quantity of matter having fixed mass. In the traditional statements of Newton's laws of motion given above, the term *body* is synonymous with a *Lagrangian system*.

The acceleration of a body's center of mass is simply the derivative of the velocity of the center of mass with respect to time. Because the mass of a Lagrangian system is constant by definition, the product of the system mass and the acceleration of the center of mass is equal to the time derivative of the product of the system mass and the velocity of the center of mass. Hence, from Eq. (2.3.1), **Newton's second law for a Lagrangian system can be written equivalently as**

$$\mathbf{F} = \frac{d}{dt}(m\bar{\mathbf{V}}) \quad (2.3.2)$$

where $\bar{\mathbf{V}}$ is the velocity vector of the center of mass and t denotes time.

The product of the system mass and the velocity of the center of mass is a vector called the *linear momentum* or simply the *momentum* of the system. Thus, Eq. (2.3.2) provides an alternate definition for **Newton's second law** that states, *the time rate of change of the momentum of a Lagrangian system is equal to the total net force acting on the system*. Although the total mass included in any Lagrangian system remains constant, this mass need not be distributed uniformly over the system, and the velocity need not be the same at all points within the system.

Consider the arbitrary Lagrangian system of fixed mass that is shown in Fig. 2.3.1. This system may move and deform with time, but no mass crosses the boundary of this system. Let $d\mathcal{V}$ denote a differential volume centered at a point within the system where the instantaneous local density is ρ and the instantaneous local velocity vector is $\mathbf{V}$. In the limit as the differential volume $d\mathcal{V}$ approaches zero, the mass contained within $d\mathcal{V}$ is the product $\rho d\mathcal{V}$, and momentum of the mass contained within $d\mathcal{V}$ is the product $\rho \mathbf{V} d\mathcal{V}$. Thus, integrating over the instantaneous system volume $\mathcal{V}$, **the total momentum of the mass contained within any finite volume at any instant in time is**

$$m\bar{\mathbf{V}} = \iiint_{\mathcal{V}} \rho \mathbf{V} d\mathcal{V} \quad (2.3.3)$$

Similarly, from Eq. (1.5.8), the gravitational force per unit volume acting on any substance is $-\rho g \nabla H$, where ∇H is the gradient of the geometric altitude. Multiplying by the differential volume and integrating over the instantaneous volume $\mathcal{V}$, **the total gravitational force vector acting on the mass contained within any finite volume is**

$$\mathbf{F}_g = - \iiint_{\mathcal{V}} \rho g \nabla H d\mathcal{V} \quad (2.3.4)$$

Pressure also acts on any surface surrounded by a fluid. Because pressure is a force per unit area that acts in the direction opposite to the local outward surface normal $\mathbf{n}$, the pressure force vector acting on the differential surface area $d\mathcal{S}$ is $-p \mathbf{n} d\mathcal{S}$. Integrating over the entire instantaneous surface area $\mathcal{S}$, **the total resultant pressure force vector that acts on any finite surface area is**

$$\mathbf{F}_p = - \iint_{\mathcal{S}} p \mathbf{n} d\mathcal{S} \quad (2.3.5)$$

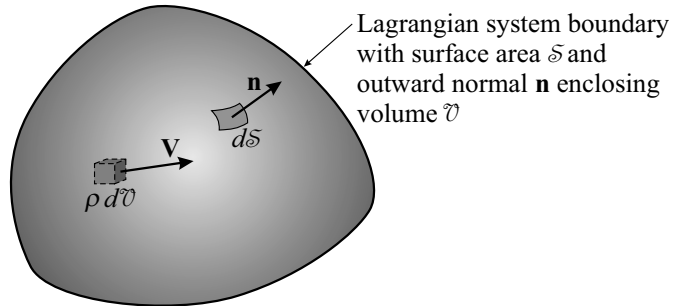


Figure 2.3.1. Lagrangian system in a fluid flow field.

Using Eq. (2.3.3) in Eq. (2.3.2) provides a more detailed and rigorous statement of **Newton's second law for any Lagrangian system**

$$\frac{d}{dt} \iiint_{\mathcal{V}} \rho \mathbf{V} d\mathcal{V} = \mathbf{F} \quad (2.3.6)$$

where $\mathbf{V}$ is the velocity vector relative to an *inertial coordinate system*, which is at rest or moves with constant velocity; and $\mathbf{F}$ is the total net force vector acting on the Lagrangian system, which includes all body and surface forces combined. Reiterating, **Newton's second law states, the time rate of change of the momentum of a Lagrangian system is equal to the total net force acting on the system.** This is more consistent with Newton's original statement of the second law, and it is more convenient for use in fluid dynamics, where the mass is not necessarily distributed uniformly over the system volume, and the instantaneous velocity is not necessarily the same at all points within the system volume. In light of Eq. (2.3.6), Newton's second law is also commonly referred to as the *linear momentum equation*.

The Lagrangian description of Newton's second law stated in Eq. (2.3.6) is of limited use in fluid dynamics, because it is often difficult to distinguish and follow the same fixed quantity of fluid as it deforms and moves through a device or around an object. Hence, in the study of fluid dynamics, it is typically more convenient to employ the Eulerian description of Newton's second law. To encompass the Eulerian description, **Newton's second law could be restated as, the rate at which momentum is generated within any volume is equal to the total net force acting on the volume.** For any Lagrangian system, the momentum generated within the system remains in the system. However, when mass crosses the boundary of an Eulerian control volume it carries its momentum with it.

Consider the arbitrary Eulerian control volume shown in Fig. 2.3.2. Let $\mathbf{V}$ denote the instantaneous local fluid velocity vector relative to an inertial coordinate system and let $\mathbf{V}_s$ represent the instantaneous local velocity of the enclosing control surface relative to this same coordinate system. As shown in Fig. 2.2.2 and discussed in the development of

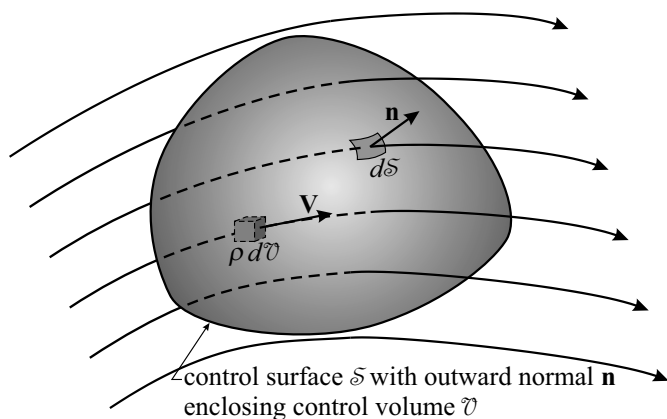


Figure 2.3.2. Momentum transport through an Eulerian control volume in a fluid flow field.

Eq. (2.2.4), the volume flow rate of the fluid that exits the control volume through the differential surface area $d\mathcal{S}$ with unit outward normal $\mathbf{n}$ is $(\mathbf{V} - \mathbf{V}_s) \cdot \mathbf{n} d\mathcal{S}$. Because ρ is the instantaneous local mass per unit volume, the mass flow rate is $\rho(\mathbf{V} - \mathbf{V}_s) \cdot \mathbf{n} d\mathcal{S}$; and because the instantaneous local fluid velocity vector $\mathbf{V}$ is the momentum per unit mass, the rate at which momentum leaves the control volume through the differential surface area $d\mathcal{S}$ is $\mathbf{V}[\rho(\mathbf{V} - \mathbf{V}_s) \cdot \mathbf{n}] d\mathcal{S}$. Integrating this differential momentum flux over the entire control surface $\mathcal{S}$ yields **the net rate at which momentum is transported out of a control volume as a result of mass flow through the control surface**

$$\text{net outward momentum flux} = \iint_{\mathcal{S}} \mathbf{V}[\rho(\mathbf{V} - \mathbf{V}_s) \cdot \mathbf{n}] d\mathcal{S} \quad (2.3.7)$$

Newton's second law requires that the momentum generated within a control volume as a result of external forces applied to the control volume must result in an increase in the momentum of the control volume and/or a net momentum flux leaving the control volume. The total momentum contained within any finite volume at any instant in time is given by Eq. (2.3.3), and the net momentum flux leaving a control volume is given by Eq. (2.3.7). Thus, **Newton's second law applied to an Eulerian control volume gives**

$$\frac{d}{dt} \iiint_{\mathcal{V}} \rho \mathbf{V} d\mathcal{V} + \iint_{\mathcal{S}} \mathbf{V}[\rho(\mathbf{V} - \mathbf{V}_s) \cdot \mathbf{n}] d\mathcal{S} = \mathbf{F} \quad (2.3.8)$$

where $\mathbf{V}$ and $\mathbf{V}_s$ are the fluid and control surface velocity vectors relative to an inertial coordinate system, and $\mathbf{F}$ is the total net force vector acting on the control volume. In this form, **Newton's second law states, the time rate of change of the momentum of an Eulerian control volume plus the net rate at which momentum is transported out of the control volume as a result of mass flow through the control surface is equal to the total net force acting on the Eulerian control volume.** The integral form of Newton's second law expressed in Eq. (2.3.8) is valid for any Eulerian control volume in any fluid flow field; steady or unsteady, compressible or incompressible, viscous or inviscid, Newtonian or non-Newtonian.

Two special cases of Eq. (2.3.8) are of particular interest. If all points on the control surface move continuously with the fluid, then $\mathbf{V}_s = \mathbf{V}$ and no mass crosses the control surfaces. Under this condition, an Eulerian control volume becomes a Lagrangian system and Eq. (2.3.8) reduces to Eq. (2.3.6). On the other hand, if the control surface is fixed relative to the coordinate system, $\mathbf{V}_s = 0$ and Eq. (2.3.8) reduces to **Newton's second law for a fixed Eulerian control volume**

$$\frac{d}{dt} \iiint_{\mathcal{V}} \rho \mathbf{V} d\mathcal{V} + \iint_{\mathcal{S}} \mathbf{V}(\rho \mathbf{V} \cdot \mathbf{n}) d\mathcal{S} = \mathbf{F} \quad (2.3.9)$$

Because both momentum and force are vector quantities, Newton's second law is a vector equation, which has three scalar components. Because all three components of any vector equation must be satisfied independently, the statements of Newton's second law presented in Eqs. (2.3.6), (2.3.8), and (2.3.9) should each be viewed as a system of three scalar equations. The components of Newton's second law are typically resolved

using some orthogonal coordinate system, such as Cartesian, cylindrical, or spherical coordinates. The forces are typically comprised of weights, pressure forces, shear forces, and reaction forces acting on support structures and connectors.

EXAMPLE 2.3.1. Consider steady horizontal flow through the turbojet engine shown in Fig. 2.3.3. Air enters the engine through inlet section i with a mass flow rate $\dot{m}_i$. Fuel also enters the engine at the mass flow rate $\dot{m}_f$ through a fuel line (not shown) and the products of combustion exit the engine through the nozzle exit section e . The engine is attached to an aircraft with a mounting structure that supports the required reaction force as shown above the engine in Fig. 2.3.3. The engine is operating at standard sea level with the following parameters:

$$\begin{aligned} \text{airspeed} &= 0.0, \quad \text{fuel-air ratio} \equiv R_f \equiv \dot{m}_f / \dot{m}_i = 0.0060, \quad p_\infty = 2,116.2 \text{ lbf/ft}^2, \\ \text{inlet area} &\equiv \mathcal{S}_i = 46.0 \text{ ft}^2, \quad T_i = 509.5 \text{ R}, \quad p_i = 1,988.6 \text{ lbf/ft}^2, \quad M_i = 0.300, \\ \text{nozzle exit area} &\equiv \mathcal{S}_e = 11.4 \text{ ft}^2, \quad T_e = 1,020.3 \text{ R}, \quad p_e = 2,128.6 \text{ lbf/ft}^2 \end{aligned}$$

Using the uniform flow approximation at the inlet and exit and assuming the gas constants for the products of combustion are the same as for air, find the velocity and Mach number at the nozzle exit and the thrust developed by the engine.

Solution. For the fixed control volume shown in Fig. 2.3.3, $\mathbf{V}_s = 0$ and $\mathbf{V} \cdot \mathbf{n}$ is zero everywhere on the control surface except in the fuel line and on sections i and e . Thus, the steady flow continuity equation given by Eq. (2.2.5) reduces to

$$\iint_{\mathcal{S}_f} \rho_f (-V_f) d\mathcal{S} + \iint_{\mathcal{S}_i} \rho_i (-V_i) d\mathcal{S} + \iint_{\mathcal{S}_e} \rho_e (V_e) d\mathcal{S} = 0$$

With the uniform flow approximation, the density and velocity are assumed to be constant over each of these three sections. Hence, the integration yields

$$-\rho_f V_f \mathcal{S}_f - \rho_i V_i \mathcal{S}_i + \rho_e V_e \mathcal{S}_e = 0 \quad (2.3.10)$$

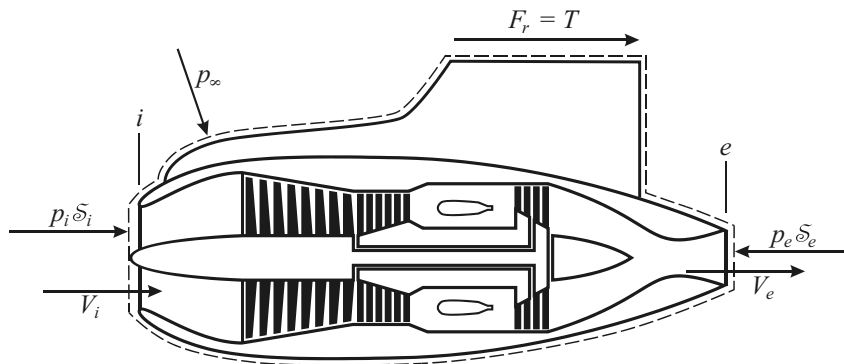


Figure 2.3.3. Eulerian control volume enclosing a turbojet engine.

From the definition of fuel-air ratio $R_f \equiv \rho_f V_f \delta_f / (\rho_i V_i \delta_i)$, the fuel flow rate is

$$\rho_f V_f \delta_f = \rho_i V_i \delta_i R_f \quad (2.3.11)$$

Using Eq. (2.3.11) to eliminate the fuel flow rate from Eq. (2.3.10) and solving for the nozzle exit velocity yields

$$V_e = V_i \frac{\rho_i}{\rho_e} \frac{\delta_i}{\delta_e} (1 + R_f) = M_i \sqrt{\gamma RT_i} \frac{p_i/RT_i}{p_e/RT_e} \frac{\delta_i}{\delta_e} (1 + R_f) = \underline{2,521.0 \text{ ft/s}}$$

and the Mach number at the nozzle exit is

$$M_e = V_e / \sqrt{\gamma RT_e} = \underline{1.610}$$

Because the flow is defined to be horizontal, the thrust is horizontal and can be evaluated from the horizontal component of Newton's second law. Because the control volume is fixed and the flow is steady, Newton's second law is obtained from Eq. (2.3.9) with the first term on the left-hand side set to zero. Hence, the horizontal component of Newton's second law for this control volume is

$$\iint_{\mathcal{S}} V_h (\rho \mathbf{V} \cdot \mathbf{n}) d\mathcal{S} = F_h \quad (2.3.12)$$

where V_h and F_h are the horizontal components of velocity and force, respectively.

The thrust is denoted as T and defined to be the forward force exerted on the aircraft by the engine. From Newton's third law, if the engine exerts a forward force T on the aircraft, then the aircraft must exert an equal and opposite force on the engine. Thus, the horizontal component of the reaction force in the mounting structure, F_r , is in the aft direction and equal in magnitude to the thrust developed by the engine. Because the airspeed is zero, there is no relative motion between the engine's outer solid surface and the surrounding air, so there can be no shear stress acting on the outer surface and the pressure on this surface is atmospheric pressure p_∞ . Hence, the total net surface force exerted on the engine is only the pressure force given by Eq. (2.3.5), which can be written as

$$\mathbf{F}_p = - \iint_{\mathcal{S}} p \mathbf{n} d\mathcal{S} = - \iint_{\mathcal{S}} (p - p_\infty) \mathbf{n} d\mathcal{S} - \iint_{\mathcal{S}} p_\infty \mathbf{n} d\mathcal{S} \quad (2.3.13)$$

From our study of fluid statics, we know that the integral of $-p_\infty \mathbf{n} d\mathcal{S}$ over any closed surface submerged in a fluid is the buoyant force, which has no component in the horizontal plane. Furthermore, because the gravitational force is normal to the horizontal plane, it does not contribute to the summation of horizontal forces acting on the engine. Thus, the total horizontal component of force acting on the engine is the sum of the horizontal components of the reaction force and the force due to the first term on the right-hand side of Eq. (2.3.13). Because $p - p_\infty$ is zero everywhere on the control surface except for the inlet and exit sections, choosing the positive direction to be aft, the net horizontal force acting on the engine is

$$F_h = T + (p_i - p_\infty) \delta_i - (p_e - p_\infty) \delta_e \quad (2.3.14)$$

Neglecting the momentum associated with the fuel entering the engine, the integral on the left-hand side of Eq. (2.3.12) becomes

$$\begin{aligned}\iint_{\mathcal{S}} V_h(\rho \mathbf{V} \cdot \mathbf{n}) d\mathcal{S} &= \iint_{\mathcal{S}_e} V_e(\rho_e V_e) d\mathcal{S} + \iint_{\mathcal{S}_i} V_i(-\rho_i V_i) d\mathcal{S} \\ &= \rho_e V_e^2 \mathcal{S}_e - \rho_i V_i^2 \mathcal{S}_i = \frac{p_e}{RT_e} M_e^2 \gamma RT_e \mathcal{S}_e - \frac{p_i}{RT_i} M_i^2 \gamma RT_i \mathcal{S}_i \quad (2.3.15) \\ &= \gamma(p_e M_e^2 \mathcal{S}_e - p_i M_i^2 \mathcal{S}_i)\end{aligned}$$

Using Eqs. (2.3.14) and (2.3.15) in Eq. (2.3.12) and solving for the thrust yields

$$T = \gamma(p_e M_e^2 \mathcal{S}_e - p_i M_i^2 \mathcal{S}_i) - (p_i - p_\infty) \mathcal{S}_i + (p_e - p_\infty) \mathcal{S}_e = \underline{82,545 \text{ lbf}}$$

Newton's Second Law at a Point in an Inviscid Flow Field

In this chapter we are concerned primarily with inviscid flow. Because there can be no shear stress in an inviscid flow field, the total surface force includes only the pressure force given by Eq. (2.3.5). Furthermore, if the only body force is that due to gravity, the total net force vector acting on a finite volume and its enclosing surface is the sum of the gravitational force given by Eq. (2.3.4) and the pressure force given by Eq. (2.3.5). Using Eqs. (2.3.4) and (2.3.5) in Eq. (2.3.8), **when the only body force is that due to gravity, Newton's second law for an arbitrary Eulerian control volume within an inviscid fluid flow field becomes**

$$\frac{d}{dt} \iiint_{\mathcal{V}} \rho \mathbf{V} d\mathcal{V} + \iint_{\mathcal{S}} \mathbf{V}[\rho(\mathbf{V} - \mathbf{V}_s) \cdot \mathbf{n}] d\mathcal{S} = - \iint_{\mathcal{S}} p \mathbf{n} d\mathcal{S} - \iiint_{\mathcal{V}} \rho g \nabla H d\mathcal{V} \quad (2.3.16)$$

where $\mathbf{V}$ and $\mathbf{V}_s$ are, respectively, the fluid and control surface velocity vectors relative to an inertial coordinate system.

For the special case of a control volume that is fixed relative to an inertial coordinate system, the local control surface velocity vector $\mathbf{V}_s$ is zero at all points and the integration limits in Eq. (2.3.16) are independent of time. Hence, **when the only body force is due to gravity, Newton's second law for any fixed Eulerian control volume in an inviscid fluid flow field can be written as**

$$\iiint_{\mathcal{V}} \frac{\partial}{\partial t} (\rho \mathbf{V}) d\mathcal{V} + \iint_{\mathcal{S}} \mathbf{V}(\rho \mathbf{V} \cdot \mathbf{n}) d\mathcal{S} = - \iint_{\mathcal{S}} p \mathbf{n} d\mathcal{S} - \iiint_{\mathcal{V}} \rho g \nabla H d\mathcal{V} \quad (2.3.17)$$

Because $\mathbf{V}$, $\mathbf{n}$, and ∇H are vectors, each term in Eq. (2.3.17) is a vector with three scalar components.

The divergence theorem can be used to express the surface integral on the left-hand side of Eq. (2.3.17) as a volume integral. For example, let $\mathbf{A}$ and $\mathbf{B}$ denote two arbitrary vector fields, which are defined over an arbitrary volume $\mathcal{V}$ enclosed by the surface $\mathcal{S}$. By expanding $\mathbf{A}$ in Cartesian coordinates, we can write

$$\iint_{\mathcal{S}} \mathbf{A}(\mathbf{B} \cdot \mathbf{n}) d\mathcal{S} = \iint_{\mathcal{S}} A_x \mathbf{B} \cdot \mathbf{n} d\mathcal{S} \mathbf{i}_x + \iint_{\mathcal{S}} A_y \mathbf{B} \cdot \mathbf{n} d\mathcal{S} \mathbf{i}_y + \iint_{\mathcal{S}} A_z \mathbf{B} \cdot \mathbf{n} d\mathcal{S} \mathbf{i}_z$$

or after applying the divergence theorem given in Eq. (2.2.20) to each scalar component

$$\begin{aligned} \iint_{\mathcal{S}} \mathbf{A}(\mathbf{B} \cdot \mathbf{n}) d\mathcal{S} &= \iiint_{\mathcal{V}} \nabla \cdot (A_x \mathbf{B}) d\mathcal{V} \mathbf{i}_x \\ &+ \iiint_{\mathcal{V}} \nabla \cdot (A_y \mathbf{B}) d\mathcal{V} \mathbf{i}_y + \iiint_{\mathcal{V}} \nabla \cdot (A_z \mathbf{B}) d\mathcal{V} \mathbf{i}_z \end{aligned} \quad (2.3.18)$$

The divergence of the product of a scalar field and a vector field obeys the associative law $\nabla \cdot (A\mathbf{B}) = \mathbf{B} \cdot \nabla A + A \nabla \cdot \mathbf{B}$. Hence, Eq. (2.3.18) can be written equivalently as

$$\begin{aligned} \iint_{\mathcal{S}} \mathbf{A}(\mathbf{B} \cdot \mathbf{n}) d\mathcal{S} &= \iiint_{\mathcal{V}} (\mathbf{B} \cdot \nabla A_x + A_x \nabla \cdot \mathbf{B}) d\mathcal{V} \mathbf{i}_x \\ &+ \iiint_{\mathcal{V}} (\mathbf{B} \cdot \nabla A_y + A_y \nabla \cdot \mathbf{B}) d\mathcal{V} \mathbf{i}_y \\ &+ \iiint_{\mathcal{V}} (\mathbf{B} \cdot \nabla A_z + A_z \nabla \cdot \mathbf{B}) d\mathcal{V} \mathbf{i}_z \end{aligned}$$

which yields the **mathematical identity**

$$\iint_{\mathcal{S}} \mathbf{A}(\mathbf{B} \cdot \mathbf{n}) d\mathcal{S} = \iiint_{\mathcal{V}} [(\mathbf{B} \cdot \nabla) \mathbf{A} + \mathbf{A} \nabla \cdot \mathbf{B}] d\mathcal{V} \quad (2.3.19)$$

Another mathematical identity known as the **gradient theorem** requires that, for any scalar field q , the integral of the gradient of q over an arbitrary volume $\mathcal{V}$ enclosed by the surface $\mathcal{S}$ must equal the product of q with the unit outward surface normal $\mathbf{n}$ integrated over the enclosing surface, i.e.,

$$\iiint_{\mathcal{V}} \nabla q d\mathcal{V} = \iint_{\mathcal{S}} q \mathbf{n} d\mathcal{S} \quad (2.3.20)$$

Applying the mathematical identities given in Eqs. (2.3.19) and (2.3.20) with $\mathbf{A} = \mathbf{V}$, $\mathbf{B} = \rho \mathbf{V}$, and $q = p$, the surface integrals in Eq. (2.3.17) can be written as volume integrals and combined with the other volume integrals in Eq. (2.3.17) to yield

$$\iiint_{\mathcal{V}} \left[\frac{\partial}{\partial t} (\rho \mathbf{V}) + (\rho \mathbf{V} \cdot \nabla) \mathbf{V} + \mathbf{V} \nabla \cdot (\rho \mathbf{V}) + \nabla p + \rho g \nabla H \right] d\mathcal{V} = 0 \quad (2.3.21)$$

As discussed in relation to Eq. (2.2.23), because the fixed volume $\mathcal{V}$ is arbitrary, the integral in Eq. (2.3.21) must be zero over any and all portions of the inviscid flow field. This is possible only if the integrand is zero at every point within the flow field. Hence, Eq. (2.3.21) can be satisfied for an arbitrary fixed volume $\mathcal{V}$ only if

$$\frac{\partial}{\partial t} (\rho \mathbf{V}) + (\rho \mathbf{V} \cdot \nabla) \mathbf{V} + \mathbf{V} \nabla \cdot (\rho \mathbf{V}) + \nabla p + \rho g \nabla H = 0 \quad (2.3.22)$$

at all points within the inviscid flow field.

Equation (2.3.22) can be further simplified by making use of the continuity equation. Expanding the time derivative using $d(\rho \mathbf{V}) = \rho d\mathbf{V} + \mathbf{V} d\rho$ and applying the associative law for the divergence $\nabla \cdot (\rho \mathbf{V}) = \mathbf{V} \cdot \nabla \rho + \rho \nabla \cdot \mathbf{V}$, Eq. (2.3.22) can be rearranged as

$$\mathbf{V} \left(\frac{\partial \rho}{\partial t} + \mathbf{V} \cdot \nabla \rho + \rho \nabla \cdot \mathbf{V} \right) + \rho \left[\frac{\partial \mathbf{V}}{\partial t} + (\mathbf{V} \cdot \nabla) \mathbf{V} \right] + \nabla p + \rho g \nabla H = 0 \quad (2.3.23)$$

From the continuity equation in Eq. (2.2.18), it can be seen that the terms in parentheses, which are multiplied by $\mathbf{V}$ on the left-hand side of Eq. (2.3.23), are identical to the left-hand side of the continuity equation given in Eq. (2.2.18). Thus, this collection of terms must always sum to zero in any flow field. Applying Eq. (2.2.18) to Eq. (2.3.23) and rearranging, **when the only body force is due to gravity, Newton's second law at any fixed point within an inviscid fluid flow field can be written in vector notation as**

$$\rho \left[\frac{\partial \mathbf{V}}{\partial t} + (\mathbf{V} \cdot \nabla) \mathbf{V} \right] = -\nabla p - \rho g \nabla H \quad (2.3.24)$$

The form of Newton's second law expressed in Eq. (2.3.24) is valid for any inviscid flow field; steady or unsteady, compressible or incompressible. Because $\mathbf{V}$, ∇p , and ∇H are vectors, the statement of Newton's second law given in Eq. (2.3.24) represents a system of three scalar equations. These coupled partial differential equations are commonly called the *Euler equations*, after the Swiss mathematician Leonhard Euler (1707–1783).

The sum of the two bracketed terms on the left-hand side of Eq. (2.3.24) is the substantial derivative of the vector $\mathbf{V}$, which is commonly denoted as

$$\frac{D\mathbf{V}}{Dt} \equiv \frac{\partial \mathbf{V}}{\partial t} + (\mathbf{V} \cdot \nabla) \mathbf{V} \quad (2.3.25)$$

where $(\mathbf{V} \cdot \nabla)$ is a differential operator formed by taking the dot product of the vector $\mathbf{V}$ with the differential operator ∇ . The components of $(\mathbf{V} \cdot \nabla) \mathbf{V}$ are defined as follows:

In **Cartesian coordinates**

$$(\mathbf{V} \cdot \nabla) \mathbf{V} \equiv \mathbf{V} \cdot \nabla V_x \mathbf{i}_x + \mathbf{V} \cdot \nabla V_y \mathbf{i}_y + \mathbf{V} \cdot \nabla V_z \mathbf{i}_z \quad (2.3.26)$$

in **cylindrical coordinates**

$$(\mathbf{V} \cdot \nabla) \mathbf{V} \equiv \left(\mathbf{V} \cdot \nabla V_r - \frac{V_\theta^2}{r} \right) \mathbf{i}_r + \left(\mathbf{V} \cdot \nabla V_\theta + \frac{V_r V_\theta}{r} \right) \mathbf{i}_\theta + \mathbf{V} \cdot \nabla V_z \mathbf{i}_z \quad (2.3.27)$$

and in **spherical coordinates**

$$(\mathbf{V} \cdot \nabla) \mathbf{V} \equiv \left(\mathbf{V} \cdot \nabla V_r - \frac{V_\theta^2 + V_\phi^2}{r} \right) \mathbf{i}_r + \left(\mathbf{V} \cdot \nabla V_\theta + \frac{V_r V_\theta - V_\phi^2 \cot \theta}{r} \right) \mathbf{i}_\theta + \left(\mathbf{V} \cdot \nabla V_\phi + \frac{V_r V_\phi + V_\theta V_\phi \cot \theta}{r} \right) \mathbf{i}_\phi \quad (2.3.28)$$

The physical significance of the substantial derivative of $\mathbf{V}$ is that it represents the time rate of change of the fluid velocity (i.e., the fluid acceleration) at a point that moves with the fluid. Because ρ is the mass per unit volume, the left-hand side of Eq. (2.3.24) is the product of the local mass per unit volume and the local fluid acceleration. As was shown in the development of Eqs. (1.5.5) and (1.5.8), the two terms on the right-hand side of Eq. (2.3.24) are the pressure and gravitational forces per unit volume, respectively.

Summary: If the only body force is due to gravity and viscous effects are negligible, which is generally true for aerodynamic flows outside the boundary layer and any regions of separated flow, Newton's second law can be enforced at any point in the flow field using Eq. (2.3.24). The three scalar components of this vector differential equation are commonly called the **Euler equations**.

The Euler equations for inviscid flow in Cartesian coordinates are

$$\begin{aligned}\rho \left[\frac{\partial V_x}{\partial t} + V_x \frac{\partial V_x}{\partial x} + V_y \frac{\partial V_x}{\partial y} + V_z \frac{\partial V_x}{\partial z} \right] &= -\frac{\partial p}{\partial x} - \rho g \frac{\partial H}{\partial x} \\ \rho \left[\frac{\partial V_y}{\partial t} + V_x \frac{\partial V_y}{\partial x} + V_y \frac{\partial V_y}{\partial y} + V_z \frac{\partial V_y}{\partial z} \right] &= -\frac{\partial p}{\partial y} - \rho g \frac{\partial H}{\partial y} \\ \rho \left[\frac{\partial V_z}{\partial t} + V_x \frac{\partial V_z}{\partial x} + V_y \frac{\partial V_z}{\partial y} + V_z \frac{\partial V_z}{\partial z} \right] &= -\frac{\partial p}{\partial z} - \rho g \frac{\partial H}{\partial z}\end{aligned}\quad (2.3.29)$$

In cylindrical coordinates the Euler equations are

$$\begin{aligned}\rho \left[\frac{\partial V_r}{\partial t} + V_r \frac{\partial V_r}{\partial r} + \frac{V_\theta}{r} \frac{\partial V_r}{\partial \theta} + V_z \frac{\partial V_r}{\partial z} - \frac{V_\theta^2}{r} \right] &= -\frac{\partial p}{\partial r} - \rho g \frac{\partial H}{\partial r} \\ \rho \left[\frac{\partial V_\theta}{\partial t} + V_r \frac{\partial V_\theta}{\partial r} + \frac{V_\theta}{r} \frac{\partial V_\theta}{\partial \theta} + V_z \frac{\partial V_\theta}{\partial z} + \frac{V_r V_\theta}{r} \right] &= -\frac{1}{r} \frac{\partial p}{\partial \theta} - \frac{\rho g}{r} \frac{\partial H}{\partial \theta} \\ \rho \left[\frac{\partial V_z}{\partial t} + V_r \frac{\partial V_z}{\partial r} + \frac{V_\theta}{r} \frac{\partial V_z}{\partial \theta} + V_z \frac{\partial V_z}{\partial z} \right] &= -\frac{\partial p}{\partial z} - \rho g \frac{\partial H}{\partial z}\end{aligned}\quad (2.3.30)$$

In spherical coordinates the Euler equations are

$$\begin{aligned}\rho \left[\frac{\partial V_r}{\partial t} + V_r \frac{\partial V_r}{\partial r} + \frac{V_\theta}{r} \frac{\partial V_r}{\partial \theta} + \frac{V_\phi}{r \sin \theta} \frac{\partial V_r}{\partial \phi} - \frac{V_\theta^2 + V_\phi^2}{r} \right] &= -\frac{\partial p}{\partial r} - \rho g \frac{\partial H}{\partial r} \\ \rho \left[\frac{\partial V_\theta}{\partial t} + V_r \frac{\partial V_\theta}{\partial r} + \frac{V_\theta}{r} \frac{\partial V_\theta}{\partial \theta} + \frac{V_\phi}{r \sin \theta} \frac{\partial V_\theta}{\partial \phi} + \frac{V_r V_\theta - V_\phi^2 \cot \theta}{r} \right] &= -\frac{1}{r} \frac{\partial p}{\partial \theta} - \frac{\rho g}{r} \frac{\partial H}{\partial \theta} \\ \rho \left[\frac{\partial V_\phi}{\partial t} + V_r \frac{\partial V_\phi}{\partial r} + \frac{V_\theta}{r} \frac{\partial V_\phi}{\partial \theta} + \frac{V_\phi}{r \sin \theta} \frac{\partial V_\phi}{\partial \phi} + \frac{V_r V_\phi + V_\theta V_\phi \cot \theta}{r} \right] &= -\frac{1}{r \sin \theta} \frac{\partial p}{\partial \phi} - \frac{\rho g}{r \sin \theta} \frac{\partial H}{\partial \phi}\end{aligned}\quad (2.3.31)$$

In Eq. (2.3.30), the radial acceleration term $-V_\theta^2/r$ is called the centripetal acceleration. The circumferential acceleration term $V_r V_\theta/r$ seen in the θ component of Eq. (2.3.30) is called the Coriolis acceleration. Similar terms can be seen in the three components of the spherical Euler equations given in Eq. (2.3.31).

2.4. Vorticity and Circulation

The Euler equations developed in Sec. 2.3 relate the translational velocity in an inviscid fluid flow field to the pressure and gravitational forces acting on the fluid. In addition to translational motion, fluid elements may rotate as they move with the flow. Furthermore, timelines oriented in different directions may rotate at different rates.

For example, consider the rotation of the timelines that form the boundaries of the two-dimensional Lagrangian fluid element shown in Fig. 2.4.1. At time t this fluid element is rectangular and at time $t+dt$ the element is rhombic. In the time increment dt , the vertical timeline of original length dy rotates clockwise through the angle $d\alpha$ and the horizontal timeline of original length dx rotates counterclockwise through the angle $d\beta$. In the limit as dt approaches zero, the angles $d\alpha$ and $d\beta$ can be written as

$$d\alpha = \lim_{dt \rightarrow 0} \frac{\frac{\partial V_x}{\partial y} dy dt}{dy + \frac{\partial V_y}{\partial y} dy dt} = \lim_{dt \rightarrow 0} \frac{\frac{\partial V_x}{\partial y} dt}{1 + \frac{\partial V_y}{\partial y} dt} = \frac{\partial V_x}{\partial y} dt \quad (2.4.1)$$

$$d\beta = \lim_{dt \rightarrow 0} \frac{\frac{\partial V_y}{\partial x} dx dt}{dx + \frac{\partial V_x}{\partial x} dx dt} = \lim_{dt \rightarrow 0} \frac{\frac{\partial V_y}{\partial x} dt}{1 + \frac{\partial V_x}{\partial x} dt} = \frac{\partial V_y}{\partial x} dt \quad (2.4.2)$$

If the x and y axes shown in Fig. 2.4.1 are elements of a right-hand Cartesian coordinate system, the z axis points out of the page. Following the right-hand rule, $d\alpha$ is a negative rotation about the z axis and $d\beta$ is a positive rotation about the z axis. Hence, in the limit as dt approaches zero, the average rate of rotation about the z axis for the fluid contained within this Lagrangian element is

$$\text{average rate of rotation about the } z \text{ axis} = \lim_{dt \rightarrow 0} \frac{[(-d\alpha) + d\beta]/2}{dt} \quad (2.4.3)$$

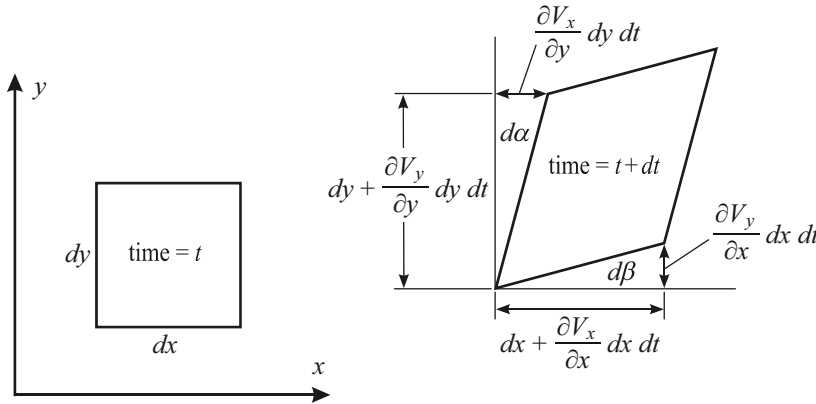


Figure 2.4.1. Rotation of the boundaries of a moving Lagrangian fluid element.

Using Eqs. (2.4.1) and (2.4.2) in Eq. (2.4.3) yields

$$\begin{aligned}\text{average rate of rotation about the } z \text{ axis} &= \frac{1}{2} \left(-\frac{\partial V_x}{\partial y} dt + \frac{\partial V_y}{\partial x} dt \right) / dt \\ &= \frac{1}{2} \left(\frac{\partial V_y}{\partial x} - \frac{\partial V_x}{\partial y} \right)\end{aligned}\quad (2.4.4)$$

In general, a Lagrangian fluid element may rotate about the x and y axes as well as the z axis. Just as the translation rate is expressed as a translational velocity vector, the rate of fluid rotation is expressed as an angular velocity vector. Flowing developments identical to that used to obtain the z component of angular velocity given in Eq. (2.4.4), the x and y components of angular velocity are found to yield

$$\text{angular velocity vector} = \frac{1}{2} \left(\frac{\partial V_z}{\partial y} - \frac{\partial V_y}{\partial z} \right) \mathbf{i}_x + \frac{1}{2} \left(\frac{\partial V_x}{\partial z} - \frac{\partial V_z}{\partial x} \right) \mathbf{i}_y + \frac{1}{2} \left(\frac{\partial V_y}{\partial x} - \frac{\partial V_x}{\partial y} \right) \mathbf{i}_z \quad (2.4.5)$$

It should be noted that the angular velocity vector given in Eq. (2.4.5) is exactly one half the curl of the translational velocity vector $\mathbf{V}$, as defined in Eq. (1.4.24). To avoid the appearance of the factor one half in certain expressions, **it is traditional to characterize fluid rotation with a vector called the vorticity, which is twice the angular velocity. The fluid vorticity vector, denoted as Ω , is related to the fluid velocity vector $\mathbf{V}$ by**

$$\Omega \equiv \nabla \times \mathbf{V} \quad (2.4.6)$$

In **Cartesian coordinates**

$$\Omega = \left(\frac{\partial V_z}{\partial y} - \frac{\partial V_y}{\partial z} \right) \mathbf{i}_x + \left(\frac{\partial V_x}{\partial z} - \frac{\partial V_z}{\partial x} \right) \mathbf{i}_y + \left(\frac{\partial V_y}{\partial x} - \frac{\partial V_x}{\partial y} \right) \mathbf{i}_z \quad (2.4.7)$$

in **cylindrical coordinates**

$$\Omega = \left(\frac{1}{r} \frac{\partial V_z}{\partial \theta} - \frac{\partial V_\theta}{\partial z} \right) \mathbf{i}_r + \left(\frac{\partial V_r}{\partial z} - \frac{\partial V_z}{\partial r} \right) \mathbf{i}_\theta + \left(\frac{\partial V_\theta}{\partial r} - \frac{1}{r} \frac{\partial V_r}{\partial \theta} + \frac{V_\theta}{r} \right) \mathbf{i}_z \quad (2.4.8)$$

and in **spherical coordinates**

$$\begin{aligned}\Omega &= \left(\frac{1}{r} \frac{\partial V_\phi}{\partial \theta} - \frac{1}{r \sin \theta} \frac{\partial V_\theta}{\partial \phi} + \frac{V_\phi \cos \theta}{r \sin \theta} \right) \mathbf{i}_r \\ &+ \left(\frac{1}{r \sin \theta} \frac{\partial V_r}{\partial \phi} - \frac{\partial V_\phi}{\partial r} - \frac{V_\phi}{r} \right) \mathbf{i}_\theta + \left(\frac{\partial V_\theta}{\partial r} - \frac{1}{r} \frac{\partial V_r}{\partial \theta} + \frac{V_\theta}{r} \right) \mathbf{i}_\phi\end{aligned}\quad (2.4.9)$$

For a very important class of inviscid fluid flows, the vorticity vector is identically zero at every point within the flow field. In such a flow field, the fluid elements do not rotate as they move with the flow. Hence, any flow field with zero vorticity throughout is commonly called an *irrotational flow*.

Another important fluid flow parameter called the *circulation* is related to the fluid vorticity. The circulation is a scalar, usually denoted as Γ and defined in terms of a line integral around a closed curve in the flow field. For example, consider the simple closed curve C , which encloses an arbitrary open surface $\mathcal{S}$ in some fluid flow field, as shown in Fig. 2.4.2. Let $\mathbf{V}$ be the local fluid velocity vector, and let $d\mathbf{s} \equiv dx\mathbf{i}_x + dy\mathbf{i}_y + dz\mathbf{i}_z$ denote a directed differential vector segment of C at some arbitrary point on C . The positive direction for the tangent vector $d\mathbf{s}$ is in the counterclockwise direction, when viewed from the “outside” of the enclosed surface. This is the direction that keeps the enclosed surface $\mathcal{S}$ on the left of the boundary curve C , when viewed from outside $\mathcal{S}$. With these definitions, the **circulation** of the vector field $\mathbf{V}$ about the closed curve C is defined as

$$\Gamma \equiv -\oint_C \mathbf{V} \cdot d\mathbf{s} \quad (2.4.10)$$

Traditional mathematical convention defines a line integral to be positive in the counterclockwise direction. However, circulation is traditionally defined to be positive in the clockwise direction. This accounts for the negative sign in the definition of circulation.

A mathematical identity known as **Stokes’ theorem** relates the surface integral of the curl of an arbitrary vector field $\mathbf{V}$ over an arbitrary open surface $\mathcal{S}$ to the line integral of $\mathbf{V}$ about the closed boundary curve C enclosing the open surface $\mathcal{S}$, i.e.,

$$\iint_{\mathcal{S}} (\nabla \times \mathbf{V}) \cdot \mathbf{n} d\mathcal{S} = \oint_C \mathbf{V} \cdot d\mathbf{s} \quad (2.4.11)$$

Using the mathematical identity given in Eq. (2.4.11) and the definition of vorticity given in Eq. (2.4.6), the definition of circulation from Eq. (2.4.10) yields

$$\Gamma = -\iint_{\mathcal{S}} \boldsymbol{\Omega} \cdot \mathbf{n} d\mathcal{S} \quad (2.4.12)$$

It should be noted that, when a fluid flow field has finite circulation about some closed curve, as defined by either Eq. (2.4.10) or Eq. (2.4.12), the fluid particles do not necessarily move in a circuit around the curve. For example, consider the closed curve C

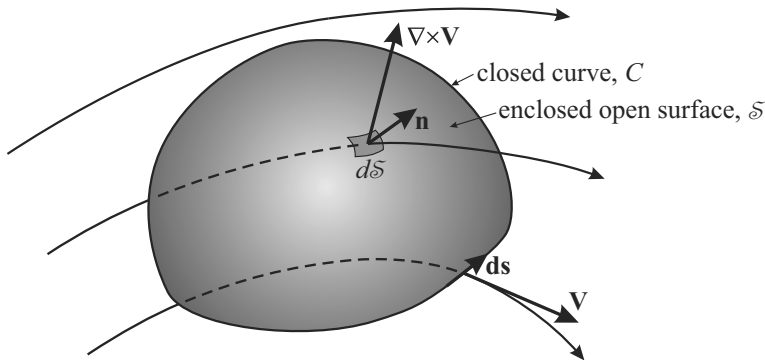


Figure 2.4.2. Circulation around a closed curve C enclosing the open surface $\mathcal{S}$.

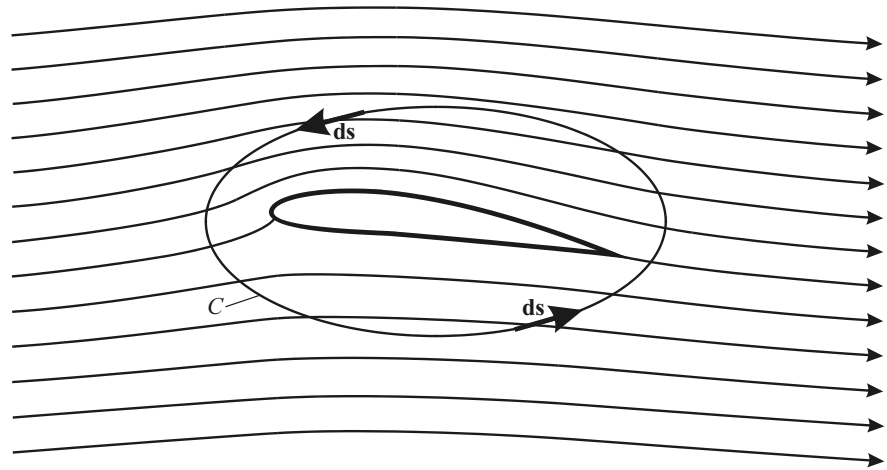


Figure 2.4.3. Circulation around a closed curve C enclosing a lifting airfoil.

in the two-dimensional flow field about an airfoil, which is shown in Fig. 2.4.3. The fluid flow follows the streamlines shown in this figure, moving from left to right at all points in the flow field. Because the directed differential vector segment $\mathbf{ds}$ is defined positive in the counterclockwise direction at all points on C , $\mathbf{ds}$ points from left to right below the airfoil and from right to left above the airfoil. Hence, because the velocity vector $\mathbf{V}$ points from left to right at all points within the flow field, the scalar $\mathbf{V} \cdot \mathbf{ds}$ is positive below the airfoil and negative above the airfoil. As a result, the line integral of $\mathbf{V} \cdot \mathbf{ds}$ along the curve C shown in Fig. 2.4.3 has a positive contribution over the lower half of C and a negative contribution over the upper half of C . If this velocity field were symmetric above and below the airfoil, the net line integral of $\mathbf{V} \cdot \mathbf{ds}$ around the complete closed curve C would be identically zero. However, we see from the streamline spacing above and below the airfoil, that the airspeed above the airfoil is greater than that below the airfoil. This requires that the negative contribution to the line integral of $\mathbf{V} \cdot \mathbf{ds}$ over the upper half of the closed curve C is greater in magnitude than the positive contribution over the lower half of C . Hence, simple inspection of Fig. 2.4.3 reveals that the line integral of $\mathbf{V} \cdot \mathbf{ds}$ around the closed curve C is negative and the circulation as defined by Eq. (2.4.10) is greater than zero, in spite of the fact that the fluid is not flowing in a circuit around the closed curve C .

EXAMPLE 2.4.1. Consider the steady inviscid flow normal to a long circular cylinder, which is described in Example 1.14.1. The velocity field for this flow is expressed in cylindrical coordinates as

$$\mathbf{V} = V_{\infty}(1 - R^2/r^2)\cos\theta\mathbf{i}_r - V_{\infty}(1 + R^2/r^2)\sin\theta\mathbf{i}_{\theta} + 0\mathbf{i}_z$$

where V_{∞} and R are constants denoting the freestream airspeed and the radius of the cylinder, respectively. Find the vorticity vector at an arbitrary point in this flow field. Also find the circulation as defined by Eq. (2.4.10) around each of the

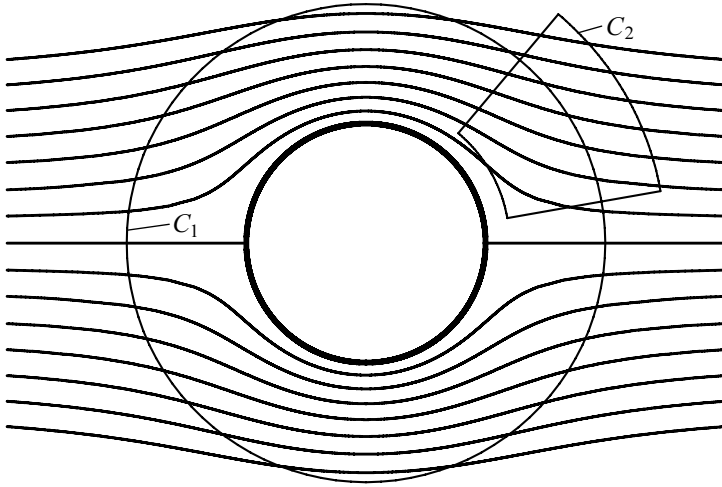


Figure 2.4.4. Circulation around two closed curves in the inviscid flow field about a cylinder.

closed paths labeled C_1 and C_2 in Fig. 2.4.4. Path C_1 is a closed circle of arbitrary radius R_1 centered about the cylinder. Path C_2 encloses a section of an annular area in the flow field and is composed of four segments; a radial line extending from coordinates $(R_1, \theta_1, 0)$ to $(R_2, \theta_1, 0)$, a circular arc extending from coordinates $(R_2, \theta_1, 0)$ to $(R_2, \theta_2, 0)$, a radial line extending from coordinates $(R_2, \theta_2, 0)$ to $(R_1, \theta_2, 0)$, and a circular arc extending from coordinates $(R_1, \theta_2, 0)$ to $(R_1, \theta_1, 0)$, which is the point of beginning.

Solution. The vorticity field in cylindrical coordinates is obtained by using the given equation for the velocity field in Eq. (2.4.8)

$$\begin{aligned}
 \boldsymbol{\Omega} &= \left(\frac{1}{r} \frac{\partial}{\partial \theta} (0) - \frac{\partial}{\partial z} [-V_\infty (1 + R^2/r^2) \sin \theta] \right) \mathbf{i}_r \\
 &\quad + \left(\frac{\partial}{\partial z} [V_\infty (1 - R^2/r^2) \cos \theta] - \frac{\partial}{\partial r} (0) \right) \mathbf{i}_\theta \\
 &\quad + \left(\frac{\partial}{\partial r} [-V_\infty (1 + R^2/r^2) \sin \theta] - \frac{1}{r} \frac{\partial}{\partial \theta} [V_\infty (1 - R^2/r^2) \cos \theta] \right. \\
 &\quad \quad \quad \left. + \frac{1}{r} [-V_\infty (1 + R^2/r^2) \sin \theta] \right) \mathbf{i}_z \\
 &= (0) \mathbf{i}_r + (0) \mathbf{i}_\theta + [-V_\infty (-2R^2/r^3) \sin \theta] \mathbf{i}_z - (1/r) [V_\infty (1 - R^2/r^2) (-\sin \theta)] \mathbf{i}_z \\
 &\quad \quad \quad + (1/r) [-V_\infty (1 + R^2/r^2) \sin \theta] \mathbf{i}_z \\
 &= \underline{(0) \mathbf{i}_r + (0) \mathbf{i}_\theta + (0) \mathbf{i}_z}
 \end{aligned}$$

Hence, we see that this inviscid flow about a circular cylinder is irrotational.

In cylindrical coordinates, the directed differential vector segment $\mathbf{ds}$ along an arbitrary path is $\mathbf{ds} = dr \mathbf{i}_r + r d\theta \mathbf{i}_\theta + dz \mathbf{i}_z$. Therefore, for this velocity field

$$\begin{aligned}\mathbf{V} \cdot d\mathbf{s} &= (V_r \mathbf{i}_r + V_\theta \mathbf{i}_\theta + V_z \mathbf{i}_z) \cdot (dr \mathbf{i}_r + r d\theta \mathbf{i}_\theta + dz \mathbf{i}_z) = V_r dr + V_\theta r d\theta + V_z dz \\ &= V_\infty (1 - R^2/r^2) \cos \theta dr - V_\infty (r + R^2/r) \sin \theta d\theta\end{aligned}$$

At all points on the closed circular path C_1 , $r = R_1$, and because the line integral of $\mathbf{V} \cdot d\mathbf{s}$ is positive in the counterclockwise direction, the integration limits in θ are from 0 to 2π . Thus, the circulation about C_1 as defined by Eq. (2.4.10) is

$$\begin{aligned}\Gamma &\equiv -\oint_{C_1} \mathbf{V} \cdot d\mathbf{s} = -\int_{\theta=0}^{2\pi} [-V_\infty (R_1 + R^2/R_1) \sin \theta] d\theta \\ &= V_\infty (R_1 + R^2/R_1) [-\cos(2\pi) + \cos(0)] = 0\end{aligned}$$

Notice that the circulation of this velocity field about the closed circular path C_1 is exactly zero, completely independent of the path radius R_1 . This should not be too surprising in view of the symmetry of the streamlines relative to C_1 , which can be seen in Fig. 2.4.4. However, the flow need not be symmetric relative to the path of integration to produce zero circulation.

For the path labeled C_2 in Fig. 2.4.4, the line integral of $\mathbf{V} \cdot d\mathbf{s}$ can be broken into four parts. Starting at the lower left-hand corner of C_2 and integrating in the counterclockwise direction, along the first radial segment from the point $(R_1, \theta_1, 0)$ to $(R_2, \theta_1, 0)$, θ remains fixed at θ_1 as r increases from R_1 to R_2 . The next segment is a circular arc extending from $(R_2, \theta_1, 0)$ to $(R_2, \theta_2, 0)$, where r remains constant at R_2 and θ increases from θ_1 to θ_2 . Similarly, in the third segment, θ is fixed at θ_2 while r is decreased from R_2 to R_1 . Segment four is another circular arc, where r is held constant at R_1 and θ is decreased from θ_2 to θ_1 , which brings us back to the point of beginning. Thus, from Eq. (2.4.10), the circulation about C_2 is

$$\begin{aligned}\Gamma &\equiv -\oint_{C_2} \mathbf{V} \cdot d\mathbf{s} \\ &= -\int_{r=R_1}^{R_2} [V_\infty (1 - R^2/r^2) \cos \theta_1] dr - \int_{\theta=\theta_1}^{\theta_2} [-V_\infty (R_2 + R^2/R_2) \sin \theta] d\theta \\ &\quad - \int_{r=R_2}^{R_1} [V_\infty (1 - R^2/r^2) \cos \theta_2] dr - \int_{\theta=\theta_2}^{\theta_1} [-V_\infty (R_1 + R^2/R_1) \sin \theta] d\theta \\ &= -[V_\infty (R_2 - R_1 + R^2/R_2 - R^2/R_1) \cos \theta_1] \\ &\quad - [-V_\infty (R_2 + R^2/R_2) (-\cos \theta_2 + \cos \theta_1)] \\ &\quad - [V_\infty (R_1 - R_2 + R^2/R_1 - R^2/R_2) \cos \theta_2] \\ &\quad - [-V_\infty (R_1 + R^2/R_1) (-\cos \theta_1 + \cos \theta_2)] \\ &= V_\infty (-R_2 + R_1 - R^2/R_2 + R^2/R_1 + R_2 + R^2/R_2 - R_1 - R^2/R_1) \cos \theta_1 \\ &\quad + V_\infty (-R_1 + R_2 - R^2/R_1 + R^2/R_2 - R_2 - R^2/R_2 + R_1 + R^2/R_1) \cos \theta_2 = 0\end{aligned}$$

Here again, we see that the circulation of this velocity field about the closed path C_2 is zero, independent of the arbitrary path limits R_1 , θ_1 , R_2 , and θ_2 . Any closed path in this flow field would yield exactly the same result. The circulation about any closed path in this velocity field is zero, because this flow is irrotational at every point. The mathematical identity known as Stokes' theorem, which is given in Eq. (2.4.11), requires that the circulation about any closed path is exactly zero, provided that the flow is irrotational at every point enclosed by the path.

2.5. Irrotational Flow and the Velocity Potential

For aerodynamic flow over streamlined bodies at low angles of attack and high Reynolds numbers, the only body force is due to gravity and viscous effects are negligible outside the boundary layer and any regions of separated flow. For such conditions, which occur commonly with an aircraft in flight, Newton's second law reduces to the Euler equations developed in Sec. 2.3 and written in vector notation in Eq. (2.3.24). The related concepts of vorticity and circulation presented in Sec. 2.4 are also very important in aerodynamic flows. In this section we will develop an important simplification to the Euler equations, which results from a consideration of fluid vorticity.

Starting with Eq. (2.3.24), we divide through by the fluid density ρ and rearrange the resulting terms to yield an alternate form of the Euler equations in vector notation,

$$\frac{\partial \mathbf{V}}{\partial t} + (\mathbf{V} \cdot \nabla) \mathbf{V} + \frac{\nabla p}{\rho} + g \nabla H = 0 \quad (2.5.1)$$

The second term on the left-hand side of Eq. (2.5.1) can be expressed in terms of vorticity by using the important **mathematical identity**

$$(\mathbf{V} \cdot \nabla) \mathbf{V} = \nabla \left(\frac{1}{2} V^2 \right) - \mathbf{V} \times (\nabla \times \mathbf{V}) \quad (2.5.2)$$

where $\mathbf{V}$ is any arbitrary vector field and V denotes the magnitude of $\mathbf{V}$. The verification of this mathematical identity is left as an exercise for the student. Using Eq. (2.5.2) to replace the second term on the left-hand side of Eq. (2.5.1), the **Euler equations for inviscid flow when the only body force is due to gravity can be rearranged as**

$$\frac{\partial \mathbf{V}}{\partial t} + \nabla \left(\frac{1}{2} V^2 \right) + \frac{\nabla p}{\rho} + g \nabla H = \mathbf{V} \times \boldsymbol{\Omega} \quad (2.5.3)$$

where $\boldsymbol{\Omega}$ is the vorticity vector, which is defined in Eq. (2.4.6) to be the curl of the velocity vector $\boldsymbol{\Omega} \equiv \nabla \times \mathbf{V}$.

For compressible fluids like air, a change in pressure can produce a corresponding change in density. As discussed in Sec. 1.11, when the density of a gas changes with pressure, the temperature changes as well. Thus, in a compressible fluid such as air, the pressure gradients that accompany velocity gradients also produce temperature gradients. In general, temperature gradients result in heat transfer, but the thermal conductivity of air is extremely low. Hence, compressible flow outside the boundary layer is accurately modeled as being adiabatic as well as inviscid. Inviscid, adiabatic flow is isentropic. From the fundamentals of thermodynamics, the reader should recall that for isentropic flow of an ideal gas, density is related to pressure according to the relation

$$\rho = \rho_{\text{ref}} \left(\frac{p}{p_{\text{ref}}} \right)^{1/\gamma} \quad (2.5.4)$$

where ρ_{ref} and p_{ref} are, respectively, the density and pressure at a fixed reference point in the flow field, and γ is the ratio of specific heats defined in Eq. (1.2.15), which yields

$\gamma = 1.4$ for standard atmospheric air. From Eq. (2.5.4), the third term on the left-hand side of Eq. (2.5.3) can be written as

$$\frac{\nabla p}{\rho} = \frac{p_{\text{ref}}^{1/\gamma}}{\rho_{\text{ref}}} \frac{\nabla p}{p^{1/\gamma}} \quad (2.5.5)$$

which can be rearranged as

$$\frac{\nabla p}{\rho} = \frac{1}{1-1/\gamma} \frac{p_{\text{ref}}^{1/\gamma}}{\rho_{\text{ref}}} \left(\frac{1-1/\gamma}{p^{1/\gamma}} \nabla p \right) = \frac{\gamma}{\gamma-1} \frac{p_{\text{ref}}^{1/\gamma}}{\rho_{\text{ref}}} \nabla p^{1-1/\gamma} \quad (2.5.6)$$

Because γ , ρ_{ref} , and p_{ref} are constants, Eq. (2.5.6) can be written equivalently as

$$\frac{\nabla p}{\rho} = \nabla \left(\frac{\gamma}{\gamma-1} \frac{p_{\text{ref}}^{1/\gamma}}{\rho_{\text{ref}}} p^{1-1/\gamma} \right) = \nabla \left(\frac{\gamma}{\gamma-1} \frac{p}{\rho_{\text{ref}} p^{1/\gamma} / p_{\text{ref}}^{1/\gamma}} \right) \quad (2.5.7)$$

or after applying Eq. (2.5.4)

$$\frac{\nabla p}{\rho} = \nabla \left(\frac{\gamma}{\gamma-1} \frac{p}{\rho} \right) \quad (2.5.8)$$

From Newton's law of gravitation given in Eq. (1.5.11), the fourth term on the left-hand side of Eq. (2.5.3) can be written in terms of the acceleration of gravity at mean sea level, $g_o = 9.806645 \text{ m/s}^2$, and the radius of the Earth at mean sea level, $R_E = 6,356,766 \text{ m}$,

$$g \nabla H = g_o \left(\frac{R_E}{R_E + H} \right)^2 \nabla H \quad (2.5.9)$$

which can be rearranged as

$$g \nabla H = g_o \frac{(R_E + H)(R_E \nabla H) - (R_E H) \nabla H}{(R_E + H)^2} = g_o \nabla \left(\frac{R_E H}{R_E + H} \right) \quad (2.5.10)$$

Because g_o is a constant, Eq. (2.5.10) can be written as

$$g \nabla H = \nabla (g_o Z) \quad (2.5.11)$$

where $Z \equiv R_E H / (R_E + H)$ is the geopotential altitude defined in Eq. (1.6.5).

Using Eqs. (2.5.8) and (2.5.11) in Eq. (2.5.3), the **Euler equations for inviscid, compressible flow when the only body force is due to gravity** can be written as

$$\frac{\partial \mathbf{V}}{\partial t} + \nabla \left(\frac{1}{2} V^2 + \frac{\gamma}{\gamma-1} \frac{p}{\rho} + g_o Z \right) = \mathbf{V} \times \boldsymbol{\Omega} \quad (2.5.12)$$

where again $\boldsymbol{\Omega} \equiv \nabla \times \mathbf{V}$ is the vorticity vector. The third and fourth terms on the left-hand side of Eq. (2.5.3) are the pressure and gravitational forces per unit mass, respectively.

From Eq. (2.5.4), the density of an ideal gas in an inviscid, adiabatic flow depends only on the pressure. This allowed us to express the fluid pressure force per unit mass in Eq. (2.5.12) as the gradient of a scalar field, which is sometimes called a *potential field*. Any flow in which the density depends only on the pressure is called a *barotropic flow*. The pressure force per unit mass can be expressed as the gradient of a scalar potential field in any barotropic flow. Similarly, the gravitational body force per unit mass in Eq. (2.5.3) was expressed in Eq. (2.5.12) as the gradient of the scalar potential field $g_o Z$. Any body force per unit mass that can be expressed as the gradient of some scalar potential field is called a *conservative body force*.

Taking the curl of Eq. (2.5.12) yields what is called the *vorticity transport equation* for inviscid, adiabatic, flow of an ideal gas,

$$\nabla \times \frac{\partial \mathbf{V}}{\partial t} + \nabla \times \nabla \left(\frac{1}{2} V^2 + \frac{\gamma}{\gamma-1} \frac{p}{\rho} + g_o Z \right) = \nabla \times (\mathbf{V} \times \boldsymbol{\Omega}) \quad (2.5.13)$$

Because time and position are independent variables, we can interchange the order of differentiation in the first term on the left-hand side of Eq. (2.5.13). With the definition of vorticity, this yields

$$\nabla \times \frac{\partial \mathbf{V}}{\partial t} = \frac{\partial}{\partial t} (\nabla \times \mathbf{V}) = \frac{\partial \boldsymbol{\Omega}}{\partial t} \quad (2.5.14)$$

It can be shown that the curl of the gradient of any scalar function is zero, which yields the **mathematical identity**

$$\nabla \times \nabla \psi = 0 \quad (2.5.15)$$

where ψ denotes any arbitrary scalar field. Hence, applying Eq. (2.5.15) to the second term on the left-hand side of Eq. (2.5.13) requires

$$\nabla \times \nabla \left(\frac{1}{2} V^2 + \frac{\gamma}{\gamma-1} \frac{p}{\rho} + g_o Z \right) = 0 \quad (2.5.16)$$

The curl of the cross product of two arbitrary vector fields $\mathbf{A}$ and $\mathbf{B}$ can be expanded using the **mathematical identity**

$$\nabla \times (\mathbf{A} \times \mathbf{B}) = \mathbf{A}(\nabla \cdot \mathbf{B}) + (\mathbf{B} \cdot \nabla) \mathbf{A} - (\mathbf{A} \cdot \nabla) \mathbf{B} - \mathbf{B}(\nabla \cdot \mathbf{A}) \quad (2.5.17)$$

Furthermore, it can be shown that the divergence of the curl of any vector field is zero, which yields the **mathematical identity**

$$\nabla \cdot (\nabla \times \mathbf{V}) = 0 \quad (2.5.18)$$

where $\mathbf{V}$ denotes any arbitrary vector field. Using Eq. (2.5.17) with $\mathbf{A} = \mathbf{V}$, $\mathbf{B} = \boldsymbol{\Omega} \equiv \nabla \times \mathbf{V}$, and applying Eq. (2.5.18) to eliminate the first term on the right-hand side of Eq. (2.5.17), the right-hand side of Eq. (2.5.13) can be written as

$$\nabla \times (\mathbf{V} \times \boldsymbol{\Omega}) = (\boldsymbol{\Omega} \cdot \nabla) \mathbf{V} - (\mathbf{V} \cdot \nabla) \boldsymbol{\Omega} - \boldsymbol{\Omega}(\nabla \cdot \mathbf{V}) \quad (2.5.19)$$

Using Eqs. (2.5.14), (2.5.16), and (2.5.19) in Eq. (2.5.13) and rearranging, yields the **vorticity transport equation for inviscid flow**

$$\frac{\partial \mathbf{\Omega}}{\partial t} + (\mathbf{V} \cdot \nabla) \mathbf{\Omega} = (\mathbf{\Omega} \cdot \nabla) \mathbf{V} - \mathbf{\Omega} (\nabla \cdot \mathbf{V}) \quad (2.5.20)$$

It should be noted that Eq. (2.5.20) was developed here for inviscid, adiabatic, flow of an ideal gas. **However, it can be shown that the vorticity transport equation given by Eq. (2.5.20) applies to any barotropic, inviscid flow with conservative body forces.** For example, Eq. (2.5.20) is sometimes developed from Eq. (2.5.3) by neglecting changes in density and the acceleration of gravity. Using these approximations in Eq. (2.5.3), the **Euler equations for inviscid, incompressible flow when the only body force is that due to a constant gravitational acceleration can be written as**

$$\frac{\partial \mathbf{V}}{\partial t} + \nabla \left(\frac{1}{2} V^2 + \frac{p}{\rho} + gH \right) = \mathbf{V} \times \mathbf{\Omega} \quad (2.5.21)$$

The only difference between Eq. (2.5.21) and Eq. (2.5.12) is in the form of the gradient term. Because the curl of the gradient of any scalar function is zero, taking the curl of Eq. (2.5.21) yields exactly the same inviscid vorticity transport equation that is given in Eq. (2.5.20). However, for the case of incompressible flow, the last term on the right-hand side of Eq. (2.5.20) is usually eliminated by invoking the incompressible continuity equation given in Eq. (2.2.26).

The left-hand side of Eq. (2.5.20) is the substantial derivative of the fluid vorticity vector $\mathbf{\Omega}$. This substantial derivative is the time rate of change of vorticity at a point that moves with the fluid along a pathline. The components of both terms on the right-hand side of Eq. (2.5.20) are proportional to the components of the vorticity vector. Hence, at any point in an inviscid flow field where the vorticity vector is zero, the right-hand side of Eq. (2.5.20) and the substantial derivative of the vorticity vector are zero as well. At any point in a flow field where the substantial derivative of vorticity is zero, the vorticity does not change as we move in either direction along a pathline from this irrotational point. This leads to an important consequence of Eq. (2.5.20).

In any barotropic, inviscid flow field with conservative body forces, if the vorticity is zero at any point on a pathline, the vorticity is zero at every point on the pathline.

Both *inviscid, incompressible flow* and *inviscid, adiabatic flow of an ideal gas* satisfy these conditions, when the only body force is that due to gravity.

Many problems of importance in fluid mechanics and aerodynamics involve flow about some object like a ship or an airplane moving through a static fluid. In such flows, the components of the fluid velocity vector far upstream from the object are constant. Any flow field with constant velocity components is called a *uniform flow*, and all spatial derivatives of velocity are zero in a uniform flow. From the definition of vorticity, this requires that the vorticity is zero everywhere far upstream from an object moving through a static fluid, and all pathlines in the flow field pass through a region of zero vorticity.

For inviscid flows of this type, Eq. (2.5.20) requires that the vorticity must be zero at every point in the flow field, i.e., the flow field is irrotational throughout. This leads to another important consequence of Eq. (2.5.20).

Inviscid flow about any object moving through a static fluid is also irrotational flow, provided that all body forces are conservative like that due to gravity.

This has important implications for flow about an aircraft in flight. Because viscous effects are negligible outside the boundary layer, Eq. (2.5.20) requires that the vorticity must be zero at all points in the inviscid flow field outside the boundary layer. Vorticity is generated in the viscous boundary layer next to the skin of the aircraft, and thus there is also vorticity in the boundary layer wake trailing behind an aircraft in flight. However, outside the boundary layer and trailing wake, the flow can be assumed to be inviscid and free of vorticity, so the right-hand side of either Eq. (2.5.12) or (2.5.21) is exactly zero. This type of flow is called *inviscid, irrotational flow*. For Mach numbers less than about 0.3, the flow can be assumed to be *incompressible* as well. In addition, the acceleration of gravity is virtually constant in the flow field about an aircraft in flight.

When the body force is due to constant gravitational acceleration, the equations of motion for inviscid, incompressible, irrotational flow can be written as follows:

The **continuity equation** from Eq. (2.2.26)

$$\nabla \cdot \mathbf{V} = 0 \quad (2.5.22)$$

and **Newton's second law** from Eq. (2.5.21) with the vorticity set to zero

$$\frac{\partial \mathbf{V}}{\partial t} + \nabla \left(\frac{1}{2} V^2 + \frac{p}{\rho} + gH \right) = 0 \quad (2.5.23)$$

Because the fluid density ρ is a known constant for incompressible flow, Eq. (2.5.22) combined with the three components of Eq. (2.5.23) provide four partial differential equations, which contain four scalar unknowns; the pressure p and the three components of the velocity vector $\mathbf{V}$.

Stokes' theorem provides an important simplification that can be applied to the equations of motion for inviscid, irrotational flow. From Eq. (2.4.11) **for irrotational flow, Stokes' theorem requires**

$$\oint_C \mathbf{V} \cdot d\mathbf{s} = 0 \quad (2.5.24)$$

where C is a completely arbitrary closed path in the flow field. However, a line integral about a closed path can be independent of the path of integration, only if the integrand is an exact differential. Hence, we can write

$$\mathbf{V} \cdot d\mathbf{s} = d\phi \quad (2.5.25)$$

where $d\phi$ is the exact differential of some scalar function ϕ . In Cartesian coordinates, an exact differential is written as

$$d\phi = \frac{\partial\phi}{\partial x}dx + \frac{\partial\phi}{\partial y}dy + \frac{\partial\phi}{\partial z}dz \quad (2.5.26)$$

Also in Cartesian coordinates,

$$\mathbf{V} \equiv V_x \mathbf{i}_x + V_y \mathbf{i}_y + V_z \mathbf{i}_z \quad (2.5.27)$$

$$d\mathbf{s} \equiv dx \mathbf{i}_x + dy \mathbf{i}_y + dz \mathbf{i}_z \quad (2.5.28)$$

and

$$\mathbf{V} \cdot d\mathbf{s} = V_x dx + V_y dy + V_z dz \quad (2.5.29)$$

Using Eqs. (2.5.26) and (2.5.29) in Eq. (2.5.25) requires

$$V_x = \frac{\partial\phi}{\partial x}, \quad V_y = \frac{\partial\phi}{\partial y}, \quad V_z = \frac{\partial\phi}{\partial z} \quad (2.5.30)$$

Notice that these three components of velocity are exactly equal to the three components of the gradient of the scalar function ϕ , as defined in Eq. (1.4.22).

In any coordinate system, the exact differential of an arbitrary scalar function ϕ can be written in vector notation as $d\phi = \nabla\phi \cdot d\mathbf{s}$. Hence, Eqs. (2.5.24) and (2.5.25) lead to an important consequence of Stokes' theorem.

In any irrotational flow field, the velocity vector can be expressed as the gradient of a scalar function, which is commonly called the velocity potential.

$$\mathbf{V} = \nabla\phi \quad (2.5.31)$$

In Cartesian coordinates

$$\mathbf{V} = \frac{\partial\phi}{\partial x} \mathbf{i}_x + \frac{\partial\phi}{\partial y} \mathbf{i}_y + \frac{\partial\phi}{\partial z} \mathbf{i}_z \quad (2.5.32)$$

in cylindrical coordinates

$$\mathbf{V} = \frac{\partial\phi}{\partial r} \mathbf{i}_r + \frac{1}{r} \frac{\partial\phi}{\partial \theta} \mathbf{i}_\theta + \frac{\partial\phi}{\partial z} \mathbf{i}_z \quad (2.5.33)$$

and in spherical coordinates

$$\mathbf{V} = \frac{\partial\phi}{\partial r} \mathbf{i}_r + \frac{1}{r} \frac{\partial\phi}{\partial \theta} \mathbf{i}_\theta + \frac{1}{r \sin \theta} \frac{\partial\phi}{\partial \varphi} \mathbf{i}_\varphi \quad (2.5.34)$$

where ϕ is the scalar *velocity potential*, which in general may depend on all three position coordinates and time. Any velocity field that can be expressed as the gradient of a scalar function is called a *potential flow*. Stokes' theorem requires that **any irrotational flow is a potential flow**; and the mathematical identity given in Eq. (2.5.15) requires that **any potential flow is an irrotational flow**.

EXAMPLE 2.5.1. For a certain steady, inviscid, irrotational fluid flow field, the velocity potential is given in cylindrical coordinates as

$$\phi = V_{\infty}(r + R^2/r)\cos\theta + C$$

where V_{∞} , R , and C are constants. Find the velocity vector at an arbitrary point in this flow field.

Solution. In any irrotational flow field, the velocity vector can be computed from the gradient of the scalar velocity potential. Thus, from Eq. (2.5.33)

$$\begin{aligned}\mathbf{V} &= \frac{\partial\phi}{\partial r}\mathbf{i}_r + \frac{1}{r}\frac{\partial\phi}{\partial\theta}\mathbf{i}_{\theta} + \frac{\partial\phi}{\partial z}\mathbf{i}_z \\ &= \frac{\partial}{\partial r}[V_{\infty}(r + R^2/r)\cos\theta + C]\mathbf{i}_r + \frac{1}{r}\frac{\partial}{\partial\theta}[V_{\infty}(r + R^2/r)\cos\theta + C]\mathbf{i}_{\theta} \\ &\quad + \frac{\partial}{\partial z}[V_{\infty}(r + R^2/r)\cos\theta + C]\mathbf{i}_z \\ &= \underline{V_{\infty}(1 - R^2/r^2)\cos\theta\mathbf{i}_r - V_{\infty}(1 + R^2/r^2)\sin\theta\mathbf{i}_{\theta} + 0\mathbf{i}_z}\end{aligned}$$

Notice that all three components of this velocity field are completely independent of the constant C in the velocity potential. In general, all three components of the velocity field for any potential flow can be determined, if the velocity potential is known to within an arbitrary constant.

2.6. Laplace's Equation and Boundary Conditions

Inviscid flow about an object moving through a static fluid with conservative body forces is irrotational. Furthermore, the velocity field in any irrotational flow can be expressed as the gradient of a scalar potential field. The differential equations of motion for inviscid, incompressible, irrotational flow, which are given by Eqs. (2.5.22) and (2.5.23), can be simplified significantly by introducing the velocity potential ϕ .

The incompressible continuity equation can be expressed in terms of the velocity potential by using Eq. (2.5.31) in Eq. (2.5.22), which yields

$$\nabla \cdot (\nabla\phi) = 0 \quad (2.6.1)$$

The divergence of the gradient of any scalar function is a well known scalar differential operator, traditionally called the *Laplacian* after the renowned French mathematician Pierre S. De Laplace (1749–1827). The Laplacian is commonly designated using the shorthand notation $\nabla^2\phi \equiv \nabla \cdot (\nabla\phi)$. Employing this notation, **the continuity equation for incompressible, irrotational flow can be written in terms of the velocity potential,**

$$\nabla^2\phi = 0 \quad (2.6.2)$$

Equation (2.6.2) is called *Laplace's equation*, and it is quite possibly the most extensively studied partial differential equation in mathematical physics. This second-order linear

partial differential equation is relatively simple, even when compared with the inviscid Euler equations that are given in Eq. (2.3.34). **Laplace's equation** is easily written in the commonly used orthogonal coordinate systems as follows:

In **Cartesian coordinates**

$$\frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} + \frac{\partial^2 \phi}{\partial z^2} = 0 \quad (2.6.3)$$

in **cylindrical coordinates**

$$\frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial \phi}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2 \phi}{\partial \theta^2} + \frac{\partial^2 \phi}{\partial z^2} = 0 \quad (2.6.4)$$

and in **spherical coordinates**

$$\frac{1}{r^2} \left[\frac{\partial}{\partial r} \left(r^2 \frac{\partial \phi}{\partial r} \right) + \frac{1}{\sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial \phi}{\partial \theta} \right) + \frac{1}{\sin^2 \theta} \frac{\partial^2 \phi}{\partial \varphi^2} \right] = 0 \quad (2.6.5)$$

Many physical phenomena in addition to incompressible, irrotational fluid flow are governed by Laplace's equation. These include heat conduction, elasticity, electricity, and magnetism. Hence, a great volume of literature has been devoted to methods for obtaining solutions to Laplace's equation. Numerous solutions to this second-order linear partial differential equation have been developed and are well understood.

Laplace's equation for inviscid, incompressible, irrotational flow provides a great simplification over the more general Euler equations for inviscid flow, which are given in vector notation in Eq. (2.3.34). Even for the case of incompressible flow where density is a known constant, the components of Eq. (2.3.34) are three coupled partial differential equations containing four scalar unknowns; the pressure p and the three components of the velocity vector $\mathbf{V}$. To complete the incompressible Euler formulation we include the continuity equation from Eq. (2.2.26). Thus, obtaining solutions to the Euler formulation for inviscid, incompressible flow necessitates solving a system of four coupled partial differential equations containing four unknown scalar variables. In contrast, Laplace's equation given in Eq. (2.6.2) is a single partial differential equation containing only one unknown scalar variable, i.e., the velocity potential ϕ . Once a solution for ϕ has been obtained, the three components of velocity are easily found from Eq. (2.5.31). With all components of velocity known, the pressure is the only remaining unknown, which can be obtained from Eq. (2.5.23).

There is another advantage of the potential flow equations over the Euler equations. Notice from the second term in the square brackets on the left-hand side of Eq. (2.3.34) that the inviscid Euler equations are nonlinear, whereas Laplace's equation for potential flow is linear. If two different solutions satisfy the same linear differential equation, the sum of these two solutions will also satisfy the differential equation. This provides the important advantage of allowing superposition of known solutions to be used to produce another solution satisfying a new set of boundary conditions. Hence, a rich collection of solutions to Laplace's equation has evolved.

The potential flow problem of greatest interest in aerodynamics is the flow about a solid object moving through a static fluid. Because Laplace's equation is elliptic, two boundary conditions are required to obtain a unique solution. One boundary condition is required at the surface of the object and a second boundary condition is required at the outer boundary of the fluid. It is usually convenient to choose a coordinate system fixed to the object, and the surrounding fluid is typically assumed to extend an infinite distance from the object in all directions, as shown in Fig. 2.6.1.

Relative to any object-fixed coordinate system, the velocity of the object's surface is zero. However, because the flow field is inviscid, we cannot apply the no-slip boundary condition, which means that we cannot impose zero fluid velocity at the object's surface. Nevertheless, even an inviscid fluid does not penetrate a solid boundary. Thus, the fluid velocity at the surface must be tangent to the surface, i.e., the normal component of fluid velocity must be zero at all points on the object's surface. Because the fluid velocity vector is equal to the gradient of the velocity potential, **to ensure that the fluid does not penetrate a solid boundary, we use the *inviscid wall boundary condition***

$$\nabla\phi \cdot \mathbf{n} = 0, \text{ at all points on the solid surface} \quad (2.6.6)$$

where $\mathbf{n}$ is typically defined to be the unit outward normal to the surface.

For flow about an object moving through an infinite static fluid, as the distance from the object approaches infinity, the fluid velocity relative to any object-fixed coordinate system must approach the negative of the object's velocity vector, which is called the *relative wind* or *freestream velocity*. In terms of velocity potential, **to ensure uniform relative wind far from the object, we use the *far-field boundary condition***

$$\nabla\phi = \mathbf{V}_\infty, \text{ as the distance from the object approaches } \infty \quad (2.6.7)$$

where $\mathbf{V}_\infty$ is the constant freestream velocity vector for the far-field uniform flow.

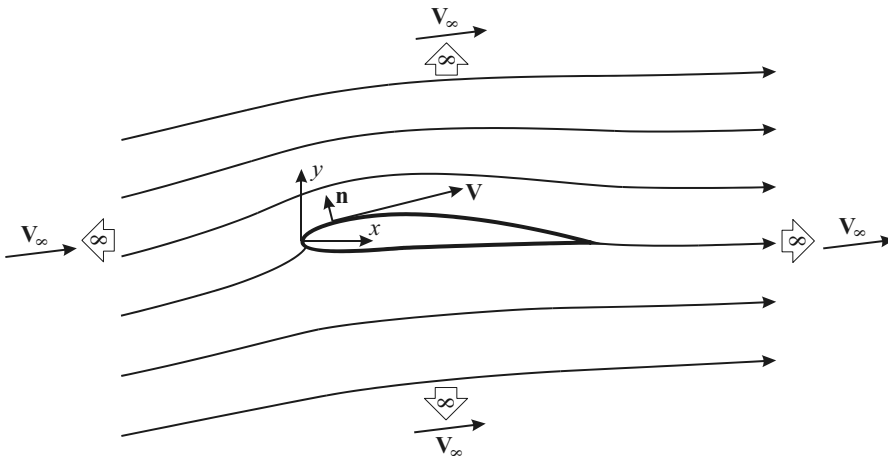


Figure 2.6.1. Potential flow about an object moving through a static fluid.

The far-field boundary condition given in Eq. (2.6.7) is commonly presented in terms of a Cartesian coordinate system having the x axis aligned with the freestream velocity vector. Clearly, the orientation of the coordinate system is arbitrary and could be chosen to have this alignment. However, the wall boundary condition in Eq. (2.6.6) is normally much more complex than far-field boundary condition. Thus, we usually find it more convenient to choose and align the coordinate system to simplify the implementation of Eq. (2.6.6). Although this might complicate implementation of the far-field boundary condition somewhat, Eq. (2.6.7) can be implemented quite easily for any orientation of any coordinate system.

2.7. Bernoulli's Equation

After a solution for the velocity potential has been obtained from Laplace's equation and the velocity components are found from Eq. (2.5.31), the fluid pressure remains the only unknown. Because we have obtained a solution for the velocity field after enforcing only the incompressible continuity equation subject to the velocity boundary conditions and the constraint of irrotational flow, Newton's second law as expressed in Eq. (2.5.23) is still available for evaluating the pressure. In this section we will examine such pressure solutions for the important special case of steady flow.

Setting the time derivative in Eq. (2.5.23) to zero, Newton's second law for steady, inviscid, incompressible, irrotational flow with constant gravitational acceleration yields

$$\nabla \left(\frac{1}{2} V^2 + \frac{p}{\rho} + gH \right) = 0 \quad (2.7.1)$$

Recall that the gradient of any scalar function is a vector with a magnitude equal to the maximum derivative of the function with respect to distance, considering all directions. Furthermore, this gradient vector points in the direction of the maximum increase in the function with respect to distance. For example, the gradient of geometric altitude H points in the direction of the maximum increase in altitude with respect to distance, which is always *vertically up* regardless of the coordinate system or its orientation. The magnitude of ∇H is the maximum derivative of altitude with respect to distance, which occurs in the direction of ∇H and is always 1 foot per foot regardless of the chosen coordinate system or its orientation. Hence, at any point in space where the gradient of any scalar function is zero, the function does not change as we move in any direction from that point. Moreover, if the gradient of a scalar function is zero at all points within some spatial domain, the function must remain constant over the entire domain.

Because Eq. (2.7.1) applies **at all points in any steady, inviscid, incompressible, irrotational flow with the only body force due to constant gravitational acceleration, the general solution is**

$$\frac{1}{2} V^2 + \frac{p}{\rho} + gH = C \quad (2.7.2)$$

where C is the constant of integration. Equation (2.7.2) is called *Bernoulli's Equation*, named after the Swiss mathematician Daniel Bernoulli (1700–1782).

The physical significance of Bernoulli's Equation is important in the aerodynamics of flight. Although both pressure and velocity vary from point to point in the flow field about an aircraft in flight, Bernoulli's Equation provides a very simple relation between pressure, velocity, and altitude, which applies in the inviscid region outside the boundary layer. Rearranging Eq. (2.7.2) to solve for the pressure yields

$$p = \rho(C - \frac{1}{2}V^2 - gH) \quad (2.7.3)$$

This form of Bernoulli's Equation exposes an important fundamental principle of fluid mechanics that lie at the foundation of the science of flight.

In a fluid flow field where the velocity of the fluid increases, the pressure decreases, and where the velocity decreases, the pressure increases.

Although Bernoulli's Equation does not apply in the viscous boundary layer next to the skin of an aircraft, for the conditions normally encountered in flight, the boundary layer is extremely thin and there is no significant pressure change across its thickness. Hence, Bernoulli's Equation accurately predicts the pressure acting on the surface of an aircraft at low Mach numbers and high Reynolds numbers, in the absence of the boundary layer separation phenomenon known as stall.

Examination of Eq. (2.7.3) reveals that, at Mach numbers less than about 0.3 where the air density is virtually constant, there are only two things that result in a change in air pressure, i.e., a change in velocity and a change in altitude. The difference in altitude between the upper and lower surfaces of an aircraft gives rise to the buoyant force, which is discussed in Sec. 1.10. Likewise, a difference in velocity above and below the aircraft results in the production of aerodynamic lift, which will occupy our attention throughout a significant portion of this textbook.

Because geometric altitude is always a known function of the spatial coordinates, the unknown constant C in Bernoulli's Equation can be evaluated from known values of velocity and pressure at any point in an irrotational flow field. For the important case of steady, inviscid flow about an object moving through an infinite static fluid, as the distance from the object approaches infinity, the relative fluid velocity approaches the uniform freestream velocity V_∞ , and the pressure approaches the freestream hydrostatic pressure p_∞ , which depends only on the altitude H . In this case, the constant C in Bernoulli's Equation can be evaluated from these known conditions in the far field,

$$C = \frac{1}{2}V_\infty^2 + \frac{p_\infty}{\rho} + gH \quad (2.7.4)$$

where V_∞ is the magnitude of the uniform relative wind far from the object and p_∞ is the far-field hydrostatic pressure at the altitude H . Using Eq. (2.7.4) in Eq. (2.7.2) yields

$$\frac{1}{2}V^2 + \frac{p}{\rho} + gH = \frac{1}{2}V_\infty^2 + \frac{p_\infty}{\rho} + gH \quad (2.7.5)$$

The pressure p that appears in Eqs. (2.7.1), (2.7.2), (2.7.3), and (2.7.5) is commonly referred to as the *static pressure*. The term *static pressure* should not be confused with

the term *hydrostatic pressure*. **The term *static pressure* does not refer to the pressure in a static fluid.** As discussed in Sec. 1.8, the pressure in a static fluid is traditionally called the *hydrostatic pressure*. **The term *static pressure* refers to the actual absolute thermodynamic pressure in the moving fluid at any point in a flow field.**

The difference between the local static pressure and the far-field hydrostatic pressure at the same altitude is an important parameter in aerodynamic flows. Solving Eq. (2.7.5) for the pressure difference $p - p_\infty$ yields

$$p - p_\infty = \frac{1}{2}\rho(V_\infty^2 - V^2) \quad (2.7.6)$$

As discussed in Sec. 1.3, this pressure difference is traditionally nondimensionalized in terms of what is called the *pressure coefficient*, denoted as C_p and defined in Eq. (1.3.5). After rearranging Eq. (2.7.6), we see that Bernoulli's equation yields a remarkably simple relation for the **pressure coefficient in an inviscid, incompressible flow field about an object in steady motion through an infinite static fluid**,

$$C_p \equiv \frac{p - p_\infty}{\frac{1}{2}\rho V_\infty^2} = 1 - \left(\frac{V}{V_\infty}\right)^2 \quad (2.7.7)$$

The development of Eq. (2.7.2) presented here has shown that Bernoulli's equation applies throughout any steady, inviscid, incompressible, irrotational flow field. However, Bernoulli's equation is slightly more general. For any steady, inviscid, incompressible, flow (rotational or irrotational), the Euler equations expressed in Eq. (2.5.21) require

$$\nabla \left(\frac{1}{2}V^2 + \frac{p}{\rho} + gH \right) = \mathbf{V} \times \boldsymbol{\Omega} \quad (2.7.8)$$

Hence, the gradient of Bernoulli's function ($\frac{1}{2}V^2 + p/\rho + gH$) is equal to the cross product of the velocity vector and the vorticity vector. Recall from vector algebra that the cross product of any two vectors is perpendicular to both vectors. Furthermore, recall from vector calculus that the derivative of any function with respect to distance along any path is equal to the dot product of the gradient of the function with a unit vector tangent to the path. Because the gradient of Bernoulli's function is orthogonal to both $\mathbf{V}$ and $\boldsymbol{\Omega}$, and because the dot product of two orthogonal vectors is zero, Bernoulli's function does not change along any path that is tangent to either the velocity vector or the vorticity vector. Because streamlines are tangent to the velocity vector everywhere, **Bernoulli's equation holds along a streamline even for rotational flows.**

In general, Bernoulli's equation ($\frac{1}{2}V^2 + p/\rho + gH = C$) holds along any streamline in any steady, inviscid, incompressible flow field, but the value of the constant C may be different for different streamlines. If the flow is also irrotational, the vorticity vector $\boldsymbol{\Omega}$ on the right-hand side of Eq. (2.7.8) is zero everywhere, and the constant C is the same for every streamline in the flow field. **Remember that Bernoulli's equation applies only in regions of the flow field where viscous effects are negligible; it does not apply in the boundary layer next to a solid surface, and it does not apply in the wake downstream from any region of separated flow.**

EXAMPLE 2.7.1. For the steady, inviscid, irrotational flow field that is described in Example 2.5.1, use Bernoulli's equation to find the pressure coefficient at an arbitrary point in the flow field.

Solution. From the solution to Example 2.5.1, the local fluid velocity vector at an arbitrary point in this flow field was found to be

$$\mathbf{V} = V_\infty(1 - R^2/r^2)\cos\theta\mathbf{i}_r - V_\infty(1 + R^2/r^2)\sin\theta\mathbf{i}_\theta + 0\mathbf{i}_z \quad (2.7.9)$$

Hence, the magnitude of the local fluid velocity vector $V(r, \theta, z)$ is obtained from

$$\begin{aligned} V^2 &= \mathbf{V} \cdot \mathbf{V} = V_r^2 + V_\theta^2 + V_z^2 = V_\infty^2[(1 - R^2/r^2)^2 \cos^2\theta + (1 + R^2/r^2)^2 \sin^2\theta] \\ &= V_\infty^2[(1 - 2R^2/r^2 + R^4/r^4)\cos^2\theta + (1 + 2R^2/r^2 + R^4/r^4)\sin^2\theta] \\ &= V_\infty^2[(1 - 2R^2/r^2 + R^4/r^4)\cos^2\theta + (1 - 2R^2/r^2 + R^4/r^4 + 4R^2/r^2)\sin^2\theta] \\ &= V_\infty^2[(1 - 2R^2/r^2 + R^4/r^4)(\cos^2\theta + \sin^2\theta) + 4(R^2/r^2)\sin^2\theta] \end{aligned}$$

or after applying the trigonometric identity $\cos^2\theta + \sin^2\theta = 1$,

$$V(r, \theta, z) = V_\infty \left(1 - 2\frac{R^2}{r^2} + \frac{R^4}{r^4} + 4\frac{R^2}{r^2}\sin^2\theta \right)^{1/2} \quad (2.7.10)$$

From this result, the magnitude of the far-field velocity in the limit as r approaches infinity is $V(\infty, \theta, z) = V_\infty$. The pressure coefficient found from Bernoulli's equation is given by Eq. (2.7.7),

$$\begin{aligned} C_p(r, \theta, z) &\equiv \frac{p - p_\infty}{\frac{1}{2}\rho V_\infty^2} = 1 - \left(\frac{V}{V_\infty} \right)^2 \\ &= 1 - \left(1 - 2\frac{R^2}{r^2} + \frac{R^4}{r^4} + 4\frac{R^2}{r^2}\sin^2\theta \right) \\ &= \frac{R^2}{r^2} \left(2 - \frac{R^2}{r^2} - 4\sin^2\theta \right) \end{aligned} \quad (2.7.11)$$

The velocity and pressure fields presented in Examples 2.5.1 and 2.7.1 are those for steady, inviscid flow normal to a long circular cylinder of radius R . The relative velocity far from the cylinder is uniform with magnitude V_∞ , and the pressure far from the cylinder is the hydrostatic pressure p_∞ , which depends only on the altitude H . The streamlines for this flow field were considered in Example 1.14.1.

From Eq. (2.7.9), we see that the radial component of relative velocity is zero at the surface of the cylinder where $r=R$; and because $\mathbf{i}_r$ is the surface normal, we see that the flow does not penetrate the surface of the cylinder. However, the tangential component of relative velocity remains finite at the surface. From either Eq. (2.7.9) or Eq. (2.7.10), we see that the magnitude of the relative velocity at the surface of the cylinder is

$$V(R, \theta, z) = 2V_\infty |\sin\theta| \quad (2.7.12)$$

Note from Eq. (2.7.12) that the surface velocity reaches a maximum value of $2V_\infty$ at the top and bottom of the cylinder ($\theta=\pi/2$ and $\theta=3\pi/2$, respectively). Also notice that the surface velocity is zero at the leading and trailing edges ($\theta=\pi$ and $\theta=0$, respectively), as shown in Fig. 2.7.1.

A point of zero velocity is called a *stagnation point* and the pressure at a stagnation point is called the *stagnation pressure*. From Eq. (2.7.6), the stagnation pressure is the sum of the far-field hydrostatic pressure p_∞ and the *dynamic pressure* $\frac{1}{2}\rho V_\infty^2$,

$$\text{stagnation pressure} \equiv p_0 = p_\infty + \frac{1}{2}\rho V_\infty^2 \quad (2.7.13)$$

Using the definition of stagnation pressure from Eq. (2.7.13) in the definition of pressure the coefficient from Eq. (1.3.5) or Eq. (2.7.7), **the reader should note that the pressure coefficient is always 1.0 at a stagnation point.**

From Eq. (2.7.11) for a circular cylinder in cross flow, the pressure coefficient at the surface of the cylinder ($r=R$) is

$$C_p(R, \theta, z) = 1 - 4\sin^2\theta \quad (2.7.14)$$

As shown in Fig. 2.7.1, the surface pressure on the upstream side of the cylinder remains above ambient from the leading-edge stagnation point to a point 30 degrees aft of the leading edge where the pressure is equal to the ambient pressure. Downstream from this point the surface pressure continues to decrease below ambient, reaching a minimum value of $p_\infty - \frac{3}{2}\rho V_\infty^2$ at the top and bottom of the cylinder where velocity is maximum.

Inviscid flow theory cannot predict the boundary layer separation that occurs on the lee side of a circular cylinder in cross flow. Hence, due to the cylinder's geometric symmetry, both the velocity magnitude and the pressure coefficient are symmetric fore and aft, as well as above and below the cylinder. From this symmetric pressure field, the net aerodynamic force acting on the cylinder's surface is zero, and the only pressure force predicted on the cylinder is the buoyant force, resulting from the hydrostatic pressure p_∞ .

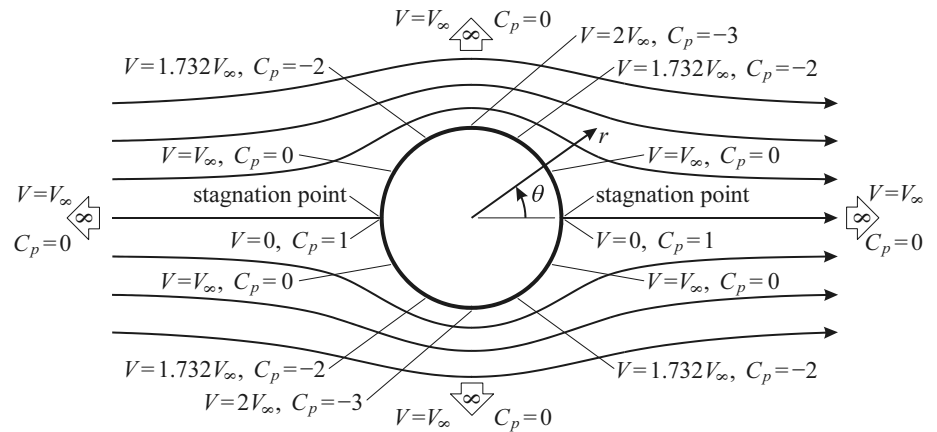


Figure 2.7.1. Velocity and pressure coefficient in a potential flow field about a circular cylinder.

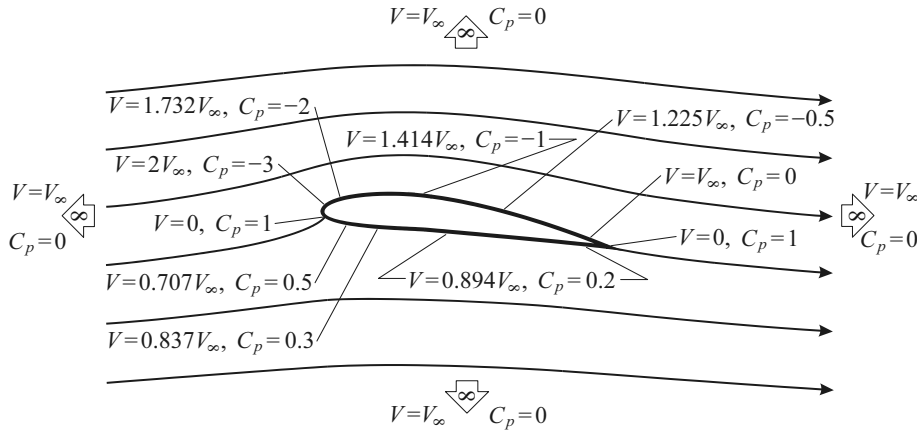


Figure 2.7.2. Velocity and pressure coefficient in a potential flow field about an airfoil.

Whereas potential flow about the symmetric geometry of a circular cylinder yields symmetric velocity and pressure fields, flow about an asymmetric object will result in asymmetric velocity and pressure fields. For the case of potential flow about an airfoil like that shown in Fig. 2.7.2, leading- and trailing-edge stagnation points are observed as was the case for flow about a circular cylinder. However, in this case the magnitude of the relative velocity at the surface of the airfoil remains below that of the freestream over the entire lower surface between the stagnation points. This results in a surface pressure that is above ambient on the lower surface. On the other hand, the magnitude of the surface velocity remains above the freestream value and the surface pressure is below ambient over most of the upper surfaces of the airfoil. This pressure difference between the upper and lower surfaces is the origin of the lift produced on the airfoil.

Compressibility Effects

Strictly speaking, the name *Bernoulli's equation* refers to Newton's second law as expressed in Eq. (2.7.2), which applies to steady, inviscid flow under the approximations of constant density and constant gravitational acceleration. However, the mathematical model used to develop Bernoulli's equation from the Euler equations is easily applied to compressible flow without the approximation of constant gravitational acceleration.

From Eq. (2.5.12), **when the only body force is due to gravity, the Euler equations for steady, inviscid, adiabatic flow of an ideal gas can be written as**

$$\nabla \left(\frac{1}{2} V^2 + \frac{\gamma}{\gamma-1} \frac{p}{\rho} + g_o Z \right) = \mathbf{V} \times \boldsymbol{\Omega} \quad (2.7.15)$$

For any case where the right-hand side of Eq. (2.7.15) is zero, integration yields

$$\frac{1}{2} V^2 + \frac{\gamma}{\gamma-1} \frac{p}{\rho} + g_o Z = C \quad (2.7.16)$$

where C is the constant of integration.

Equation (2.7.16) can be thought of simply as a *compressible version of Bernoulli's equation* that does not impose the approximation of constant gravitational acceleration. Like Bernoulli's equation, Eq. (2.7.16) holds along any streamline in a steady, inviscid flow field, but the value of the constant C may be different for different streamlines. If the flow is also irrotational, the vorticity vector $\mathbf{\Omega}$ on the right-hand side of Eq. (2.7.15) is zero everywhere, and the constant C is the same for every streamline in the flow field.

The development of Eq. (2.7.16) shows that the gravitational term gH on the left-hand side of Bernoulli's equation could have been written more precisely as $g_o Z$, where g_o is the constant gravitational acceleration at standard sea level and $Z \equiv R_E H / (R_E + H)$ is the geopotential altitude as defined in Eq. (1.6.5). For most aerodynamics flows of practical interest, the variation in altitude is insignificant compared to the radius of the Earth, and $g_o Z \cong gH$. However, for large-scale atmospheric flows, Eq. (2.7.16) should be used in place of Bernoulli's equation.

It is quite often convenient to express the constant C in Eq. (2.7.16) in terms of the stagnation conditions. By definition, the relative velocity is always zero at stagnation and Eq. (2.7.16) requires

$$C = \frac{\gamma}{\gamma-1} \frac{p_0}{\rho_0} + g_o Z \quad (2.7.17)$$

where p_0 and ρ_0 are, respectively, the stagnation pressure and density at the geopotential altitude Z . Using Eq. (2.7.17) in Eq. (2.7.16) and rearranging yields

$$\begin{aligned} \frac{1}{2} V^2 + \frac{\gamma}{\gamma-1} \frac{p}{\rho} + g_o Z &= \frac{\gamma}{\gamma-1} \frac{p_0}{\rho_0} + g_o Z \\ p + \frac{\gamma-1}{2\gamma} \rho V^2 &= p_0 \frac{\rho}{\rho_0} \end{aligned} \quad (2.7.18)$$

Inviscid, adiabatic flow is isentropic, and for isentropic flow of an ideal gas, density is related to pressure according to Eq. (2.5.4), which requires

$$\rho = \frac{\rho_{\text{ref}}}{p_{\text{ref}}^{1/\gamma}} p^{1/\gamma} \quad \text{and} \quad \rho_0 = \frac{\rho_{\text{ref}}}{p_{\text{ref}}^{1/\gamma}} p_0^{1/\gamma} \quad (2.7.19)$$

Using the ideal gas law from Eq. (1.2.6) in Eq. (2.7.19) and rearranging yields

$$\begin{aligned} \frac{p}{RT} &= \frac{\rho_{\text{ref}}}{p_{\text{ref}}^{1/\gamma}} p^{1/\gamma} \quad \text{and} \quad \frac{p_0}{RT_0} = \frac{\rho_{\text{ref}}}{p_{\text{ref}}^{1/\gamma}} p_0^{1/\gamma} \\ T &= \frac{p_{\text{ref}}^{1/\gamma}}{R \rho_{\text{ref}}} p^{(\gamma-1)/\gamma} \quad \text{and} \quad T_0 = \frac{p_{\text{ref}}^{1/\gamma}}{R \rho_{\text{ref}}} p_0^{(\gamma-1)/\gamma} \end{aligned} \quad (2.7.20)$$

From Eqs. (2.7.19) and (2.7.20), **for inviscid, adiabatic flow of an ideal gas**

$$\frac{\rho}{\rho_0} = \left(\frac{p}{p_0} \right)^{1/\gamma} \quad \text{and} \quad \frac{T}{T_0} = \left(\frac{p}{p_0} \right)^{(\gamma-1)/\gamma} \quad (2.7.21)$$

Using the ideal gas law to eliminate the density from the second term on the left-hand side of Eq. (2.7.18), and applying Eq. (2.7.21) to express the density ratio on the right-hand side of Eq. (2.7.18) in terms of a pressure ratio, we obtain the relation

$$\begin{aligned} p + \frac{\gamma-1}{2\gamma} \rho V^2 &= p_0 \frac{\rho}{\rho_0} \\ p + \frac{\gamma-1}{2\gamma} \frac{p}{RT} V^2 &= p_0 \left(\frac{p}{p_0} \right)^{1/\gamma} \\ \frac{p}{p_0} \left(1 + \frac{\gamma-1}{2} \frac{V^2}{\gamma RT} \right) &= \left(\frac{p}{p_0} \right)^{1/\gamma} \end{aligned} \quad (2.7.22)$$

From Eqs. (1.2.16) and (1.3.1), we see that $V^2/(\gamma RT)$ is the Mach number squared. Thus, solving Eq. (2.7.22) for the pressure ratio p/p_0 yields an important relation that **applies to steady, inviscid, adiabatic flow of an ideal gas**,

$$\frac{p}{p_0} = \left(1 + \frac{\gamma-1}{2} M^2 \right)^{-\gamma/(\gamma-1)} \quad (2.7.23)$$

Using Eq. (2.7.21) in Eq. (2.7.23) leads to the additional isentropic relations

$$\frac{\rho}{\rho_0} = \left(1 + \frac{\gamma-1}{2} M^2 \right)^{-1/(\gamma-1)} \quad (2.7.24)$$

$$\frac{T}{T_0} = \left(1 + \frac{\gamma-1}{2} M^2 \right)^{-1} \quad (2.7.25)$$

Equations (2.7.23)–(2.7.25) apply along the streamlines of any steady, inviscid, adiabatic flow of an ideal gas, rotational or irrotational. If the flow is irrotational, these isentropic relations apply throughout the flow field.

The stagnation pressure provides a convenient reference pressure for inviscid flow, because it combines the effects of static pressure and velocity into a single variable. For the important case of steady, inviscid flow about an object moving through an infinite static ideal gas, the flow is irrotational and the stagnation conditions are readily evaluated by applying Eqs. (2.7.23)–(2.7.25) to the freestream flow:

$$p_0 = p_\infty \left(1 + \frac{\gamma-1}{2} M_\infty^2 \right)^{\gamma/(\gamma-1)} \quad (2.7.26)$$

$$\rho_0 = \rho_\infty \left(1 + \frac{\gamma-1}{2} M_\infty^2 \right)^{1/(\gamma-1)} \quad (2.7.27)$$

$$T_0 = T_\infty \left(1 + \frac{\gamma-1}{2} M_\infty^2 \right) \quad (2.7.28)$$

EXAMPLE 2.7.2. Use Bernoulli's equation for incompressible flow to plot the ratio of the static pressure p to the stagnation pressure p_0 as a function of Mach number over the range $0 \leq M \leq 1$. Assume that the fluid density and the speed of sound remain constant at the values obtained for standard atmospheric air at the stagnation state. Compare the results with results obtained from Eq. (2.7.23).

Solution. The constant C in Bernoulli's equation can be expressed in terms of the stagnation pressure. By definition, the relative velocity is zero at stagnation and Eq. (2.7.2) requires

$$C = \frac{p_0}{\rho} + gH \quad (2.7.29)$$

where p_0 is the stagnation pressure at the altitude H . Using Eq. (2.7.29) for the constant C in Eq. (2.7.2) and rearranging yields

$$\begin{aligned} \frac{1}{2}V^2 + \frac{p}{\rho} + gH &= \frac{p_0}{\rho} + gH \\ p &= p_0 - \frac{1}{2}\rho V^2 \end{aligned} \quad (2.7.30)$$

Expressing the assumed constant density in Eq. (2.7.30) using the ideal gas law at the stagnation state and evaluating the assumed constant speed of sound from the ideal gas relation given by Eq. (1.2.16) leads to

$$\begin{aligned} p &= p_0 - \frac{1}{2}\frac{p_0}{RT_0}V^2 = p_0\left(1 - \frac{1}{2}\gamma\frac{V^2}{\gamma RT_0}\right) \\ &= p_0\left(1 - \frac{1}{2}\gamma\frac{V^2}{a^2}\right) = p_0(1 - \frac{1}{2}\gamma M^2) \end{aligned} \quad (2.7.31)$$

For example, using $\gamma=1.4$ and $M=0.3$ in Eq. (2.7.31) yields

$$\frac{p}{p_0} = 1 - \frac{1}{2}\gamma M^2 = 1 - \frac{1}{2}(1.4)(0.3)^2 = 0.937$$

For the same conditions, Eq. (2.7.23) yields

$$\frac{p}{p_0} = \left(1 + \frac{\gamma-1}{2}M^2\right)^{-\gamma/(\gamma-1)} = \left(1 + \frac{1.4-1}{2}0.3^2\right)^{-1.4/(1.4-1)} = 0.939$$

At higher Mach numbers the difference is more significant. For example,

$$\text{at } M=0.6: 1 - \frac{1}{2}\gamma M^2 = 0.748 \text{ and } \left(1 + \frac{\gamma-1}{2}M^2\right)^{-\gamma/(\gamma-1)} = 0.784$$

Using these and similar calculations for other Mach numbers produces the results shown in Fig. 2.7.3. Notice that for Mach numbers less than about 0.3, the constant density approximation that is used in Bernoulli's equation yields results that are in good agreement with compressible flow results obtained from Eq. (2.7.23).

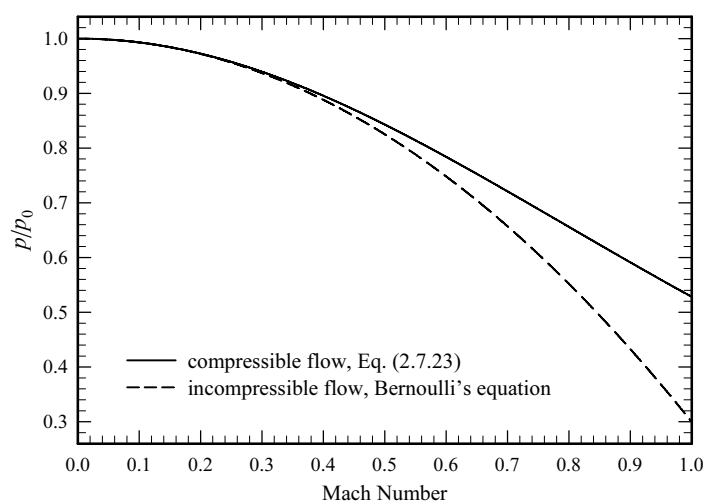


Figure 2.7.3. Comparison of results obtained from Bernoulli's equation for incompressible flow with results obtained from the compressible flow relation given by Eq. (2.7.23).

2.8. Airspeed Measurement

The physical and mathematical principles used to develop Bernoulli's equation provide the analytical foundation for one of the most frequently used methods for measuring fluid velocities. For measuring velocities in water or other incompressible liquids, Bernoulli's equation can be used directly. Expressing the constant C in Bernoulli's equation in terms of the stagnation pressure leads to Eq. (2.7.30), which is easily solved for the magnitude of the fluid velocity V ,

$$V = \sqrt{\frac{2(p_0 - p)}{\rho}} \quad (2.8.1)$$

where p_0 is the stagnation pressure (also commonly referred to as the *total pressure*), p is the local static pressure, and ρ is the constant density of the fluid. From Eq. (2.8.1), we see that **the velocity of an incompressible fluid of known density can be determined from the difference between the stagnation pressure and the static pressure.**

A typical apparatus for the determination of fluid velocity is shown schematically in Fig. 2.8.1. The device used to sense the stagnation pressure is called a *Pitot tube*, named after its French inventor Henri Pitot (1695–1771). The Pitot tube is inserted into the fluid with its open end facing directly into the flow; i.e., the plane of the opening in the tube is perpendicular to the flow. The other end of the Pitot tube's channel is closed by sealing it to one of the closed chambers of a differential pressure gage, as shown in Fig. 2.8.1. When the Pitot tube is first inserted into the flow, the fluid flows into the open end of the tube, filling the channel. However, because the tube is closed at the other end, all fluid inside the tube will soon come to rest or *stagnate*, including that at the open end of the tube. Thus, the open end of the Pitot tube is a stagnation point for the flow field about the tube, and the pressure inside the tube is the stagnation pressure for the flow.

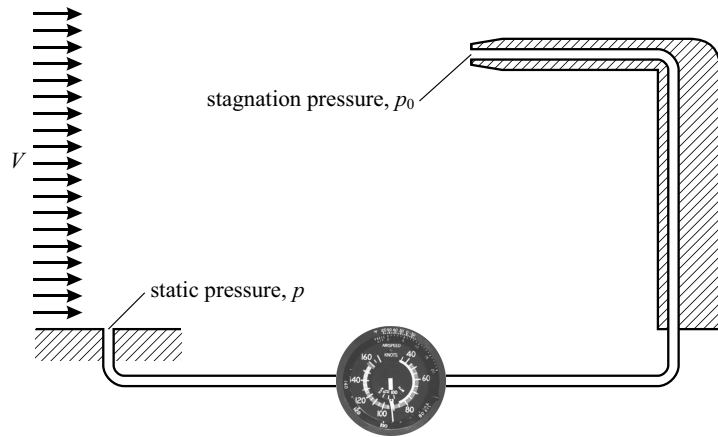


Figure 2.8.1. Pitot tube and static-pressure orifice used to experimentally determine fluid velocity.

As shown in Fig. 2.8.1, the other side of the differential pressure gage is connected to a small hole that is drilled perpendicular to a solid surface, which is aligned with the direction of flow; i.e., the plane of the opening is parallel with the flow. Such a hole is commonly called a *static-pressure orifice* or a *static-pressure tap*. If the hole is properly drilled and located, the pressure at the opening and throughout the channel connecting the orifice to the pressure gage is equal to the static pressure in the fluid above the orifice. Although the opening of the static-pressure orifice is located on the surface beneath the viscous boundary layer, where Bernoulli's equation does not apply, the boundary layer is thin and there is no significant pressure change across its thickness.

Because the Pitot tube and the static-pressure orifice are connected to opposite sides of a differential pressure gage, this gage indicates the difference between the stagnation pressure and the static pressure ($p_0 - p$). For an incompressible fluid of known density, this pressure difference can be used in Eq. (2.8.1) to determine the relative velocity V .

A Pitot tube is also commonly used for determining the airspeed of an airplane. For low-speed aircraft, a wing-mounted Pitot tube, like that shown in Fig. 2.8.2, is typically combined with a static-pressure tap located on the side of the fuselage at a point where



Figure 2.8.2. Wing-mounted Pitot tube on a Cessna 172.

the local static pressure is equal to the freestream static pressure. The exact location of the Pitot tube is not critical, because the stagnation pressure is the same at all locations in the inviscid flow field outside the boundary layer and any regions of separated flow. Furthermore, a properly designed Pitot tube is fairly insensitive to misalignment. For a Pitot tube with a flat face like that shown in Fig. 2.8.1, the measured stagnation pressure varies only about 1 percent for misalignments as large as 20 degrees. On the other hand, a satisfactory location for the static-pressure orifice sometimes can be difficult to find, because at most locations on the surface of an airplane, the local static pressure is not equal to the freestream static pressure. Moreover, the static pressure at most locations on the surface of an airplane will vary significantly with angle of attack.

For airplanes that do not exceed a Mach number of about 0.3, the Bernoulli-based relation given by Eq. (2.8.1) could be used to convert a differential pressure measurement to velocity, if the local air density were known. Unfortunately, even for flight at low Mach numbers, Eq. (2.8.1) cannot be used to mark the face of a differential pressure gage in terms of velocity, because local air density varies with altitude, time, and location.

It is actually easier to account for compressibility effects at higher Mach numbers than it is to account for variations in local air density. From the compressible version of Bernoulli's equation, airspeed is related to the static and stagnation pressures according to Eq. (2.7.22), which is easily solved for the freestream airspeed V_∞ ,

$$V_\infty = \sqrt{\frac{2\gamma RT_\infty}{\gamma-1} \left[\left(\frac{p_0}{p_\infty} \right)^{(\gamma-1)/\gamma} - 1 \right]} \quad (2.8.2)$$

To improve accuracy, the stagnation pressure is typically measured relative to the static pressure using a differential pressure gage. Hence, Eq. (2.8.2) is commonly written as

$$V_\infty = \sqrt{\frac{2\gamma RT_\infty}{\gamma-1} \left[\left(\frac{p_0 - p_\infty}{p_\infty} + 1 \right)^{(\gamma-1)/\gamma} - 1 \right]} \quad (2.8.3)$$

where p_0 is the stagnation pressure, p_∞ is the freestream static pressure, and T_∞ is the freestream static temperature. Within the accuracy of the adiabatic ideal gas relations, the airspeed computed from Eq. (2.8.3) is the actual speed of the aircraft relative to the surrounding air, which is commonly called the **true airspeed** (TAS). If the freestream values of p_0 , p_∞ , and T_∞ are determined accurately from any experimental measurements, **the true airspeed can be computed accurately from Eq. (2.8.3)**. Formally, the true airspeed is defined in the US Federal Aviation Regulations (FAR) Part 1, *Definitions and Abbreviations* as:

True airspeed (TAS or V_∞) is defined to be “the airspeed of an aircraft relative to undisturbed air.”

Equation (2.8.3) cannot be used to mark the face of the differential pressure gage in terms of airspeed, because the scale would depend on the ambient static temperature and pressure. In lieu of an airspeed scale that is valid at any temperature and pressure, the

output of the differential pressure gage is converted to a *pseudo airspeed*, which is called the **indicated airspeed** (IAS) and is denoted here as V_i ,

$$V_i \equiv \sqrt{\frac{2\gamma RT_{ssl}}{\gamma-1} \left[\left(\frac{\Delta p}{p_{ssl}} + 1 \right)^{(\gamma-1)/\gamma} - 1 \right]} \quad (2.8.4)$$

where $T_{ssl} \equiv 288.150 \text{ K}$ and $p_{ssl} \equiv 101,325 \text{ N/m}^2$ are the standard atmospheric temperature and pressure at sea level, and Δp is the Pitot-static pressure difference obtained from the differential pressure gage, uncorrected for airspeed system errors. Because the indicated airspeed obtained from Eq. (2.8.4) is not the true airspeed, the cockpit instrument that displays an airplane's Pitot-statically determined airspeed is commonly referred to as an *airspeed indicator*. Formally, the indicated airspeed is defined in FAR Part 1 as:

Indicated airspeed (IAS or V_i) is defined as “the speed of an aircraft as shown on its Pitot-static airspeed indicator calibrated to reflect standard atmosphere adiabatic compressible flow at sea level, uncorrected for airspeed system errors.”

The measured Pitot-static pressure difference Δp in Eq. (2.8.4) is not precisely equal to the freestream Pitot-static pressure difference ($p_0 - p_\infty$) in Eq. (2.8.3). There are two major classifications of airspeed system errors in a Pitot-static system. The first is called *position error*. This is the error associated with locating the static-pressure tap at a point where the local static pressure is not equal to the freestream static pressure. This error often can be all but eliminated by choosing a proper location for the static-pressure tap. The second classification of Pitot-static airspeed system errors is called *instrument error*. This is simply precision error in the differential pressure gage and readout device.

When certified, an airplane's airspeed system must be calibrated to assure that the true airspeed can be determined from the indicated airspeed when suitably corrected. The calibration provides means to obtain a relation between the measured Pitot-static pressure difference Δp and the freestream Pitot-static pressure difference ($p_0 - p_\infty$). This calibration corrects for both position and instrument errors, and allows us to evaluate the freestream Pitot-static pressure difference ($p_0 - p_\infty$) from the differential pressure gage reading Δp and other instrument readings. The calibrated Pitot-static pressure difference is related to what is called the **calibrated airspeed** (CAS), which is denoted here as V_c ,

$$V_c \equiv \sqrt{\frac{2\gamma RT_{ssl}}{\gamma-1} \left[\left(\frac{\Delta p_c}{p_{ssl}} + 1 \right)^{(\gamma-1)/\gamma} - 1 \right]} \quad (2.8.5)$$

where Δp_c is the calibrated Pitot-static pressure difference. A proper calibration of the Pitot-static airspeed system will yield $\Delta p_c = p_0 - p_\infty$. Formally, the calibrated airspeed is defined in FAR Part 1 as:

Calibrated airspeed (CAS or V_c) is defined as “the indicated airspeed of an aircraft, corrected for position and instrument error. Calibrated airspeed is equal to true airspeed in the standard atmosphere at sea level.”

Comparing Eqs. (2.8.3) and (2.8.5), we see that the last statement in the FAR definition of calibrated airspeed requires $\Delta p_c = p_0 - p_\infty$. Typically, the relation between Δp_c and Δp is not obtained directly, but it is obtained in the form of a relation between V_i and V_c . Hence, it may be helpful to think of Eq. (2.8.5), not as a definition of V_c , but rather as a definition of Δp_c .

The relation between calibrated airspeed and true airspeed is often misunderstood. The true airspeed can be obtained from the calibrated airspeed by solving Eq. (2.8.5) for Δp_c and using the result in Eq. (2.8.3),

$$\Delta p_c \equiv p_{\text{ssl}} \left[\left(\frac{\gamma-1}{2} \frac{V_c^2}{\gamma R T_{\text{ssl}}} + 1 \right)^{(\gamma/(\gamma-1))} - 1 \right] \quad (2.8.6)$$

$$V_\infty = \sqrt{\gamma R T_\infty} \sqrt{\frac{2}{\gamma-1} \left[\left(\frac{\Delta p_c}{p_\infty} + 1 \right)^{(\gamma-1)/\gamma} - 1 \right]} \quad (2.8.7)$$

These two relations are based solely on the FAR definitions for the calibrated and true airspeeds, and they allow us to obtain the true airspeed V_∞ as a function of the calibrated airspeed V_c , the ambient temperature T_∞ , and the ambient pressure p_∞ .

Because an airplane's airspeed indicator is calibrated to read the true airspeed only in the standard atmosphere at sea level, it is sometimes convenient to define another pseudo airspeed, which is the airspeed in the standard atmosphere at sea level that would produce the same freestream dynamic pressure as that produced by flight at the true airspeed and the correct density altitude. This pseudo airspeed is called the **equivalent airspeed** (EAS) and is denoted here as V_e . The equivalent airspeed is defined to give $\frac{1}{2} \rho_{\text{ssl}} V_e^2 \equiv \frac{1}{2} \rho_\infty V_\infty^2$ or

$$V_e \equiv V_\infty \sqrt{\frac{\rho_\infty}{\rho_{\text{ssl}}}} \quad (2.8.8)$$

Formally, the equivalent airspeed is defined in FAR Part 1 to give:

Equivalent airspeed (EAS or V_e) is defined such that “true airspeed is equal to equivalent airspeed multiplied by $(\rho_{\text{ssl}}/\rho_\infty)^{1/2}$.” In other words, *equivalent airspeed is equal to the true airspeed multiplied by $(\rho_\infty/\rho_{\text{ssl}})^{1/2}$* . The equivalent airspeed is equal to the calibrated airspeed only in the standard atmosphere at sea level.

The last statement in this definition is important, because the true airspeed is sometimes computed by assuming that the equivalent airspeed is equal to the calibrated airspeed at any temperature and pressure. This leads to the **approximate relation for true airspeed**

$$V_\infty \cong V_c \sqrt{\frac{\rho_{\text{ssl}}}{\rho_\infty}} \quad (2.8.9)$$

This relation is sometimes presented in the literature as being exact. However, it is exact only at standard sea level and **it can produce significant error at higher altitudes**.

EXAMPLE 2.8.1. An airplane is flying at an altitude where ambient temperature and pressure are the same as those in the standard atmosphere at 40,000 ft. If the calibrated airspeed is 240 knots, find the true airspeed and Mach number. Also find the approximate true airspeed from Eq. (2.8.9) and the equivalent airspeed.

Solution. Calibrated airspeed is $V_c = 240$ knots $= 405.07$ ft/s, at standard sea level ambient temperature is $T_{ssl} = 518.67$ R, and ambient pressure is $p_{ssl} = 2116.2$ lbf/ft². Thus, the calibrated Pitot-static pressure difference obtained from Eq. (2.8.6) is

$$\begin{aligned}\Delta p_c &\equiv p_{ssl} \left[\left(\frac{\gamma-1}{2} \frac{V_c^2}{\gamma R T_{ssl}} + 1 \right)^{\gamma/(\gamma-1)} - 1 \right] \\ &= 2116.2 \text{ lbf/ft}^2 \left[\left(\frac{0.4}{2} \frac{405.07^2}{1116.45^2} + 1 \right)^{1.4/0.4} - 1 \right] = 201.503 \text{ lbf/ft}^2\end{aligned}$$

At the flight altitude, ambient temperature is $T_\infty = 389.97$ R and ambient pressure is $p_\infty = 393.13$ lbf/ft². Thus, the true airspeed obtained from Eq. (2.8.7) is

$$\begin{aligned}V_\infty &= \sqrt{\gamma R T_\infty} \sqrt{\frac{2}{\gamma-1} \left[\left(\frac{\Delta p_c}{p_\infty} + 1 \right)^{(\gamma-1)/\gamma} - 1 \right]} \\ &= 968.08 \text{ ft/s} \sqrt{\frac{2}{0.4} \left[\left(\frac{201.503}{393.13} + 1 \right)^{0.4/1.4} - 1 \right]} = \underline{766.87 \text{ ft/s} = 454.36 \text{ knots}}\end{aligned}$$

and the Mach number is

$$M_\infty = \frac{V_\infty}{\sqrt{\gamma R T_\infty}} = \frac{766.87}{968.08} = \underline{0.79216}$$

The air density at standard sea level is $\rho_{ssl} = 2.3769 \times 10^{-3}$ slugs/ft³ and the air density at the flight altitude is $\rho_\infty = 5.8728 \times 10^{-4}$ slugs/ft³. Thus, the approximate true airspeed obtained from Eq. (2.8.9) is

$$V_\infty \cong V_c \sqrt{\frac{\rho_{ssl}}{\rho_\infty}} = 405.07 \text{ ft/s} \sqrt{\frac{2.3769 \times 10^{-3}}{5.8728 \times 10^{-4}}} = \underline{814.92 \text{ ft/s} = 482.83 \text{ knots}}$$

which is 6.3% above the true airspeed. Results obtained from similar calculations over a range of altitude and Mach number are shown in Fig. 2.8.3. Note that the approximation given by Eq. (2.8.9) is accurate for low Mach numbers. Even at 100,000 ft, the error is only 1.1% for a Mach number of 0.3.

From Eq. (2.8.8) the equivalent airspeed is

$$V_e \equiv V_\infty \sqrt{\frac{\rho_\infty}{\rho_{ssl}}} = 766.87 \text{ ft/s} \sqrt{\frac{5.8728 \times 10^{-4}}{2.3769 \times 10^{-3}}} = \underline{381.19 \text{ ft/s} = 225.85 \text{ knots}}$$

This is 5.9% below the calibrated airspeed.

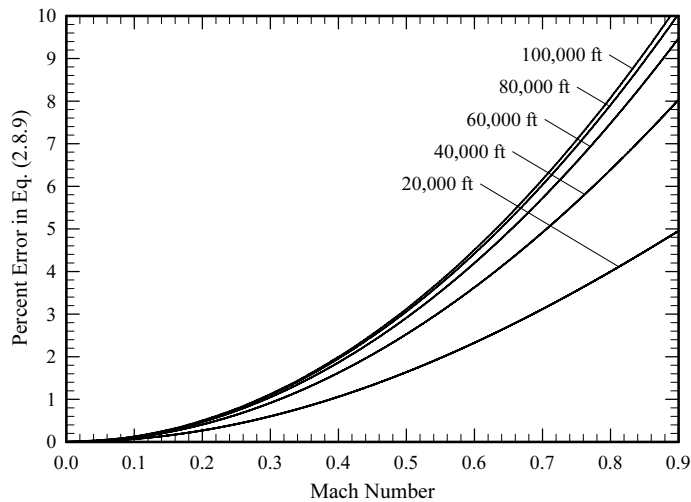


Figure 2.8.3. Error in Eq. (2.8.9) as a function of Mach number and altitude.

It is possible to combine measurement of the stagnation and static pressures into a single instrument, which is commonly called a *Pitot-static probe*. A typical apparatus of this type is shown schematically in Fig. 2.8.4. The device senses stagnation pressure at the nose of the probe just as is done with a Pitot tube. To ensure accurate static pressure readings, the nose is usually hemispherical and static pressure taps are located around the circumference of the probe 8 to 16 diameters downstream from the nose and at least 16 diameters upstream from the support stem. This results in a long-thin probe that is fragile and more sensitive to misalignment than the simple Pitot tube. For this reason, many modern airplanes use a static-pressure tap located on the side of the fuselage combined with a separate Pitot tube, which can be much shorter than a Pitot-static probe.

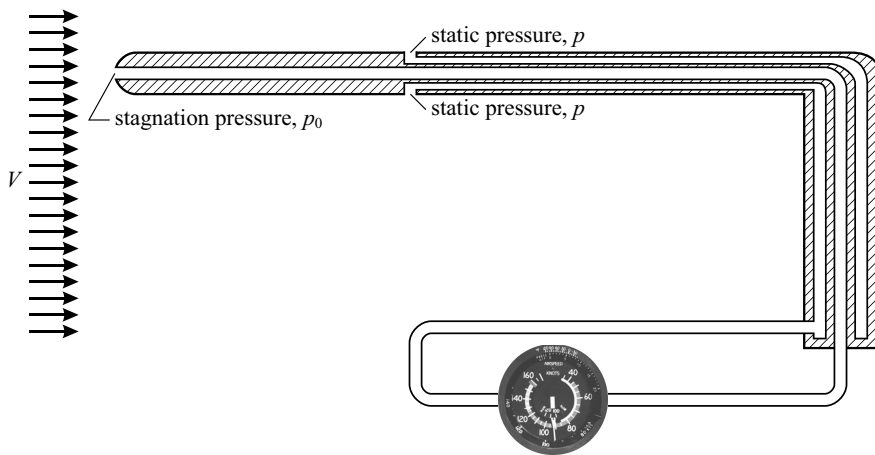


Figure 2.8.4. Pitot-static probe used to experimentally determine fluid velocity.

2.9. Stream Functions

Many problems of interest in fluid mechanics and aerodynamics are two dimensional in nature; i.e., *the fluid velocity and pressure fields depend on only two of the three spatial coordinates, and the component of velocity in the third coordinate direction is zero.* For instance, from Examples 1.14.1, 2.4.1, and 2.7.1, in cylindrical coordinates the velocity and pressure fields for inviscid flow normal to a long circular cylinder depend on only the radial coordinate r and the angular coordinate θ . This flow is independent of the axial coordinate z , and the z component of fluid velocity is zero.

Consider an arbitrary steady two-dimensional flow field in Cartesian coordinates with a velocity vector that lies in the x - y plane,

$$\mathbf{V}(x, y) = V_x(x, y)\mathbf{i}_x + V_y(x, y)\mathbf{i}_y + 0\mathbf{i}_z \quad (2.9.1)$$

From Eq. (1.14.2), the streamlines for this two-dimensional flow field are governed by

$$\frac{dy}{dx} = \frac{V_y(x, y)}{V_x(x, y)}, \quad \frac{dz}{dx} = 0 \quad (2.9.2)$$

If V_x and V_y were known functions of x and y , Eq. (2.9.2) could be integrated to yield the algebraic equations for the two-dimensional streamlines, which are of the form:

$$\psi(x, y) = C_1, \quad z = C_2 \quad (2.9.3)$$

where C_1 and C_2 are arbitrary constants of integration, which have different values for different streamlines. From Eq. (2.9.3), we see that both ψ and z remain constant along the streamlines. The function $\psi(x, y)$ is called the *stream function*.

Consider a differential vector segment of some streamline $d\mathbf{s} = dx\mathbf{i}_x + dy\mathbf{i}_y + dz\mathbf{i}_z$. Because the stream function ψ is constant along any streamline, the dot product of the gradient of ψ with $d\mathbf{s}$ is zero. That is, along any streamline Eq. (2.9.3) requires

$$\nabla\psi \cdot d\mathbf{s} = 0 \quad (2.9.4)$$

Similarly, because z remains constant along any streamline, Eq. (2.9.3) also requires

$$\nabla z \cdot d\mathbf{s} = 0 \quad (2.9.5)$$

The dot product of two nonzero vectors is zero if and only if the two vectors are normal (i.e., the two vectors are mutually orthogonal or perpendicular to each other). Thus, the mathematical relations given in vector notation by Eqs. (2.9.4) and (2.9.5) require that the local streamline segment $d\mathbf{s}$ is normal to both $\nabla\psi$ and ∇z at every point in the flow field. Furthermore, by definition, the local velocity vector is parallel with the local streamline at every point in the flow field. Consequently, the velocity vector $\mathbf{V}$ is also normal to both $\nabla\psi$ and ∇z at every point in the flow field. In other words, at any point in the flow field, $\mathbf{V}$ is normal to the plane formed by the two vectors $\nabla\psi$ and ∇z .

Recall from our review of vector mathematics in Sec. 1.4, that the cross product of two vectors is normal to both vectors. That is, if $\mathbf{A}$ and $\mathbf{B}$ are two nonparallel vectors, $\mathbf{A} \times \mathbf{B}$ is a vector that is normal to both $\mathbf{A}$ and $\mathbf{B}$ in the direction of the axis of a right-hand rotation from $\mathbf{A}$ to $\mathbf{B}$. In other words, if the fingers of your right hand are curled in the

direction of the rotation from **A** to **B**, your right thumb will point in the direction of the vector $\mathbf{A} \times \mathbf{B}$. Hence, because the local streamline segment $d\mathbf{s}$ is parallel with the local velocity vector $\mathbf{V}$, Eqs. (2.9.4) and (2.9.5) require that the velocity vector $\mathbf{V}$ is parallel with the cross product of the vectors $\nabla\psi$ and ∇z at every point in the flow field. The stream function $\psi(x, y)$ does not directly specify the magnitude of $\mathbf{V}$, only its direction. Accordingly, when relating $\mathbf{V}$ to the cross product of $\nabla\psi$ and ∇z , we may include an unknown function to account for magnitude, i.e.,

$$\chi(x, y)\mathbf{V}(x, y) = \nabla\psi \times \nabla z \quad (2.9.6)$$

where $\chi(x, y)$ is a scalar function of position, which is as yet unknown. Here we shall refer to $\chi(x, y)$ as the *magnitude function*.

There is a degree of arbitrariness associated with the magnitude function $\chi(x, y)$ and the stream function $\psi(x, y)$. Clearly, the magnitude of the fluid velocity vector is not arbitrary; it depends on the equations of motion. However, from Eq. (2.9.6), we see that the magnitude of $\mathbf{V}$ is given by $V(x, y) = |\nabla\psi \times \nabla z|/\chi(x, y)$. From Eq. (1.4.22), the vector ∇z is simply equal to $\mathbf{i}_z$, which is of unit magnitude. Thus, the magnitude of $\mathbf{V}$ depends on the ratio $|\nabla\psi|/\chi(x, y)$. We may choose $\chi(x, y)$ as needed to help satisfy the equations of motion. However, once $\chi(x, y)$ is chosen, $\psi(x, y)$ is fixed by Eq. (2.9.6).

To see how $\chi(x, y)$ can be chosen to help satisfy the equations of motion, Eq. (2.9.6) can be expanded in Cartesian coordinates using Eqs. (1.4.15) and (1.4.22). For the steady two-dimensional flow field specified by Eqs. (2.9.1)–(2.9.3), this yields

$$\begin{aligned} \chi(x, y)\mathbf{V}(x, y) &= \chi(x, y)V_x(x, y)\mathbf{i}_x + \chi(x, y)V_y(x, y)\mathbf{i}_y + 0\mathbf{i}_z \\ &= \nabla\psi \times \nabla z \\ &= \begin{vmatrix} \mathbf{i}_x & \mathbf{i}_y & \mathbf{i}_z \\ \frac{\partial\psi}{\partial x} & \frac{\partial\psi}{\partial y} & 0 \\ 0 & 0 & 1 \end{vmatrix} = \frac{\partial\psi}{\partial y}\mathbf{i}_x - \frac{\partial\psi}{\partial x}\mathbf{i}_y + 0\mathbf{i}_z \end{aligned} \quad (2.9.7)$$

From Eq. (1.4.23), taking the divergence of Eq. (2.9.7) yields

$$\nabla \cdot (\chi\mathbf{V}) = \mathbf{V} \cdot (\nabla\chi) + \chi\nabla \cdot \mathbf{V} = \frac{\partial^2\psi}{\partial x\partial y} - \frac{\partial^2\psi}{\partial y\partial x} = 0 \quad (2.9.8)$$

Note that defining the stream function according to Eq. (2.9.6) forces the divergence of the product $\chi\mathbf{V}$ to zero, independent of our choice of $\chi(x, y)$. For incompressible flow, the continuity equation from Eq. (2.2.26) requires $\nabla \cdot \mathbf{V} = 0$; so choosing $\chi=1$ will enforce the incompressible continuity equation for any stream function. Likewise, for steady compressible flow, the continuity equation from Eq. (2.2.18) requires $\nabla \cdot (\rho\mathbf{V}) = 0$; and choosing $\chi=\rho$ enforces the compressible continuity equation for any stream function. Thus, **for steady two-dimensional incompressible flow in Cartesian coordinates**, we use $\chi=1$ and the velocity vector is related to the stream function by

$$\mathbf{V} = V_x\mathbf{i}_x + V_y\mathbf{i}_y + 0\mathbf{i}_z = \frac{\partial\psi}{\partial y}\mathbf{i}_x - \frac{\partial\psi}{\partial x}\mathbf{i}_y + 0\mathbf{i}_z \quad (2.9.9)$$

For steady two-dimensional compressible flow in Cartesian coordinates, we use $\chi=\rho$ and the velocity vector is related to the stream function by

$$\rho \mathbf{V} = \rho V_x \mathbf{i}_x + \rho V_y \mathbf{i}_y + 0 \mathbf{i}_z = \frac{\partial \psi}{\partial y} \mathbf{i}_x - \frac{\partial \psi}{\partial x} \mathbf{i}_y + 0 \mathbf{i}_z \quad (2.9.10)$$

Similarly, consider a steady two-dimensional flow in cylindrical coordinates with a velocity vector that lies in the r - θ plane,

$$\mathbf{V}(r, \theta) = V_r(r, \theta) \mathbf{i}_r + V_\theta(r, \theta) \mathbf{i}_\theta + 0 \mathbf{i}_z \quad (2.9.11)$$

The streamlines for this two-dimensional flow as obtained from Eq. (1.14.3) are given by the algebraic equations

$$\psi(r, \theta) = C_1, \quad z = C_2 \quad (2.9.12)$$

where C_1 and C_2 are arbitrary constants of integration, which have different values for different streamlines. Because ψ and z remain constant along these streamlines, which are everywhere tangent to the velocity vector, from Eq. (1.4.27)

$$\begin{aligned} \chi(r, \theta) \mathbf{V}(r, \theta) &= \chi(r, \theta) V_r(r, \theta) \mathbf{i}_r + \chi(r, \theta) V_\theta(r, \theta) \mathbf{i}_\theta + 0 \mathbf{i}_z \\ &= \nabla \psi \times \nabla z \\ &= \begin{vmatrix} \mathbf{i}_r & \mathbf{i}_\theta & \mathbf{i}_z \\ \frac{\partial \psi}{\partial r} & \frac{1}{r} \frac{\partial \psi}{\partial \theta} & 0 \\ 0 & 0 & 1 \end{vmatrix} = \frac{1}{r} \frac{\partial \psi}{\partial \theta} \mathbf{i}_r - \frac{\partial \psi}{\partial r} \mathbf{i}_\theta + 0 \mathbf{i}_z \end{aligned} \quad (2.9.13)$$

From Eq. (1.4.28), taking the divergence of Eq. (2.9.13) yields

$$\begin{aligned} \nabla \cdot (\chi \mathbf{V}) &= \mathbf{V} \cdot (\nabla \chi) + \chi \nabla \cdot \mathbf{V} \\ &= \frac{1}{r} \frac{\partial}{\partial r} \left[r \left(\frac{1}{r} \frac{\partial \psi}{\partial \theta} \right) \right] + \frac{1}{r} \frac{\partial}{\partial \theta} \left(-\frac{\partial \psi}{\partial r} \right) \\ &= \frac{1}{r} \frac{\partial^2 \psi}{\partial r \partial \theta} - \frac{1}{r} \frac{\partial^2 \psi}{\partial \theta \partial r} = 0 \end{aligned} \quad (2.9.14)$$

As was the case for Eq. (2.9.8), this satisfies the incompressible continuity equation for $\chi=1$ and it satisfies the steady compressible continuity equation for $\chi=\rho$. Thus, **for steady two-dimensional incompressible flow in polar cylindrical coordinates**, we use $\chi=1$ and the velocity vector is related to the stream function by

$$\mathbf{V} = V_r \mathbf{i}_r + V_\theta \mathbf{i}_\theta + 0 \mathbf{i}_z = \frac{1}{r} \frac{\partial \psi}{\partial \theta} \mathbf{i}_r - \frac{\partial \psi}{\partial r} \mathbf{i}_\theta + 0 \mathbf{i}_z \quad (2.9.15)$$

and **for steady two-dimensional compressible flow in polar cylindrical coordinates**, we use $\chi=\rho$ and the velocity vector is related to the stream function by

$$\rho \mathbf{V} = \rho V_r \mathbf{i}_r + \rho V_\theta \mathbf{i}_\theta + 0 \mathbf{i}_z = \frac{1}{r} \frac{\partial \psi}{\partial \theta} \mathbf{i}_r - \frac{\partial \psi}{\partial r} \mathbf{i}_\theta + 0 \mathbf{i}_z \quad (2.9.16)$$

From a similar development for steady axisymmetric flow in cylindrical coordinates the velocity vector lies in the r - z plane, and we obtain

$$\mathbf{V}(r, z) = V_r(r, z)\mathbf{i}_r + 0\mathbf{i}_\theta + V_z(r, z)\mathbf{i}_z \quad (2.9.17)$$

$$\psi(r, z) = C_1, \quad \theta = C_2 \quad (2.9.18)$$

Because ψ and θ remain constant along these streamlines, from Eq. (1.4.27)

$$\begin{aligned} \chi(r, z)\mathbf{V}(r, z) &= \chi(r, z)V_r(r, z)\mathbf{i}_r + 0\mathbf{i}_\theta + \chi(r, z)V_z(r, z)\mathbf{i}_z \\ &= \nabla\psi \times \nabla\theta \\ &= \begin{vmatrix} \mathbf{i}_r & \mathbf{i}_\theta & \mathbf{i}_z \\ \frac{\partial\psi}{\partial r} & 0 & \frac{\partial\psi}{\partial z} \\ 0 & 1/r & 0 \end{vmatrix} = -\frac{1}{r}\frac{\partial\psi}{\partial z}\mathbf{i}_r + 0\mathbf{i}_\theta + \frac{1}{r}\frac{\partial\psi}{\partial r}\mathbf{i}_z \end{aligned} \quad (2.9.19)$$

Once again from Eq. (1.4.28), it can be shown that taking the divergence of Eq. (2.9.19) yields $\nabla \cdot (\chi\mathbf{V}) = 0$, which satisfies the continuity equation with $\chi=1$ for incompressible flow and with $\chi=\rho$ for compressible flow.

From a similar development in spherical coordinates, steady two-dimensional flow in the r - θ plane yields

$$\mathbf{V}(r, \theta) = V_r(r, \theta)\mathbf{i}_r + V_\theta(r, \theta)\mathbf{i}_\theta + 0\mathbf{i}_\phi \quad (2.9.20)$$

$$\psi(r, \theta) = C_1, \quad \phi = C_2 \quad (2.9.21)$$

Because ψ and ϕ remain constant along these streamlines, from Eq. (1.4.33)

$$\begin{aligned} \chi(r, \theta)\mathbf{V}(r, \theta) &= \chi(r, \theta)V_r(r, \theta)\mathbf{i}_r + \chi(r, \theta)V_\theta(r, \theta)\mathbf{i}_\theta + 0\mathbf{i}_\phi \\ &= \nabla\psi \times \nabla\phi = \begin{vmatrix} \mathbf{i}_r & \mathbf{i}_\theta & \mathbf{i}_\phi \\ \frac{\partial\psi}{\partial r} & \frac{1}{r}\frac{\partial\psi}{\partial\theta} & 0 \\ 0 & 0 & 1/(r\sin\theta) \end{vmatrix} \\ &= \frac{1}{r^2\sin\theta}\frac{\partial\psi}{\partial\theta}\mathbf{i}_r - \frac{1}{r\sin\theta}\frac{\partial\psi}{\partial r}\mathbf{i}_\theta + 0\mathbf{i}_\phi \end{aligned} \quad (2.9.22)$$

From Eq. (1.4.34), it can be easily shown that taking the divergence of Eq. (2.9.22) also yields $\nabla \cdot (\chi\mathbf{V}) = 0$. Once again, this satisfies the incompressible continuity equation for $\chi=1$ and it satisfies the compressible continuity equation for $\chi=\rho$.

EXAMPLE 2.9.1. For a particular steady two-dimensional incompressible fluid flow field, the velocity vector in spherical coordinates lies in the r - θ plane and the stream function is given by

$$\psi(r, \theta) = \frac{V_\infty}{2} \left(r^2 - \frac{R^3}{r} \right) \sin^2\theta \quad (2.9.23)$$

where V_∞ and R are constants. Find the velocity vector at an arbitrary point.

Solution. The velocity vector in spherical coordinates for two-dimensional flow in the r - θ plane is related to the stream function according to Eq. (2.9.22). In any two-dimensional incompressible flow field, $\chi=1$ and the velocity vector is

$$\begin{aligned}\mathbf{V}(r, \theta) &= \frac{1}{r^2 \sin \theta} \frac{\partial \psi}{\partial \theta} \mathbf{i}_r - \frac{1}{r \sin \theta} \frac{\partial \psi}{\partial r} \mathbf{i}_\theta + 0 \mathbf{i}_\phi \\ &= \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left[\frac{V_\infty}{2} \left(r^2 - \frac{R^3}{r} \right) \sin^2 \theta \right] \mathbf{i}_r \\ &\quad - \frac{1}{r \sin \theta} \frac{\partial}{\partial r} \left[\frac{V_\infty}{2} \left(r^2 - \frac{R^3}{r} \right) \sin^2 \theta \right] \mathbf{i}_\theta + 0 \mathbf{i}_\phi \\ &= \underline{V_\infty \left(1 - \frac{R^3}{r^3} \right) \cos \theta \mathbf{i}_r - V_\infty \left(1 + \frac{R^3}{2r^3} \right) \sin \theta \mathbf{i}_\theta + 0 \mathbf{i}_\phi}\end{aligned}$$

The stream function is an important tool in fluid mechanics and aerodynamics that is sometimes used as a change of variables to simplify the equations of motion for inviscid and viscous flows. For two-dimensional flows, this change of variables can simplify the equations of motion by replacing the unknown vector $\mathbf{V}$ with the unknown scalar ψ , while simultaneously satisfying the continuity equation as a mathematical identity.

The physical significance of the stream function is often stated by saying that curves of constant ψ are streamlines of the flow. However, if we wish to be mathematically precise, setting ψ equal to a constant yields the equation of a surface, not a curve. For example, the stream function given in Eq. (2.9.23) is for steady inviscid incompressible flow about a sphere. In this case, the equation $\psi(r, \theta) = C_1$ defines a surface of revolution as shown in Fig. 2.9.1. Because ψ is equal to C_1 at all points on this surface, any curve on the surface satisfies the equation $\psi(r, \theta) = C_1$. To ensure that the curve is a streamline of the flow, we must enforce both equations in Eq. (2.9.21). The equation $\phi = C_2$ also defines a geometric surface, as shown in Fig. 2.9.1. Strictly speaking, a streamline of this flow is defined as the intersection of the two *stream surfaces*, $\psi(r, \theta) = C_1$ and $\phi = C_2$.

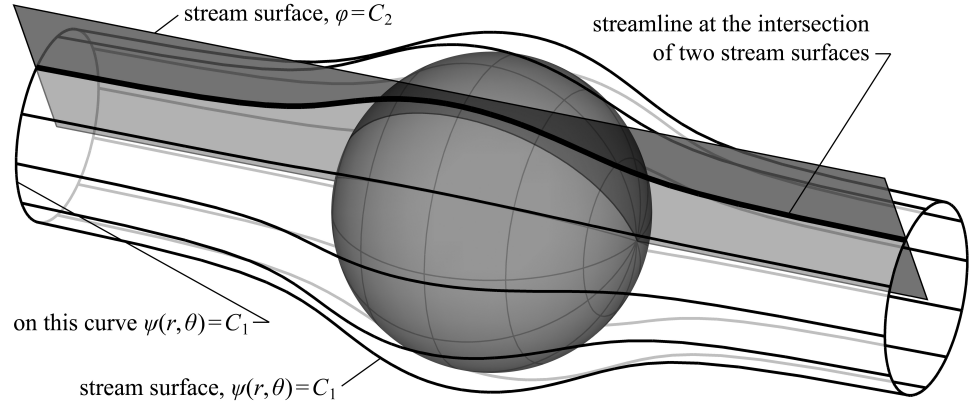


Figure 2.9.1. Streamline at the intersection of two stream surfaces in flow about a sphere.

In general, streamlines are solutions to a system of two differential equations, as presented in Sec. 1.14. These differential equations expressed in Cartesian, cylindrical, and spherical coordinates are given by Eqs. (1.14.2)–(1.14.4), which are repeated here

$$\frac{dy}{dx} = \frac{V_y}{V_x}, \quad \frac{dz}{dx} = \frac{V_z}{V_x} \quad (1.14.2)$$

$$\frac{1}{r} \frac{dr}{d\theta} = \frac{V_r}{V_\theta}, \quad \frac{1}{r} \frac{dz}{d\theta} = \frac{V_z}{V_\theta} \quad (1.14.3)$$

$$\frac{1}{r} \frac{dr}{d\theta} = \frac{V_r}{V_\theta}, \quad \sin\theta \frac{d\phi}{d\theta} = \frac{V_\phi}{V_\theta} \quad (1.14.4)$$

Because these are first-order differential equations, the general solution to each contains one arbitrary constant of integration. Hence, *the solutions to the differential equations for the streamlines of any fluid flow field can be expressed in terms of two independent functions*. For example, in spherical coordinates the solutions are of the form

$$\psi_1(r, \theta, \phi) = C_1, \quad \psi_2(r, \theta, \phi) = C_2 \quad (2.9.24)$$

where C_1 and C_2 are arbitrary constants of integration. The functions ψ_1 and ψ_2 are both stream functions, and the two algebraic equations in Eq. (2.9.24) are the equations for two nonparallel stream surfaces. The streamline that corresponds to any pair of the constants C_1 and C_2 is the intersection of the two associated stream surfaces, as shown in Fig. 2.9.2. For the special case of steady inviscid incompressible flow about a sphere, the two independent stream functions are

$$\psi_1 = \frac{V_\infty}{2} \left(r^2 - \frac{R^3}{r} \right) \sin^2\theta, \quad \psi_2 = \phi \quad (2.9.25)$$

For this fluid flow field, the streamline and associated stream surfaces corresponding to one particular pair of the constants C_1 and C_2 are shown in Fig. 2.9.1.

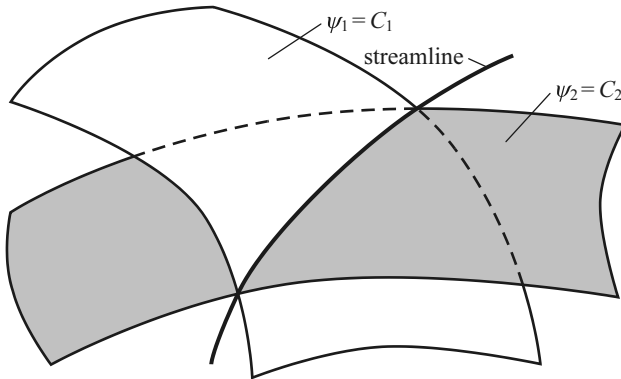


Figure 2.9.2. Streamline at the intersection of two stream surfaces in an arbitrary flow field.

At any point in a flow field, the local fluid velocity vector $\mathbf{V}$ can be related to the three-dimensional stream functions ψ_1 and ψ_2 . Because any streamline lies in the plane of two stream surfaces that are defined by the equations $\psi_1=C_1$ and $\psi_2=C_2$, the components of $\nabla\psi_1$ and $\nabla\psi_2$ in the direction of $\mathbf{V}$ are both zero, i.e.,

$$\nabla\psi_1 \cdot \mathbf{V} = 0, \quad \nabla\psi_2 \cdot \mathbf{V} = 0 \quad (2.9.26)$$

This requires that the fluid velocity vector $\mathbf{V}$ is everywhere normal to the plane formed by the two vectors $\nabla\psi_1$ and $\nabla\psi_2$. In other words, $\mathbf{V}$ is everywhere parallel with the cross product of $\nabla\psi_1$ and $\nabla\psi_2$,

$$\chi \mathbf{V} = \nabla\psi_1 \times \nabla\psi_2 \quad (2.9.27)$$

where χ is a scalar function of position, which may be chosen as needed to help satisfy the equations of motion. In **Cartesian coordinates** Eq. (2.9.27) yields

$$\chi \mathbf{V} = \begin{vmatrix} \mathbf{i}_x & \mathbf{i}_y & \mathbf{i}_z \\ \frac{\partial\psi_1}{\partial x} & \frac{\partial\psi_1}{\partial y} & \frac{\partial\psi_1}{\partial z} \\ \frac{\partial\psi_2}{\partial x} & \frac{\partial\psi_2}{\partial y} & \frac{\partial\psi_2}{\partial z} \end{vmatrix} = \begin{Bmatrix} \left(\frac{\partial\psi_1}{\partial y} \frac{\partial\psi_2}{\partial z} - \frac{\partial\psi_1}{\partial z} \frac{\partial\psi_2}{\partial y} \right) \mathbf{i}_x \\ \left(\frac{\partial\psi_1}{\partial z} \frac{\partial\psi_2}{\partial x} - \frac{\partial\psi_1}{\partial x} \frac{\partial\psi_2}{\partial z} \right) \mathbf{i}_y \\ \left(\frac{\partial\psi_1}{\partial x} \frac{\partial\psi_2}{\partial y} - \frac{\partial\psi_1}{\partial y} \frac{\partial\psi_2}{\partial x} \right) \mathbf{i}_z \end{Bmatrix} \quad (2.9.28)$$

in **Cylindrical Coordinates** we obtain

$$\chi \mathbf{V} = \begin{vmatrix} \mathbf{i}_r & \mathbf{i}_\theta & \mathbf{i}_z \\ \frac{\partial\psi_1}{\partial r} & \frac{1}{r} \frac{\partial\psi_1}{\partial \theta} & \frac{\partial\psi_1}{\partial z} \\ \frac{\partial\psi_2}{\partial r} & \frac{1}{r} \frac{\partial\psi_2}{\partial \theta} & \frac{\partial\psi_2}{\partial z} \end{vmatrix} = \begin{Bmatrix} \frac{1}{r} \left(\frac{\partial\psi_1}{\partial \theta} \frac{\partial\psi_2}{\partial z} - \frac{\partial\psi_1}{\partial z} \frac{\partial\psi_2}{\partial \theta} \right) \mathbf{i}_r \\ \left(\frac{\partial\psi_1}{\partial z} \frac{\partial\psi_2}{\partial r} - \frac{\partial\psi_1}{\partial r} \frac{\partial\psi_2}{\partial z} \right) \mathbf{i}_\theta \\ \frac{1}{r} \left(\frac{\partial\psi_1}{\partial r} \frac{\partial\psi_2}{\partial \theta} - \frac{\partial\psi_1}{\partial \theta} \frac{\partial\psi_2}{\partial r} \right) \mathbf{i}_z \end{Bmatrix} \quad (2.9.29)$$

and in **Spherical Coordinates** the expansion is

$$\chi \mathbf{V} = \begin{vmatrix} \mathbf{i}_r & \mathbf{i}_\theta & \mathbf{i}_\phi \\ \frac{\partial\psi_1}{\partial r} & \frac{1}{r} \frac{\partial\psi_1}{\partial \theta} & \frac{1}{r \sin \theta} \frac{\partial\psi_1}{\partial \phi} \\ \frac{\partial\psi_2}{\partial r} & \frac{1}{r} \frac{\partial\psi_2}{\partial \theta} & \frac{1}{r \sin \theta} \frac{\partial\psi_2}{\partial \phi} \end{vmatrix} = \begin{Bmatrix} \frac{1}{r^2 \sin \theta} \left(\frac{\partial\psi_1}{\partial \theta} \frac{\partial\psi_2}{\partial \phi} - \frac{\partial\psi_1}{\partial \phi} \frac{\partial\psi_2}{\partial \theta} \right) \mathbf{i}_r \\ \frac{1}{r \sin \theta} \left(\frac{\partial\psi_1}{\partial \phi} \frac{\partial\psi_2}{\partial r} - \frac{\partial\psi_1}{\partial r} \frac{\partial\psi_2}{\partial \phi} \right) \mathbf{i}_\theta \\ \frac{1}{r} \left(\frac{\partial\psi_1}{\partial r} \frac{\partial\psi_2}{\partial \theta} - \frac{\partial\psi_1}{\partial \theta} \frac{\partial\psi_2}{\partial r} \right) \mathbf{i}_\phi \end{Bmatrix} \quad (2.9.30)$$

Although stream functions are commonly used as a change of variables to simplify the solution of two-dimensional problems involving both viscous and inviscid fluid flows, solving three-dimensional fluid flow problems using two stream functions has not received significant attention. Nevertheless, the mathematical foundation exists for the development of such solutions.

2.10. Elementary Potential Flows

The potential flow equations are a great simplification over the equations of motion for viscous flow, and even over the inviscid Euler equations. However, solving Laplace's equation given by Eq. (2.6.2) subject to the boundary conditions given by Eqs. (2.6.6) and (2.6.7) is not trivial. Historically, potential flow theory was developed by finding functions that satisfy Laplace's equation and then determining what boundary conditions are satisfied by each function. Because Laplace's equation is linear, superposition of two functions that each satisfies Laplace's equation forms a third function that also satisfies Laplace's equation. In other words, if ϕ_1 and ϕ_2 both satisfy Laplace's equation, then the sum $\phi_3 = \phi_1 + \phi_2$ also satisfies Laplace's equation. In general, the velocity potential for a complex irrotational flow field can be synthesized by summing a number of velocity potentials for less complicated irrotational flows,

$$\phi(x, y, z) = \sum_{i=1}^n \phi_i(x, y, z) \quad (2.10.1)$$

The method of synthesizing a complex potential flow field through the superposition of several elementary potential flow fields is commonly called the *inverse method*. This method has proven to be very useful and a rich collection of elementary solutions to Laplace's equation has evolved. In this section we shall examine some of the most commonly used elementary potential flows.

Uniform Flow

The simplest potential flow is a uniform stream of fluid moving in a fixed direction at constant speed. The velocity potential for a uniform flow with constant but completely arbitrary velocity components can be expressed in Cartesian coordinates as

$$\phi(x, y, z) = V_x x + V_y y + V_z z \quad (2.10.2)$$

where V_x , V_y , and V_z are constants. It is easily shown that this velocity potential satisfies Laplace's equation. From Eq. (2.5.32), the velocity field for this potential flow is simply

$$\mathbf{V} = \nabla \phi = V_x \mathbf{i}_x + V_y \mathbf{i}_y + V_z \mathbf{i}_z \quad (2.10.3)$$

Because V_x , V_y , and V_z are constants for this potential flow, it is easily shown that this velocity field is irrotational.

An important special case of Eqs. (2.10.2) and (2.10.3) is that for a two-dimensional uniform flow in the x - y plane. For this case, Eqs. (2.10.2) and (2.10.3) can be written as

$$\phi(x, y) = V_\infty (x \cos \alpha + y \sin \alpha) \quad (2.10.4)$$

$$\mathbf{V} = \nabla \phi = V_\infty (\cos \alpha \mathbf{i}_x + \sin \alpha \mathbf{i}_y + 0 \mathbf{i}_z) \quad (2.10.5)$$

where V_∞ and α are constants; i.e., V_∞ is the constant speed of the uniform flow and α is the constant angle that the direction of the uniform flow makes with the x axis, as shown in Fig. 2.10.1. It is easily shown from Eq. (2.9.9) that the incompressible stream function for this two-dimensional uniform flow is

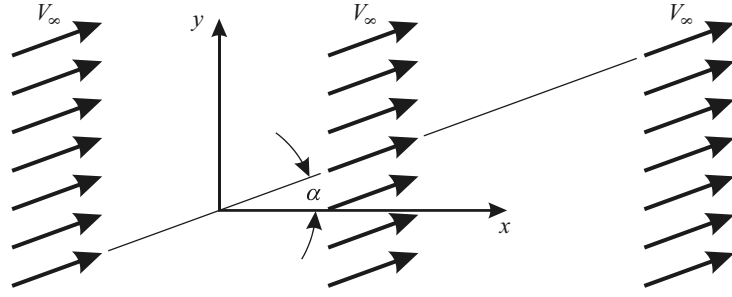


Figure 2.10.1. Uniform flow in the x - y plane at an angle α to the x axis.

$$\psi(x, y) = V_\infty(y \cos \alpha - x \sin \alpha) \quad (2.10.6)$$

As given by Eq. (2.9.3), the streamlines for this flow are obtained from $\psi(x, y) = C_1$ and $z = C_2$. Hence, the streamlines in the x - y plane are given by $V_\infty(y \cos \alpha - x \sin \alpha) = C_1$, which describes a family of straight lines all at an angle α to the x axis.

The transformation from Cartesian coordinates to cylindrical coordinates is given by the relations $x = r \cos \theta$, $y = r \sin \theta$, and $z = z$. Hence, after applying this transformation to Eqs. (2.10.4)–(2.10.6), two-dimensional uniform flow in the x - y plane can be described in cylindrical coordinates using the relations

$$\phi(r, \theta) = V_\infty r (\cos \theta \cos \alpha + \sin \theta \sin \alpha) \quad (2.10.7)$$

$$\begin{aligned} \mathbf{V} &= \nabla \phi \\ &= V_\infty [(\cos \theta \cos \alpha + \sin \theta \sin \alpha) \mathbf{i}_r - (\sin \theta \cos \alpha - \cos \theta \sin \alpha) \mathbf{i}_\theta + 0 \mathbf{i}_z] \end{aligned} \quad (2.10.8)$$

$$\psi(r, \theta) = V_\infty r (\sin \theta \cos \alpha - \cos \theta \sin \alpha) \quad (2.10.9)$$

or after applying the common trigonometric identities $\cos \theta \cos \alpha + \sin \theta \sin \alpha = \cos(\theta - \alpha)$ and $\sin \theta \cos \alpha - \cos \theta \sin \alpha = \sin(\theta - \alpha)$, **a two-dimensional uniform flow expressed in cylindrical coordinates yields**

$$\phi(r, \theta) = V_\infty r \cos(\theta - \alpha) \quad (2.10.10)$$

$$\mathbf{V} = \nabla \phi = V_\infty [\cos(\theta - \alpha) \mathbf{i}_r - \sin(\theta - \alpha) \mathbf{i}_\theta + 0 \mathbf{i}_z] \quad (2.10.11)$$

$$\psi(r, \theta) = V_\infty r \sin(\theta - \alpha) \quad (2.10.12)$$

Incompressible Line Source

Consider the possibility of an incompressible potential flow in cylindrical coordinates having a velocity potential that depends on only the radial coordinate r . If such a flow exists, the velocity potential $\phi(r)$ would satisfy Laplace's equation from Eq. (2.6.4),

$$\frac{d}{dr} \left(r \frac{d\phi}{dr} \right) = 0 \quad (2.10.13)$$

Equation (2.10.13) is easily integrated twice to yield

$$r \frac{d\phi}{dr} = A \quad \text{or} \quad \frac{d\phi}{dr} = \frac{A}{r}$$

$$\phi = A \ln r + B \quad (2.10.14)$$

where A and B are constants of integration. From Eq. (2.5.33), the velocity field for this potential flow is

$$\mathbf{V} = \nabla \phi = \frac{A}{r} \mathbf{i}_r + 0 \mathbf{i}_\theta + 0 \mathbf{i}_z \quad (2.10.15)$$

The incompressible stream function for this flow is obtained by integrating Eq. (2.9.15)

$$\frac{1}{r} \frac{d\psi}{d\theta} = \frac{A}{r} \quad \text{or} \quad \frac{d\psi}{d\theta} = A$$

$$\psi = A\theta + C \quad (2.10.16)$$

where C is another constant of integration. Using Eq. (2.10.16) in Eq. (2.9.12), we see that both θ and z remain constant along the streamlines of this flow field. That is, the streamlines are straight radial lines emanating from a *line source* that is coincident with the z axis, and the fluid velocity vector is everywhere in the radial direction normal to the line source, as shown in Fig. 2.10.2. Furthermore, we see from Eq. (2.10.15) that the velocity along each streamline varies inversely with the distance from the line source.

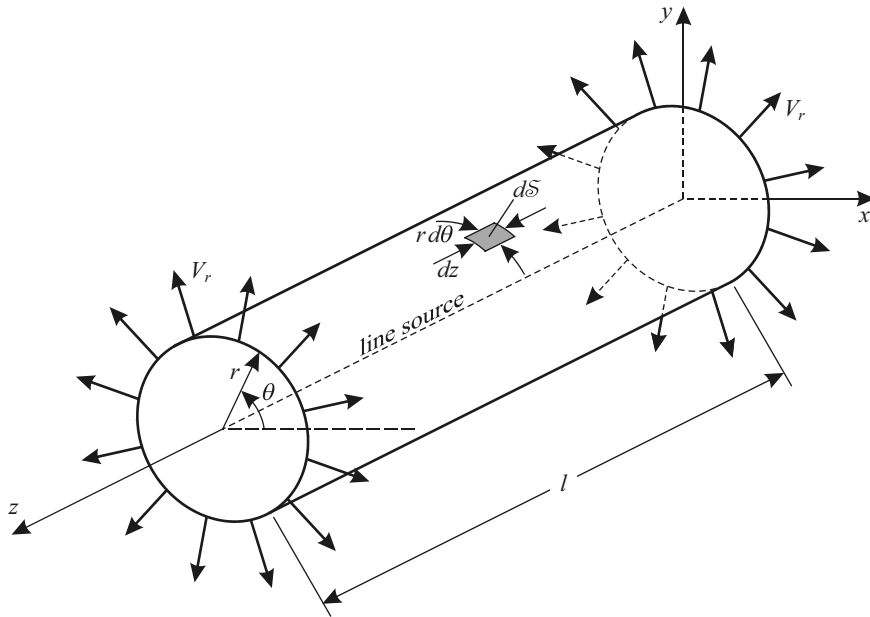


Figure 2.10.2. Flow from a segment of a line source aligned with the z axis.

The constant of integration A in Eqs. (2.10.14)–(2.10.16) is traditionally written in terms of the volume flow rate per unit length emanating from the line source, which is commonly denoted as A and referred to as the *source strength*. To see how A is related to A , consider the cylindrical control volume enclosing a length l of the line source as shown in Fig. 2.10.2. The mass flow rate exiting this control volume can be obtained from Eq. (2.2.3). Because this is a fixed control volume, $\mathbf{V}_s=0$; and because the velocity vector is in the radial direction everywhere, $\mathbf{V} \cdot \mathbf{n}=0$ on the two circular end surfaces of this cylindrical control volume. On the outer circumferential surface of the cylindrical control volume, $\mathbf{n}=\mathbf{i}_r$, and $d\mathcal{S}=r d\theta dz$. Hence, from Eq. (2.2.3), the total mass flow rate emanating from a length l of the line source is

$$\dot{m}_e = \iint_{\mathcal{S}} \rho \mathbf{V} \cdot \mathbf{n} d\mathcal{S} = \int_{z=0}^l \int_{\theta=0}^{2\pi} \rho \mathbf{V} \cdot \mathbf{i}_r r d\theta dz \quad (2.10.17)$$

Because this is an incompressible flow, using Eq. (2.10.15) in Eq. (2.10.17) yields

$$\dot{m}_e = \int_{z=0}^l \int_{\theta=0}^{2\pi} \rho \frac{A}{r} r d\theta dz = A \rho \int_{\theta=0}^{2\pi} d\theta \int_{z=0}^l dz = 2\pi A \rho l \quad (2.10.18)$$

The volume flow rate is the mass flow rate divided by the density. Hence, **the volume flow rate per unit length emanating from a line source is**

$$A \equiv \frac{\dot{m}_e}{\rho l} = 2\pi A \quad (2.10.19)$$

The remaining two constants of integration B and C in Eqs. (2.10.14) and (2.10.16) are arbitrary, since they do not affect the velocity field. Choosing B and C as zero and using Eq. (2.10.19) to eliminate A from Eqs. (2.10.14)–(2.10.16), **a two-dimensional line source in cylindrical coordinates is defined by**

$$\phi(r, \theta) = \frac{A}{2\pi} \ln r \quad (2.10.20)$$

$$\mathbf{V} = \nabla \phi = \frac{A}{2\pi r} \mathbf{i}_r + 0 \mathbf{i}_\theta + 0 \mathbf{i}_z \quad (2.10.21)$$

$$\psi(r, \theta) = \frac{A}{2\pi} \theta \quad (2.10.22)$$

where the source strength A is the constant volume flow rate per unit length emanating from the line source. This source strength may be either positive or negative. When A is negative, the resulting potential flow is referred to as a *line sink*.

It should be noted that, although this velocity potential satisfies Laplace's equation for any finite value of r , no matter how small, the solution is singular along the z axis at $r=0$. In spite of this singularity, the two-dimensional line source plays an important role in synthesizing potential flow about objects of complex shape. As we shall see, one method for handling this singularity is to place the line source inside the object, which is outside the region of physical interest. Typically, line sources are used in combination with line sinks, such that the net generation of fluid within the flow field is zero.

Incompressible Point Source

Using Laplace's equation for potential flow in spherical coordinates given by Eq. (2.6.5), and following a procedure similar to that used to obtain Eqs. (2.10.20)–(2.10.22) for the two-dimensional line source in cylindrical coordinates, we obtain another important incompressible potential flow. **The point source is defined in axisymmetric spherical coordinates by the relations**

$$\phi(r, \theta) = -\frac{Q}{4\pi r} \quad (2.10.23)$$

$$\mathbf{V} = \nabla \phi = \frac{Q}{4\pi r^2} \mathbf{i}_r + 0 \mathbf{i}_\theta + 0 \mathbf{i}_\phi \quad (2.10.24)$$

$$\psi(r, \theta) = -\frac{Q}{4\pi} \cos \theta \quad (2.10.25)$$

where Q is the *source strength* defined to be the constant volume flow rate emanating from the point source. As was the case for a line source, the strength of a point source may be either positive or negative, and when Q is negative, the resulting potential flow is called a *point sink*.

Using Eq. (2.10.25) in Eq. (2.9.21), we see that both θ and ϕ remain constant along the streamlines of this flow field. In other words, the streamlines are straight radial lines emanating outward from a single point at the origin $r=0$. For positive source strength, the fluid velocity vector at any point in the flow field is directly away from the point source, as shown in Fig. 2.10.3. The magnitude of the velocity vector at any point in the flow field varies inversely with the square of the distance from the point source. Like the line source, the point source is singular at the origin $r=0$. The point source is commonly used to synthesize potential flow about bodies of revolution.

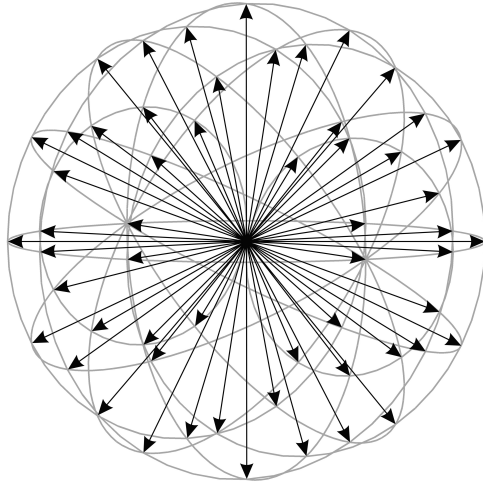


Figure 2.10.3. Flow emanating outward in all directions from a point source.

Incompressible Line Vortex

Consider the possibility of an incompressible potential flow in cylindrical coordinates having a velocity potential that depends on only the angular coordinate θ . If this flow exists, the velocity potential $\phi(\theta)$ must satisfy Laplace's equation from Eq. (2.6.4),

$$\frac{1}{r^2} \frac{d^2 \phi}{d\theta^2} = 0 \quad (2.10.26)$$

which is easily integrated twice to yield

$$\phi = A\theta + B \quad (2.10.27)$$

where A and B are constants of integration. From Eq. (2.5.33), the velocity is

$$\mathbf{V} = \nabla \phi = 0\mathbf{i}_r + \frac{A}{r}\mathbf{i}_\theta + 0\mathbf{i}_z \quad (2.10.28)$$

The incompressible stream function for this flow is obtained by integrating Eq. (2.9.15)

$$\psi = -A \ln r + C \quad (2.10.29)$$

where C is another constant of integration. Using Eq. (2.10.29) in Eq. (2.9.12), we see that both r and z remain constant along the streamlines of this flow field. That is, the streamlines are circles centered about and normal to the *line vortex*, which is coincident with the z axis as seen in Fig. 2.10.4. Because r remains constant along the streamlines, we see from Eq. (2.10.28) that the velocity also remains constant along the streamlines and varies inversely with the normal distance from the streamline to the line vortex.

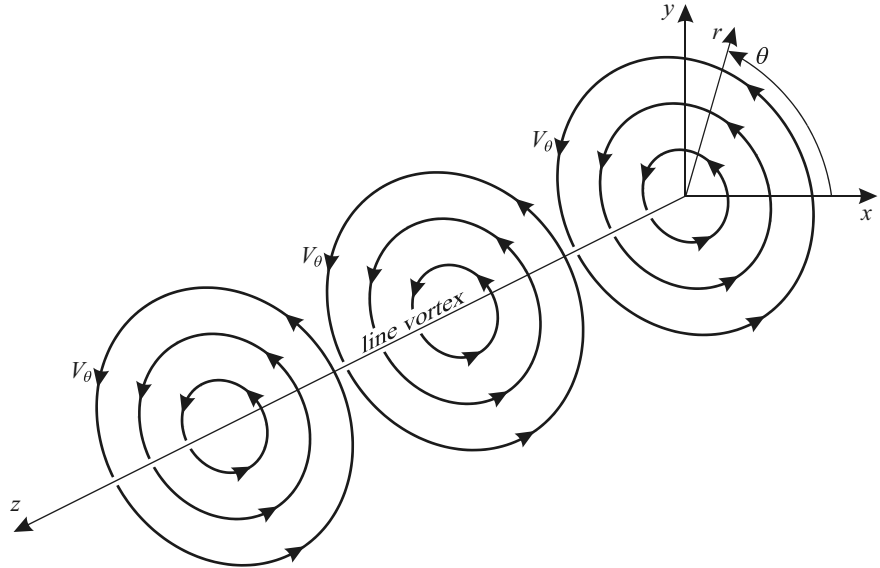


Figure 2.10.4. Streamlines for flow about a line vortex aligned with the z axis.

This potential line vortex satisfies Laplace's equation at any point in the flow field where r is finite. Hence, the flow is irrotational wherever r is finite, no matter how small r becomes. However, the solution is singular at all points along the z axis where $r=0$. It is easily shown that the circulation is zero about any closed path in this flow field that does not enclose the vortex. Conversely, **the circulation is not zero about any path that completely encloses a line vortex**. For example, consider the circulation about any of the closed streamlines shown in Fig. 2.10.4. Along any of these closed circular paths, $\mathbf{ds}=r d\theta \mathbf{i}_\theta$, and using Eq. (2.10.28) in Eq. (2.4.10) yields

$$\Gamma \equiv -\oint_C \mathbf{V} \cdot \mathbf{ds} = -\int_{\theta=0}^{2\pi} \mathbf{V} \cdot \mathbf{i}_\theta r d\theta = -\int_{\theta=0}^{2\pi} \frac{A}{r} r d\theta = -2\pi A \quad (2.10.30)$$

Notice that this circulation is completely independent of the r and z coordinates along the path of integration. The circulation associated with any potential line vortex is a constant property of the vortex. Hence, the constant of integration A in Eqs. (2.10.27)–(2.10.29) is traditionally written in terms of this circulation. The remaining constants B and C in Eqs. (2.10.27) and (2.10.29) are arbitrary, because they do not affect the velocity field. Thus, choosing B and C to be zero and using Eq. (2.10.30) to eliminate the constant A from Eqs. (2.10.27)–(2.10.29), **the two-dimensional potential line vortex is defined in cylindrical coordinates by**

$$\phi(r, \theta) = -\frac{\Gamma}{2\pi} \theta \quad (2.10.31)$$

$$\mathbf{V} = \nabla \phi = 0 \mathbf{i}_r - \frac{\Gamma}{2\pi r} \mathbf{i}_\theta + 0 \mathbf{i}_z \quad (2.10.32)$$

$$\psi(r, \theta) = \frac{\Gamma}{2\pi} \ln r \quad (2.10.33)$$

where Γ is the constant circulation of the vortex, which is also commonly called the *vortex strength*. This vortex strength may be either positive or negative. Notice from Eq. (2.10.32) that V_θ is negative when Γ is positive; i.e., a vortex of positive strength rotates clockwise. This is simply a result of the sign convention used in the definition of circulation, which is given in Eq. (2.4.10).

Incompressible Flow at an Angle to a Wall

As previously mentioned, potential flow theory was developed historically by finding functions that satisfy Laplace's equation and then determining what boundary conditions are satisfied by each function. As an example of another useful potential flow obtained from this inverse method, consider the incompressible potential flow with the following two-dimensional velocity potential in cylindrical coordinates:

$$\phi(r, \theta) = Ar^a \cos(a\theta) \quad (2.10.34)$$

where A and a are constants. It is easily shown that this velocity potential satisfies Laplace's equation in cylindrical coordinates, which is given by (2.6.4). Taking the Laplacian of this velocity potential yields

$$\begin{aligned}
\nabla^2 \phi &= \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial \phi}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2 \phi}{\partial \theta^2} + \frac{\partial^2 \phi}{\partial z^2} \\
&= \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial}{\partial r} [Ar^a \cos(a\theta)] \right) + \frac{1}{r^2} \frac{\partial}{\partial \theta} \left(\frac{\partial}{\partial \theta} [Ar^a \cos(a\theta)] \right) + 0 \\
&= \frac{1}{r} \frac{\partial}{\partial r} \{r[Aar^{a-1} \cos(a\theta)]\} + \frac{1}{r^2} \frac{\partial}{\partial \theta} [-Aar^a \sin(a\theta)] \\
&= \frac{1}{r} \frac{\partial}{\partial r} [Aar^a \cos(a\theta)] - \frac{1}{r^2} \frac{\partial}{\partial \theta} [Aar^a \sin(a\theta)] \\
&= \frac{1}{r} [Aa^2 r^{a-1} \cos(a\theta)] - \frac{1}{r^2} [Aa^2 r^a \cos(a\theta)] = 0
\end{aligned}$$

Hence, we see that $\phi(r, \theta) = Ar^a \cos(a\theta)$ is a velocity potential corresponding to a valid incompressible potential flow. From Eq. (2.5.33), the velocity for this flow field is

$$\mathbf{V} = \nabla \phi = Aar^{a-1} \cos(a\theta) \mathbf{i}_r - Aar^{a-1} \sin(a\theta) \mathbf{i}_\theta + 0 \mathbf{i}_z \quad (2.10.35)$$

and from Eq. (2.9.15) it can be shown that the stream function for this flow field is

$$\psi(r, \theta) = Ar^a \sin(a\theta) \quad (2.10.36)$$

Using Eq. (2.10.36) in Eq. (2.9.12), any streamline in this two-dimensional flow field is the intersection of two stream surfaces, which are defined by the algebraic equations

$$Ar^a \sin(a\theta) = C_1, \quad z = C_2 \quad (2.10.37)$$

where C_1 and C_2 are constants, which have different values for different streamlines. For given values of A , a , C_1 , and C_2 , the streamline can be plotted from Eq. (2.10.37). Notice that, for $C_1=0$, the stream surfaces given by $Ar^a \sin(a\theta)=0$ in Eq. (2.10.37) are planar surfaces defined by $\theta=n\pi/a$, where n is any integer. For instance, when $a=2$, the planar stream surfaces are located at $\theta=0$, $\theta=\pi/2$, $\theta=\pi$, and $\theta=3\pi/2$. Because any stream surface can be regarded as a rigid boundary of the flow field, the flow field between $\theta=0$ and $\theta=\pi/a$ is equivalent to flow around a corner of angle π/a . For example, Figs. 2.10.5 and 2.10.6 show streamlines for flow around corners of angles 60° , 120° , and 200° .

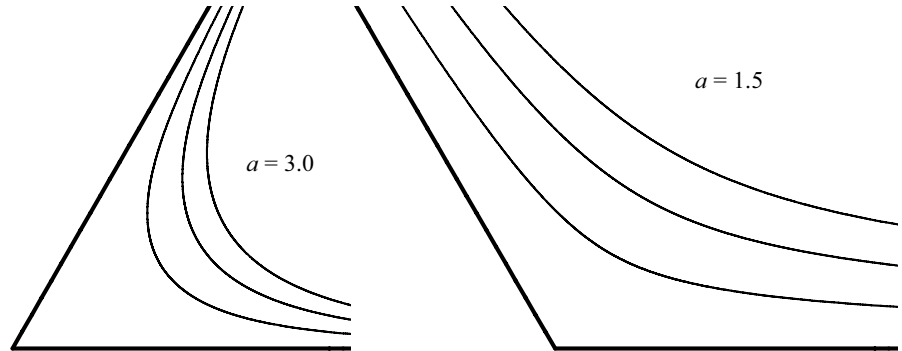


Figure 2.10.5. Streamlines for flow around 60° and 120° corners, $Ar^a \sin(a\theta) = C_1$ and $z = C_2$.

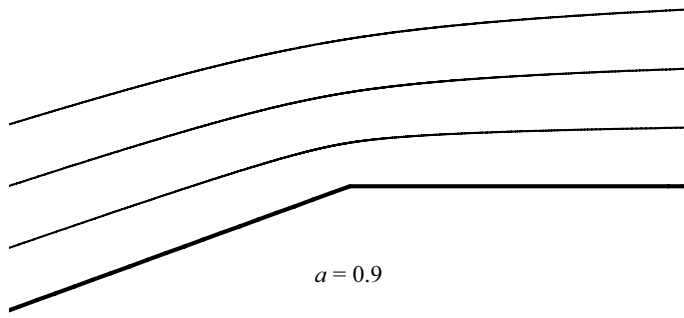


Figure 2.10.6. Streamlines for flow around a 200° corner, $Ar^a \sin(a\theta) = C_1$ and $z = C_2$.

Note from Eq. (2.10.35) that for $a > 1$ and $r = 0$, the velocity is zero; whereas for $a < 1$ and $r = 0$, the velocity is infinite. In other words, **a corner is a stagnation point for wall angles less than 180° ; whereas a corner is a point of infinite velocity for wall angles greater than 180° .** Hence, a real flow will separate for wall angles greater than 180° .

When $a = 2$, the flow in any quadrant of the r - θ plane can be regarded as flow around a 90-degree corner, as shown in Fig. 2.10.7. Furthermore, by including the flow in any two adjacent quadrants, the flow field for $a = 2$ can be regarded as flow near a stagnation point on a flat wall or a bluff body. This is commonly called *stagnation flow*.

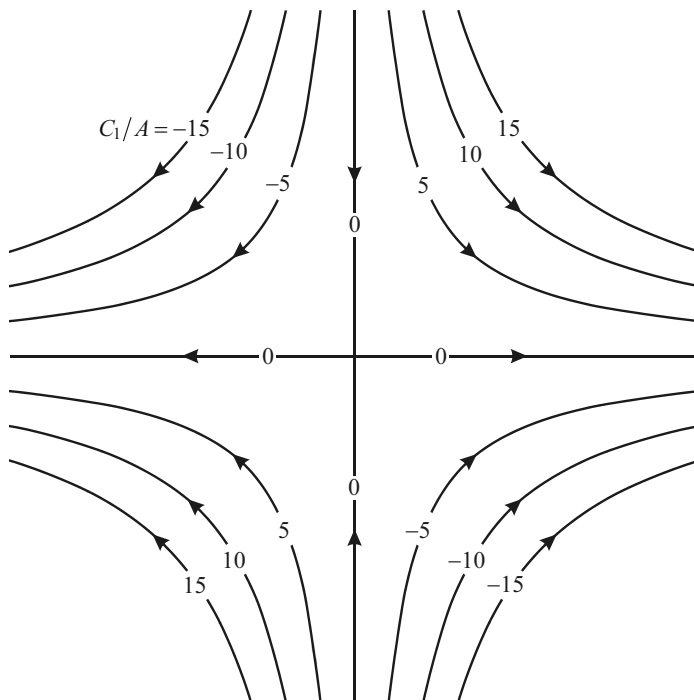


Figure 2.10.7. Streamlines in the r - θ plane for stagnation flow, $Ar^2 \sin(2\theta) = C_1$ and $z = C_2$.

2.11. Superposition of Potential Flows

Because Laplace's equation is linear, superposition of any two velocity potentials that satisfy Laplace's equation forms a third velocity potential, which also satisfies Laplace's equation. In general, the velocity potential for a complex irrotational flow field can be synthesized by summing a number of velocity potentials for less complicated irrotational flows. In this section we shall examine some simple examples of such synthesis.

Consider uniform flow parallel with the x axis superimposed on an incompressible line source that is aligned with the z axis. From Eqs. (2.10.10)–(2.10.12) combined with Eqs. (2.10.20)–(2.10.22), the synthesized flow field at any value of z yields

$$\phi(r, \theta) = V_\infty r \cos \theta + \Lambda \ln r / (2\pi) \quad (2.11.1)$$

$$\mathbf{V}(r, \theta) = \nabla \phi = [V_\infty \cos \theta + \Lambda / (2\pi r)] \mathbf{i}_r - V_\infty \sin \theta \mathbf{i}_\theta + 0 \mathbf{i}_z \quad (2.11.2)$$

$$\psi(r, \theta) = V_\infty r \sin \theta + \Lambda \theta / (2\pi) \quad (2.11.3)$$

Because the velocity potentials in Eqs. (2.10.10) and (2.10.20) satisfy Laplace's equation, we know that the velocity potential in Eq. (2.11.1) satisfies Laplace's equation as well. Therefore, Eqs. (2.11.1)–(2.11.3) describe a valid incompressible irrotational flow field.

Note that for any value of z and $\theta = \pi$, the fluid velocity vector in this flow field is

$$\mathbf{V}(r, \pi) = [\Lambda / (2\pi r) - V_\infty] \mathbf{i}_r + 0 \mathbf{i}_\theta + 0 \mathbf{i}_z \quad (2.11.4)$$

We see from Eq. (2.11.4) that, at each value of z , there is a point in this flow field where all three components of the velocity vector are zero

$$\mathbf{V}(\Lambda / 2\pi V_\infty, \pi) = 0 \mathbf{i}_r + 0 \mathbf{i}_\theta + 0 \mathbf{i}_z \quad (2.11.5)$$

That is, at every point along a line parallel with the z axis where $r = \Lambda / (2\pi V_\infty)$ and $\theta = \pi$, the fluid velocity vector is identically zero. In other words, every point on the line at the intersection of the two surfaces defined by the two equations $r = \Lambda / (2\pi V_\infty)$ and $\theta = \pi$ is a stagnation point of the flow. Streamlines for this flow field are shown in Fig. 2.11.1.

Because we are considering inviscid flow, the velocity at the surface of any solid body is always tangent to the surface. Hence, we could replace any stream surface in an inviscid flow field with a solid surface of the same shape. Notice from Fig. 2.11.1, that

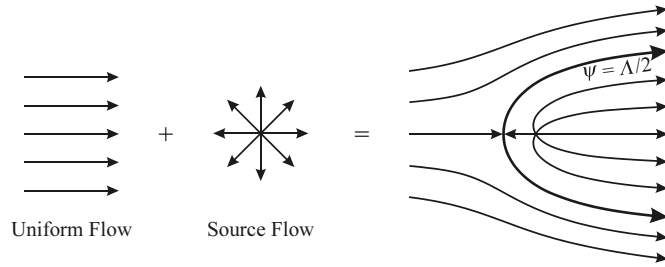


Figure 2.11.1. Streamlines for superposition of uniform flow and a line source, which are defined by the intersection of two stream surfaces, i.e., $\psi(r, \theta) = V_\infty r \sin \theta + \Lambda \theta / (2\pi) = C_1$ and $z = C_2$.

the stream surface defined by the equation $\psi(r, \theta) = V_\infty r \sin \theta + \Lambda \theta / (2\pi) = \Lambda / 2$ separates the fluid originating in the uniform flow from the fluid emitted by the line source. In other words, all of the fluid outside this stream surface comes from the freestream; and all of the fluid inside this stream surface comes from the line source. Hence, the flow field inside the stream surface defined by the equation $\psi(r, \theta) = \Lambda / 2$ could be replaced with a semi-infinite solid body of the same shape, and the flow field outside this stream surface would not be altered. This leads to the important conclusion that potential flow over a semi-infinite solid body of the shape shown in Fig. 2.11.1 can be synthesized by superimposing a uniform flow parallel with the x axis on an incompressible line source that is aligned with the z axis. Of course, the geometry of the resulting solid body is not of much practical interest, but this example demonstrates the process of flow synthesis.

Flow over a Rankine Cylinder

Consider superposition of three elementary potential flows as illustrated in Fig. 2.11.2; i.e., a uniform flow parallel with the x axis, an incompressible line source parallel with the z axis a distance d to the left of the origin, and an incompressible line sink parallel with the z axis a distance d to the right of the origin. The strengths of the source and the sink are Λ and $-\Lambda$, respectively. From Eqs. (2.10.12) and (2.10.22), the stream function for the synthesized flow field at any value of z is

$$\psi = V_\infty r \sin \theta + \Lambda \theta_1 / (2\pi) - \Lambda \theta_2 / (2\pi) \quad (2.11.6)$$

where $r = (x^2 + y^2)^{1/2}$, $\theta = \tan^{-1}(y/x)$, $\theta_1 = \tan^{-1}[y/(x+d)]$, and $\theta_2 = \tan^{-1}[y/(x-d)]$. This stream function can be written in Cartesian coordinates as

$$\psi = V_\infty y + \frac{\Lambda}{2\pi} \tan^{-1}\left(\frac{y}{x+d}\right) - \frac{\Lambda}{2\pi} \tan^{-1}\left(\frac{y}{x-d}\right) \quad (2.11.7)$$

Applying Eq. (2.9.9) to Eq. (2.11.7), the fluid velocity vector at an arbitrary point in this synthesized flow field is found to be

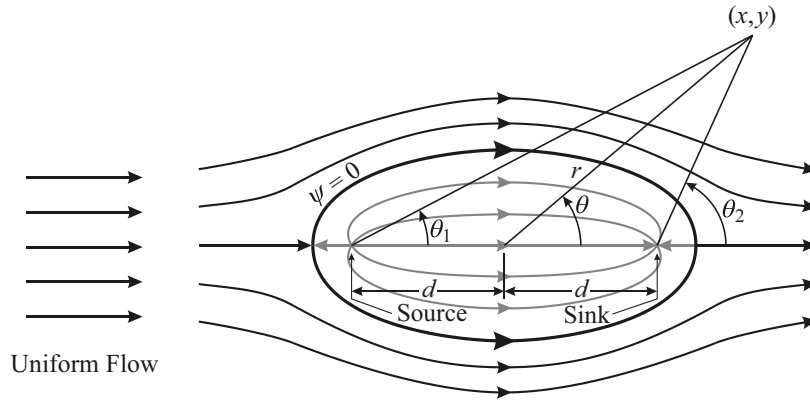


Figure 2.11.2. Potential flow over a Rankine cylinder; i.e., superposition of a uniform flow with a line source and a line sink of equal but opposite strengths.

$$\begin{aligned}
\mathbf{V} &= \frac{\partial \psi}{\partial y} \mathbf{i}_x - \frac{\partial \psi}{\partial x} \mathbf{i}_y + 0 \mathbf{i}_z \\
&= \left[V_\infty + \frac{A}{2\pi} \frac{x+d}{(x+d)^2 + y^2} - \frac{A}{2\pi} \frac{x-d}{(x-d)^2 + y^2} \right] \mathbf{i}_x \\
&\quad + \left[\frac{A}{2\pi} \frac{y}{(x+d)^2 + y^2} - \frac{A}{2\pi} \frac{y}{(x-d)^2 + y^2} \right] \mathbf{i}_y + 0 \mathbf{i}_z
\end{aligned} \tag{2.11.8}$$

Note that for any value of z and $y=0$, the fluid velocity vector in this flow field is

$$\mathbf{V}(x, 0, z) = \left(V_\infty - \frac{A}{\pi} \frac{d}{x^2 - d^2} \right) \mathbf{i}_x + 0 \mathbf{i}_y + 0 \mathbf{i}_z \tag{2.11.9}$$

We see from Eq. (2.11.9) that, at each value of z and $y=0$, there are two locations in x where all three components of the velocity vector are zero, i.e.,

$$x = \pm \sqrt{d^2 + \frac{A d}{\pi V_\infty}} \tag{2.11.10}$$

In other words, every point on the lines at the intersection of the surfaces defined by the equations $x = \pm [d^2 + A d / (\pi V_\infty)]^{1/2}$ and $y=0$ is a stagnation point of the flow field.

Notice from Fig. 2.11.2, that the stream surface defined by the equation $\psi=0$ is a closed cylinder of oval cross-section, which separates the fluid originating in the uniform flow from the fluid emitted by the line source. All of the fluid emitted by the source is consumed by the sink, and is confined entirely within the oval cylinder. In other words, all of the fluid outside this oval cylinder comes from the freestream; and all of the fluid inside this oval cylinder comes from the line source. Hence, the flow field inside the oval cylinder defined by the equation $\psi=0$ could be replaced with a solid cylinder of the same cross-section, and the flow field outside this stream surface can be interpreted as potential flow over this solid cylinder. The cross-section of any cylinder defined by setting the stream function in Eq. (2.11.7) to zero is called a *Rankine oval*, after W. J. M. Rankine; who first obtained this potential flow solution in the nineteenth century.

Incompressible Line Doublet

There is a special case of the source-sink pair shown in Fig. 2.11.2, which is of particular importance. Consider superposition of a source-sink pair with a strength that is inversely proportional to the separation distance, i.e., $A \equiv \kappa / (2d)$. From Eq. (2.10.22), the stream function for this source-sink pair without the uniform flow included is

$$\psi = \frac{\kappa}{4\pi d} (\theta_1 - \theta_2) \tag{2.11.11}$$

where

$$\tan \theta_1 = \frac{y}{x+d} \quad \text{and} \quad \tan \theta_2 = \frac{y}{x-d} \tag{2.11.12}$$

Using the trigonometric identity $\tan(\theta_1 - \theta_2) = (\tan \theta_1 - \tan \theta_2) / (1 + \tan \theta_1 \tan \theta_2)$ combined with the trigonometric relations given in Eq. (2.11.12) yields

$$\tan(\theta_1 - \theta_2) = \frac{\tan \theta_1 - \tan \theta_2}{1 + \tan \theta_1 \tan \theta_2} = \frac{\frac{y}{x+d} - \frac{y}{x-d}}{1 + \frac{y}{x+d} \frac{y}{x-d}} = \frac{-2yd}{x^2 + y^2 - d^2} \quad (2.11.13)$$

In the limit as d approaches zero, θ_1 approaches θ_2 and $\theta_1 - \theta_2$ approaches $\tan(\theta_1 - \theta_2)$. Hence, in the limit as d approaches zero, Eq. (2.11.11) yields

$$\psi = \lim_{d \rightarrow 0} \left[\frac{\kappa}{4\pi d} (\theta_1 - \theta_2) \right] = \lim_{d \rightarrow 0} \left[\frac{\kappa}{4\pi d} \frac{-2yd}{x^2 + y^2 - d^2} \right] = -\frac{\kappa}{2\pi} \frac{y}{x^2 + y^2} \quad (2.11.14)$$

In a similar manner, it can be shown that the velocity potential for this flow is

$$\phi = \frac{\kappa}{2\pi} \frac{x}{x^2 + y^2} \quad (2.11.15)$$

This potential flow field is commonly called a *line doublet* or simply *doublet flow*. Using the transformation $r^2 = x^2 + y^2$, $\cos \theta = x/r$, and $\sin \theta = y/r$, **this particular two-dimensional potential line doublet is defined in cylindrical coordinates by**

$$\phi(r, \theta) = \frac{\kappa}{2\pi} \frac{\cos \theta}{r} \quad (2.11.16)$$

$$\mathbf{V}(r, \theta) = \nabla \phi = -\frac{\kappa}{2\pi} \frac{\cos \theta}{r^2} \mathbf{i}_r - \frac{\kappa}{2\pi} \frac{\sin \theta}{r^2} \mathbf{i}_\theta + 0 \mathbf{i}_z \quad (2.11.17)$$

$$\psi(r, \theta) = -\frac{\kappa}{2\pi} \frac{\sin \theta}{r} \quad (2.11.18)$$

where κ is a constant called the *doublet strength*. Synthesis of this potential flow field is illustrated in Fig. 2.11.3. Note that in this figure the source was placed on the left and the sink was placed on the right. Hence, the flow is emitted from this doublet to the left and returns to the doublet from the right. This gives a doublet an associated flow direction, which is traditionally designated by an arrow pointing in the direction from the sink toward the source, as shown in Fig. 2.11.3.

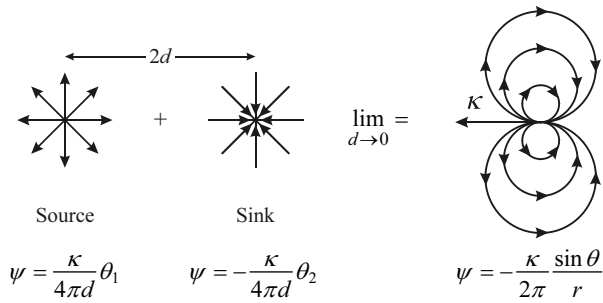


Figure 2.11.3. Line doublet; i.e., superposition of a line source of strength $\Lambda \equiv \kappa/(2d)$ and a line sink of equal but opposite strength, in the limit as the separation distance $2d$ approaches zero.

If all points in any valid potential flow field are rotated uniformly through a constant angle, the resulting flow field is also a valid potential flow. For example, the potential flow field described by Eqs. (2.11.16)–(2.11.18) and shown in Fig. 2.11.3 can be rotated about the z axis through the angle α to yield **the more general two-dimensional line doublet, which is defined in cylindrical coordinates by**

$$\phi(r, \theta) = \frac{\kappa}{2\pi} \frac{\cos(\theta - \alpha)}{r} \quad (2.11.19)$$

$$\mathbf{V}(r, \theta) = \nabla \phi = -\frac{\kappa}{2\pi} \frac{\cos(\theta - \alpha)}{r^2} \mathbf{i}_r - \frac{\kappa}{2\pi} \frac{\sin(\theta - \alpha)}{r^2} \mathbf{i}_\theta + 0 \mathbf{i}_z \quad (2.11.20)$$

$$\psi(r, \theta) = -\frac{\kappa}{2\pi} \frac{\sin(\theta - \alpha)}{r} \quad (2.11.21)$$

where κ is the doublet strength and α is the angle that the doublet flow direction makes with the radial direction at $\theta = \pi$. It is easily shown that the velocity potential given by Eq. (2.11.19) satisfies Laplace's equation for arbitrary values of the constants κ and α . This potential flow field is illustrated in Fig. 2.11.4.

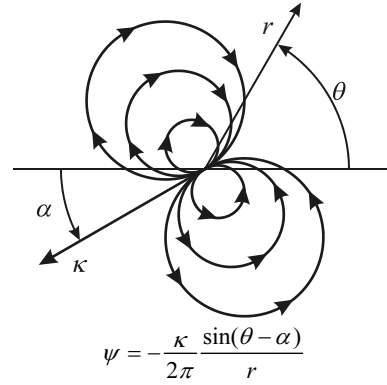


Figure 2.11.4. Two-dimensional line doublet of strength κ , similar to that shown in Fig. 2.11.3, but with the flow direction rotated through an angle α about the z axis.

2.12. Potential Flow about a Circular Cylinder

Similar to potential flow over a Rankine cylinder, potential flow over a circular cylinder can be synthesized by superposition of a uniform flow and a line doublet. Consider superposition of a uniform flow parallel with the x axis and a line doublet along the z axis pointing in the negative x direction, opposite to the uniform flow. From Eqs. (2.10.12) and (2.11.18), the stream function for the synthesized flow at any value of z is

$$\psi(r, \theta) = \left(V_\infty r - \frac{\kappa}{2\pi r} \right) \sin \theta \quad (2.12.1)$$

Note from Eq. (2.12.1) as illustrated in Fig. 2.12.1, that the stream surface defined by the equation $\psi=0$ is a closed cylinder of constant radius, which is denoted here as R . For all values of θ and z this cylinder is defined by

$$V_{\infty}r - \frac{\kappa}{2\pi r} = 0$$

or after rearranging

$$r^2 = \frac{\kappa}{2\pi V_{\infty}} \equiv R^2 \quad (2.12.2)$$

As illustrated in Fig. 2.12.1, all of the fluid emitted by the doublet is confined within this circular cylinder, and all of the fluid outside the cylinder comes from the freestream. Hence, the flow field outside this stream surface can be interpreted as potential flow over a solid circular cylinder of radius $R=[\kappa/(2\pi V_{\infty})]^{1/2}$. Using Eq. (2.12.2) to express the constant κ in Eqs. (2.11.16)–(2.11.18) in terms of R , and adding the uniform flow from Eqs. (2.10.10)–(2.10.12), **potential flow over a circular cylinder as synthesized from the superposition of uniform flow and a line doublet is defined by**

$$\phi(r, \theta) = V_{\infty}(r + R^2/r)\cos\theta \quad (2.12.3)$$

$$\mathbf{V}(r, \theta) = \nabla\phi = V_{\infty}(1 - R^2/r^2)\cos\theta\mathbf{i}_r - V_{\infty}(1 + R^2/r^2)\sin\theta\mathbf{i}_{\theta} + 0\mathbf{i}_z \quad (2.12.4)$$

$$\psi(r, \theta) = V_{\infty}(r - R^2/r)\sin\theta \quad (2.12.5)$$

The solution for potential flow over a circular cylinder, which is obtained from the superposition of a uniform flow and a line doublet, is not the only possible solution for potential flow over a long circular cylinder. Consider an incompressible line vortex superimposed along the z axis of the synthesized flow field described by Eqs. (2.12.3) through (2.12.5) and shown in Fig. 2.12.1. The resulting solution to Laplace's equation

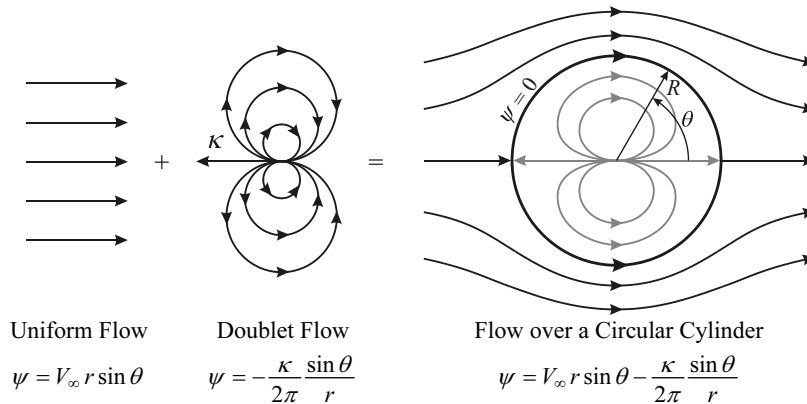


Figure 2.12.1. Flow over a circular cylinder; i.e., superposition of uniform flow and a line doublet.

is obtained by adding the velocity potential defined by Eq. (2.10.31) to that defined by Eq. (2.12.3). The velocity field for this synthesized potential flow is obtained by adding the velocity vector from Eq. (2.10.32) to that from Eq. (2.12.4). Similarly, the stream function is synthesized from Eqs. (2.10.33) and (2.12.5). The resulting synthesized flow field describes **potential flow about a circular cylinder with circulation, which is often interpreted as flow about a spinning circular cylinder, i.e.,**

$$\phi(r, \theta) = V_{\infty} \left(r + \frac{R^2}{r} \right) \cos \theta - \frac{\Gamma}{2\pi} \theta \quad (2.12.6)$$

$$\mathbf{V}(r, \theta) = \nabla \phi = V_{\infty} \left(1 - \frac{R^2}{r^2} \right) \cos \theta \mathbf{i}_r - \left[V_{\infty} \left(1 + \frac{R^2}{r^2} \right) \sin \theta + \frac{\Gamma}{2\pi r} \right] \mathbf{i}_{\theta} + 0 \mathbf{i}_z \quad (2.12.7)$$

$$\psi(r, \theta) = V_{\infty} \left(r - \frac{R^2}{r} \right) \sin \theta + \frac{\Gamma}{2\pi} \ln r \quad (2.12.8)$$

where Γ is the circulation of the line vortex. The synthesis of this potential flow field is illustrated in Fig. 2.12.2.

Notice from Eq. (2.12.7) that the r component of fluid velocity is not changed by the addition of the vortex. As r approaches infinity, the flow field still reduces to the uniform flow, and at $r=R$, the r component of fluid velocity is still zero. Thus, the potential flow solution given by Eqs. (2.12.6)–(2.12.8) satisfies Laplace's equation and all boundary conditions necessary to describe inviscid incompressible flow over a solid circular cylinder of radius R . Because the value of Γ in Eqs. (2.12.6)–(2.12.8) is arbitrary, we see that **there are an infinite number of solutions for potential flow over a long circular cylinder of any given radius**. Which of these solutions actually occurs in nature (if any) will depend on viscous effects in the boundary layer.

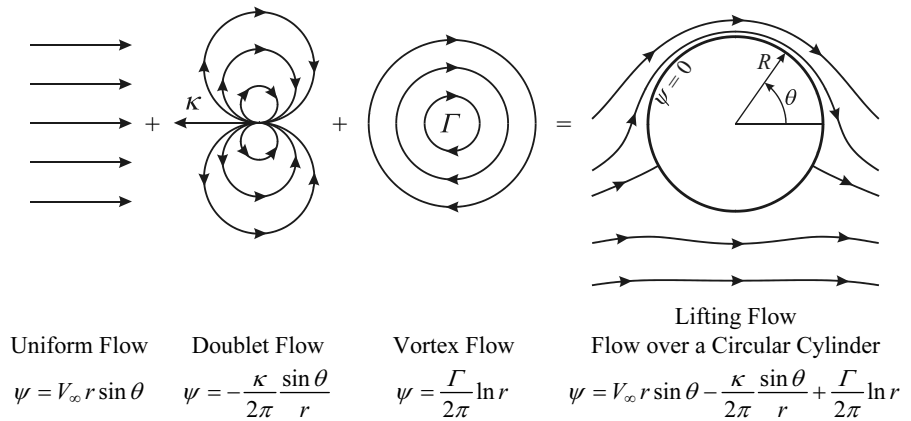


Figure 2.12.2. Flow over a circular cylinder with circulation; i.e., superposition of a uniform flow, a line doublet, and a line vortex.

Notice from Eq. (2.12.7) that the fluid velocity at the surface of the solid cylinder, where $r=R$, is given by

$$\mathbf{V}(R, \theta, z) = 0\mathbf{i}_r - \left(2V_\infty \sin \theta + \frac{\Gamma}{2\pi R} \right) \mathbf{i}_\theta + 0\mathbf{i}_z \quad (2.12.9)$$

This relation shows that a stagnation point exists on the surface of the cylinder wherever the bracketed term in Eq. (2.12.9) is equal to zero. Denoting the θ coordinate of this stagnation point as θ_0 , we obtain

$$\theta_0 = \sin^{-1} \left(-\frac{\Gamma}{4\pi R V_\infty} \right) \quad (2.12.10)$$

Because $\sin \theta$ is negative only in the third and fourth quadrants, we see that for positive values of the vortex circulation Γ , stagnation points could exist only on the lower half of the cylinder. When $\Gamma/(4\pi R V_\infty) < 1$, there are two stagnation points on the lower surface of the cylinder for each value of z , as shown in Fig. 2.12.3 (a). From Eq. (2.12.10), we also see that when $\Gamma/(4\pi R V_\infty) = 1$, there is only one stagnation point on the lower surface of the cylinder for each value of z , which is located at $(r, \theta, z) = (R, -\pi/2, z)$, as can be seen in Fig. 2.12.3 (b). Because the magnitude of $\sin \theta$ is always less than or equal to unity, when $\Gamma/(4\pi R V_\infty) > 1$, there is no stagnation point on the surface of the cylinder.

For the case $\Gamma/(4\pi R V_\infty) > 1$, in the plane $\theta = -\pi/2$, Eq. (2.12.7) reduces to

$$\mathbf{V}(r, -\pi/2, z) = 0\mathbf{i}_r + [V_\infty(1 + R^2/r^2) - \Gamma/(2\pi r)]\mathbf{i}_\theta + 0\mathbf{i}_z \quad (2.12.11)$$

Solving this for the radial coordinate where the velocity is zero, denoted as r_0 , yields

$$r_0 = \Gamma/(4\pi V_\infty) \pm \sqrt{[\Gamma/(4\pi V_\infty)]^2 - R^2} \quad (2.12.12)$$

Hence, for $\Gamma/(4\pi R V_\infty) > 1$, there is a stagnation point in the fluid a finite distance below the surface of the cylinder at $\theta = -\pi/2$, as shown in Fig. 2.12.3 (c). The other stagnation point obtained from Eq. (2.12.12) is in the nonphysical region inside the cylinder.

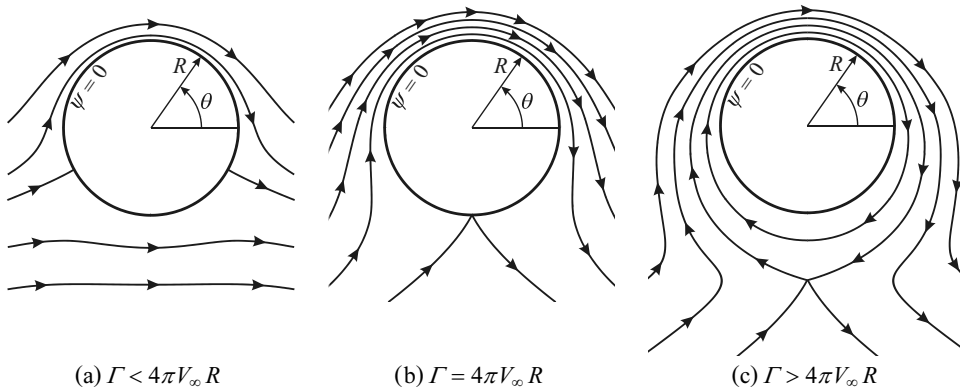


Figure 2.12.3. Examples of stagnation points in the flow over a circular cylinder with circulation.

EXAMPLE 2.12.1. For steady, inviscid, irrotational flow over a circular cylinder with circulation, use Bernoulli's equation to find the pressure coefficient at an arbitrary point in the flow field.

Solution. From Eq. (2.12.7), the local fluid velocity vector at an arbitrary point in this flow field is

$$\mathbf{V} = V_\infty \left(1 - \frac{R^2}{r^2} \right) \cos \theta \mathbf{i}_r - \left[V_\infty \left(1 + \frac{R^2}{r^2} \right) \sin \theta + \frac{\Gamma}{2\pi r} \right] \mathbf{i}_\theta + 0 \mathbf{i}_z$$

Hence, the magnitude of the local fluid velocity vector $V(r, \theta, z)$ is obtained from

$$\begin{aligned} V^2 &= \mathbf{V} \cdot \mathbf{V} = V_r^2 + V_\theta^2 + V_z^2 \\ &= V_\infty^2 \left(1 - \frac{R^2}{r^2} \right)^2 \cos^2 \theta + \left[V_\infty \left(1 + \frac{R^2}{r^2} \right) \sin \theta + \frac{\Gamma}{2\pi r} \right]^2 \\ &= V_\infty^2 \left(1 - \frac{R^2}{r^2} \right)^2 \cos^2 \theta \\ &\quad + V_\infty^2 \left(1 + \frac{R^2}{r^2} \right)^2 \sin^2 \theta + \frac{V_\infty \Gamma}{\pi r} \left(1 + \frac{R^2}{r^2} \right) \sin \theta + \left(\frac{\Gamma}{2\pi r} \right)^2 \\ &= V_\infty^2 \left[\left(1 - 2\frac{R^2}{r^2} + \frac{R^4}{r^4} \right) \cos^2 \theta + \left(1 + 2\frac{R^2}{r^2} + \frac{R^4}{r^4} \right) \sin^2 \theta \right] \\ &\quad + \frac{V_\infty \Gamma}{\pi r} \left(1 + \frac{R^2}{r^2} \right) \sin \theta + \left(\frac{\Gamma}{2\pi r} \right)^2 \end{aligned}$$

or after rearranging and applying the trigonometric identity $\cos^2 \theta + \sin^2 \theta = 1$,

$$\left(\frac{V}{V_\infty} \right)^2 = 1 - 2\frac{R^2}{r^2} + \frac{R^4}{r^4} + 4\frac{R^2}{r^2} \sin^2 \theta + \frac{\Gamma}{\pi r V_\infty} \left(1 + \frac{R^2}{r^2} \right) \sin \theta + \left(\frac{\Gamma}{2\pi r V_\infty} \right)^2 \quad (2.12.13)$$

The pressure coefficient found from Bernoulli's equation is given by Eq. (2.7.7), and using Eq. (2.12.13) in Eq. (2.7.7) yields

$$\begin{aligned} C_p(r, \theta, z) &\equiv \frac{p - p_\infty}{\frac{1}{2} \rho V_\infty^2} = 1 - \left(\frac{V}{V_\infty} \right)^2 \\ &= \frac{R^2}{r^2} \left(2 - \frac{R^2}{r^2} - 4 \sin^2 \theta \right) - \frac{\Gamma}{\pi r V_\infty} \left(1 + \frac{R^2}{r^2} \right) \sin \theta - \left(\frac{\Gamma}{2\pi r V_\infty} \right)^2 \end{aligned} \quad (2.12.14)$$

Because the value of Γ is arbitrary, the velocity and pressure fields that are given by Eqs. (2.12.7) and (2.12.14), respectively, describe an infinite number of possible solutions

for steady, inviscid flow normal to a long circular cylinder of radius R . For any value of Γ , the relative velocity far from the cylinder is uniform with magnitude V_∞ , and the pressure far from the cylinder is the hydrostatic pressure p_∞ , which depends on only the altitude H . At the surface of the cylinder, where $r=R$, the relative velocity is given by Eq. (2.12.9); and the pressure coefficient on the surface of this cylinder is obtained by setting $r=R$ in Eq. (2.12.14), which yields

$$C_p(R, \theta) = 1 - 4\sin^2\theta - \frac{2\Gamma}{\pi R V_\infty} \sin\theta - \left(\frac{\Gamma}{2\pi R V_\infty}\right)^2 \quad (2.12.15)$$

This pressure distribution is symmetric fore and aft of the cylinder's vertical mid-plane where $\theta = \pm\pi/2$. The contributions to the pressure coefficient from the first, second, and fourth terms on the right-hand side of Eq. (2.12.15) are also symmetric above and below the cylinder's horizontal mid-plane where $\theta=0$ and $\theta=\pi$. Because $\sin\theta$ is positive in the first and second quadrants and negative in the third and fourth quadrants, we see that for positive values of the vortex circulation Γ , the third of the four terms on the right-hand side of Eq. (2.12.15) decreases pressure on the upper half of the cylinder and increases pressure on the lower half. This can also be observed from the streamlines for the flow field, which are shown in Fig. 2.12.3.

As we continue with our study of inviscid aerodynamics, it will be useful to have a slightly more general solution for potential flow about a circular cylinder. By combining Eq. (2.12.2) with Eqs. (2.10.10)–(2.10.12), (2.10.31)–(2.10.33), and (2.11.19)–(2.11.21), we obtain the relations for **a more general two-dimensional potential flow about a circular cylinder with circulation**, i.e.,

$$\phi(r, \theta) = V_\infty \left(r + \frac{R^2}{r} \right) \cos(\theta - \alpha) - \frac{\Gamma}{2\pi} \theta \quad (2.12.16)$$

$$\mathbf{V} = V_\infty \left(1 - \frac{R^2}{r^2} \right) \cos(\theta - \alpha) \mathbf{i}_r - \left[V_\infty \left(1 + \frac{R^2}{r^2} \right) \sin(\theta - \alpha) + \frac{\Gamma}{2\pi r} \right] \mathbf{i}_\theta + 0 \mathbf{i}_z \quad (2.12.17)$$

$$\psi(r, \theta) = V_\infty \left(r - \frac{R^2}{r} \right) \sin(\theta - \alpha) + \frac{\Gamma}{2\pi} \ln r \quad (2.12.18)$$

where R is the radius of the cylinder, V_∞ is the freestream airspeed, α is the angle that the freestream makes with the x axis, and Γ is the circulation. The pressure coefficient in this potential flow field is

$$C_p(r, \theta) = \frac{R^2}{r^2} \left[2 - \frac{R^2}{r^2} - 4\sin^2(\theta - \alpha) \right] - \frac{\Gamma}{\pi r V_\infty} \left(1 + \frac{R^2}{r^2} \right) \sin(\theta - \alpha) - \left(\frac{\Gamma}{2\pi r V_\infty} \right)^2 \quad (2.12.19)$$

and the pressure coefficient on the surface of the cylinder is

$$C_p(R, \theta) = 1 - 4\sin^2(\theta - \alpha) - \frac{2\Gamma}{\pi R V_\infty} \sin(\theta - \alpha) - \left(\frac{\Gamma}{2\pi R V_\infty} \right)^2 \quad (2.12.20)$$

The potential flow field described by Eqs. (2.12.16)–(2.12.18) is simply the flow field described by Eqs. (2.12.6)–(2.12.8) with all points rotated about the z axis through the angle α . This describes potential flow about a long circular cylinder with the free-stream normal to the z axis and at an angle α to the x axis. Synthesis of this potential flow field is illustrated in Fig. 2.12.4.

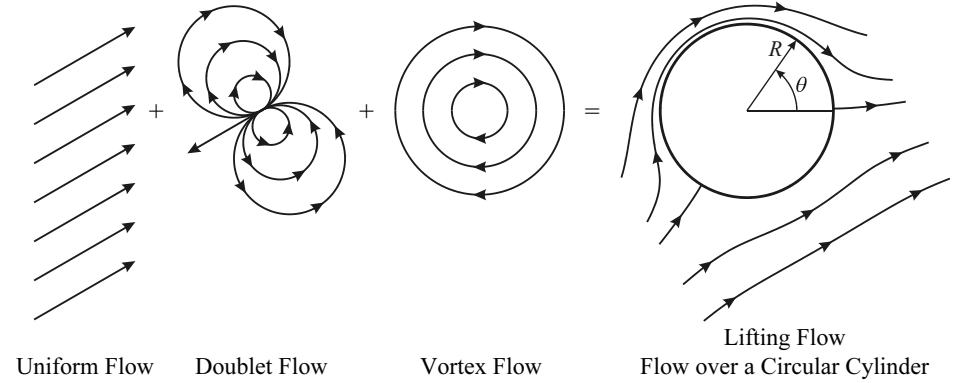


Figure 2.12.4. Synthesis of two-dimensional flow about a circular cylinder with circulation and the freestream at an angle to the x axis.

2.13. Aerodynamic Lift and Drag

As discussed in Chapter 1, the resultant surface force acting on any body immersed in a fluid originates from only two sources, *the pressure distribution integrated over the body surface* and *the shear stress distribution integrated over the body surface*. In this chapter we are concerned primarily with inviscid flow. Because there can be no shear stress in an inviscid fluid flow field, the total resultant surface force includes only the pressure force, which is given by Eq. (2.3.5). Hence, **the total resultant surface force vector that acts on any body immersed in an inviscid fluid flow field is**

$$\mathbf{F} = - \iint_{\mathcal{S}} p \mathbf{n} d\mathcal{S} \quad (2.13.1)$$

where p denotes absolute pressure, $\mathbf{n}$ is the local unit outward surface normal, $d\mathcal{S}$ is the differential surface area, and $\mathcal{S}$ is the entire surface area in contact with the fluid.

As was discussed in Sec. 1.12, the total surface force $\mathbf{F}$ exerted on an object moving through a fluid can be divided into two parts; the *hydrostatic buoyant force* $\mathbf{F}_b$ and the *aerodynamic force* $\mathbf{F}_a$. For this convenience, Eq. (2.13.1) can be written equivalently as

$$\mathbf{F} = - \iint_{\mathcal{S}} p_{\infty} \mathbf{n} d\mathcal{S} - \iint_{\mathcal{S}} (p - p_{\infty}) \mathbf{n} d\mathcal{S} \quad (2.13.2)$$

where p_{∞} denotes the freestream static pressure, which depends on only the altitude H . The first integral on the right-hand side of Eq. (2.13.2) is the **hydrostatic buoyant force**

$$\mathbf{F}_b \equiv - \iint_{\mathcal{S}} p_{\infty} \mathbf{n} d\mathcal{S} \quad (2.13.3)$$

and the second integral on the right-hand side of Eq. (2.13.2) is the **aerodynamic force that acts on any surface immersed in an inviscid fluid flow field**

$$\mathbf{F}_a \equiv - \iint_{\mathcal{S}} (p - p_\infty) \mathbf{n} d\mathcal{S} \quad (2.13.4)$$

Equation (2.13.4) can be used to obtain the resultant aerodynamic force that acts on any closed surface immersed in an inviscid fluid flow field, as shown in Fig. 2.13.1.

For flow about a wing or long cylinder of any cross-section, it is often convenient to define a *section aerodynamic force* to be the resultant aerodynamic force per unit span acting on any cross-sectional area immersed in a fluid flow field. For example, consider the cross-section of the flow field illustrated in Fig. 2.13.2. The closed curve C encloses some arbitrary area in the plane of this cross-section, ds is a differential length measured along this curve, and $\mathbf{n}$ is the local unit vector in the plane of the cross-section that is normal to the curve C pointing outward from the enclosed cross-sectional area and into the surrounding fluid. We now consider a cylindrical surface in this flow field, which is formed by extruding the curve C a differential distance $dz/2$ in both directions normal to the plane of the cross-section. The differential area in contact with the surrounding fluid on this cylindrical surface is $d\mathcal{S} = ds dz$. The section aerodynamic force acting on the cross-sectional area enclosed by the curve C is defined to be the aerodynamic force acting on this cylindrical surface divided by the length dz , taken in the limit as dz approaches zero. Hence, from Eq. (2.13.4), we obtain the **section aerodynamic force that acts on any cross-sectional area immersed in an inviscid fluid flow field**

$$\tilde{\mathbf{F}}_a \equiv - \oint_C (p - p_\infty) \mathbf{n} ds \quad (2.13.5)$$

where C is the curve enclosing the cross-sectional area, $\mathbf{n}$ is the local unit outward normal to C in the plane of the cross-section, and ds is a differential length measured along C . Equation (2.13.5) can be used to compute section aerodynamic forces in two-dimensional inviscid flows, like the potential flow about an airfoil shown in Fig. 2.7.2 or potential flow about other cylinders like those shown in Figs. 2.11.2 and 2.12.4.

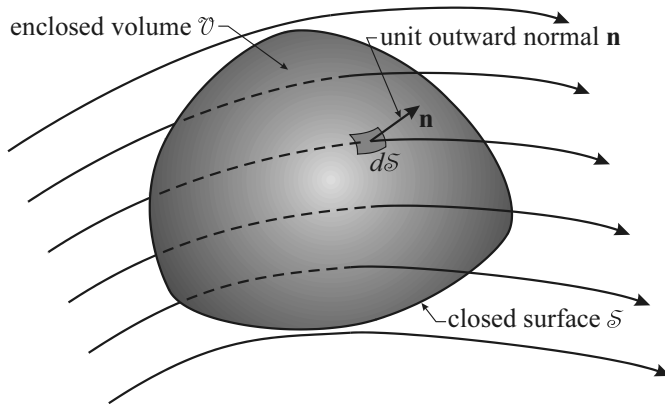


Figure 2.13.1. Closed surface $\mathcal{S}$ with local unit outward normal $\mathbf{n}$ enclosing volume $\mathcal{V}$.

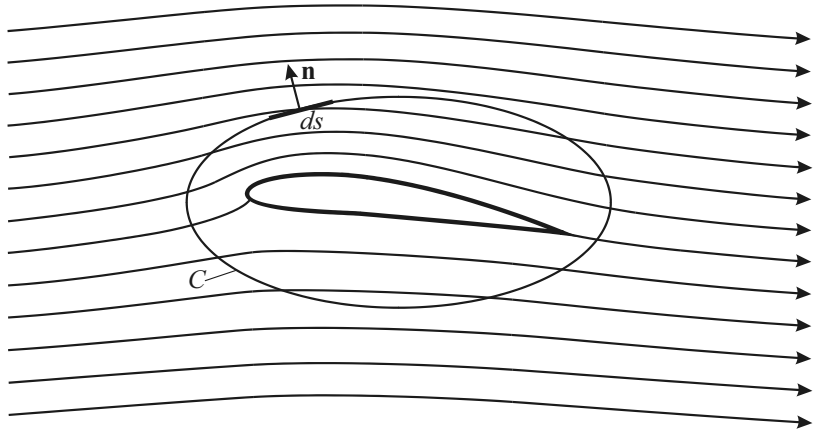


Figure 2.13.2. Closed curve C with the local unit outward normal $\mathbf{n}$ enclosing an arbitrary cross-sectional area immersed in a cross-section of an inviscid fluid flow field.

For aerodynamic flows with the relative wind normal to the span of a wing or a cylinder of any cross-section, it may be convenient to express the section aerodynamic force in terms of its Cartesian components. In traditional aerodynamic coordinates and notation, the span is aligned with the z axis and the Cartesian components of Eq. (2.13.5) are referred to as the **section axial force**

$$\tilde{A} \equiv \tilde{\mathbf{F}}_a \cdot \mathbf{i}_x = -\oint_C (p - p_\infty) \mathbf{n} \cdot \mathbf{i}_x ds \quad (2.13.6)$$

and the **section normal force**

$$\tilde{N} \equiv \tilde{\mathbf{F}}_a \cdot \mathbf{i}_y = -\oint_C (p - p_\infty) \mathbf{n} \cdot \mathbf{i}_y ds \quad (2.13.7)$$

The section aerodynamic force vector is also commonly divided into components parallel with and perpendicular to the relative wind. Using traditional aerodynamic coordinates and notation, for aerodynamic flows with the relative wind normal to the z axis, these components of Eq. (2.13.5) are the **section drag**

$$\tilde{D} \equiv \tilde{\mathbf{F}}_a \cdot \mathbf{i}_\infty = -\oint_C (p - p_\infty) \mathbf{n} \cdot \mathbf{i}_\infty ds \quad (2.13.8)$$

and the **section lift**

$$\tilde{L} \equiv \tilde{\mathbf{F}}_a \cdot (\mathbf{i}_z \times \mathbf{i}_\infty) = -\oint_C (p - p_\infty) \mathbf{n} \cdot (\mathbf{i}_z \times \mathbf{i}_\infty) ds \quad (2.13.9)$$

where $\mathbf{i}_\infty$ is the unit vector in the direction of the relative wind.

The resultant section aerodynamic force vector is completely specified in terms of either the section drag and lift, or the section axial and normal force components. These two equivalent descriptions of the resultant section aerodynamic force vector are related to each other through the angle of attack, as discussed in Sec. 1.16. Here the angle of attack is defined to be the angle that the relative wind makes with the x axis.

As discussed in Sec. 1.16, the section aerodynamic force coefficients are nondimensionalized traditionally with respect to the product of the freestream dynamic pressure $\frac{1}{2}\rho V_\infty^2$ and the chord length c . Hence, after also applying the definition of the pressure coefficient, $C_p \equiv (p - p_\infty)/(\frac{1}{2}\rho V_\infty^2)$, Eqs. (2.13.6)–(2.13.9) are nondimensionalized to yield the **inviscid flow relations for the section axial force coefficient**

$$\tilde{C}_A = -\oint_C C_p \mathbf{n} \cdot \mathbf{i}_x ds/c \quad (2.13.10)$$

the section normal force coefficient

$$\tilde{C}_N = -\oint_C C_p \mathbf{n} \cdot \mathbf{i}_y ds/c \quad (2.13.11)$$

the section drag coefficient

$$\tilde{C}_D = -\oint_C C_p \mathbf{n} \cdot \mathbf{i}_\infty ds/c \quad (2.13.12)$$

and the section lift coefficient

$$\tilde{C}_L = -\oint_C C_p \mathbf{n} \cdot (\mathbf{i}_z \times \mathbf{i}_\infty) ds/c \quad (2.13.13)$$

where C is the closed curve enclosing the cross-section.

EXAMPLE 2.13.1. For steady, inviscid, irrotational flow over a circular cylinder with circulation as described by Eqs. (2.12.16)–(2.12.20), find the section axial and normal force coefficients. For this same flow field also find the section drag and lift coefficients.

Solution. As shown in Fig. 2.12.4, the curve that encloses the cross-section of this cylinder is a circle of radius R in a plane normal to the z axis and centered about the z axis. From Eq. (2.12.20), the pressure coefficient on the surface of this cylinder is

$$C_p(R, \theta) = 1 - 4\sin^2(\theta - \alpha) - \frac{2\Gamma}{\pi R V_\infty} \sin(\theta - \alpha) - \left(\frac{\Gamma}{2\pi R V_\infty} \right)^2$$

The unit outward normal everywhere on the circumference of this circular cross-section is in the radial direction, i.e., $\mathbf{n} = \mathbf{i}_r = \cos\theta \mathbf{i}_x + \sin\theta \mathbf{i}_y$. Hence, for this cross-section we have $\mathbf{n} \cdot \mathbf{i}_x = \cos\theta$ and $\mathbf{n} \cdot \mathbf{i}_y = \sin\theta$. The differential arc length on the circumference of this circular cross-section is $ds = R d\theta$ and the section chord length is $c = 2R$. Thus, $ds/c = d\theta/2$ and the section axial force coefficient obtained from Eq. (2.13.10) is

$$\tilde{C}_A = - \int_{\theta=0}^{2\pi} \left[\frac{1}{2} - 2\sin^2(\theta - \alpha) - \frac{\Gamma}{\pi R V_\infty} \sin(\theta - \alpha) - \frac{1}{8} \left(\frac{\Gamma}{\pi R V_\infty} \right)^2 \right] \cos\theta d\theta$$

After applying the trigonometric identity $\sin(\theta - \alpha) = \sin\theta \cos\alpha - \cos\theta \sin\alpha$, this relation can be rearranged to yield

$$\begin{aligned}\tilde{C}_A = & \left[\frac{1}{8} \left(\frac{\Gamma}{\pi R V_\infty} \right)^2 - \frac{1}{2} \right] \int_{\theta=0}^{2\pi} \cos \theta d\theta + 2 \int_{\theta=0}^{2\pi} [\sin \theta \cos \alpha - \cos \theta \sin \alpha]^2 \cos \theta d\theta \\ & + \frac{\Gamma}{\pi R V_\infty} \int_{\theta=0}^{2\pi} [\sin \theta \cos \alpha - \cos \theta \sin \alpha] \cos \theta d\theta\end{aligned}$$

The first integral on the right-hand side of this equation evaluates to

$$\int_{\theta=0}^{2\pi} \cos \theta d\theta = 0$$

and the remaining two integrals can be further rearranged to give

$$\begin{aligned}\tilde{C}_A = & 2 \cos^2 \alpha \int_{\theta=0}^{2\pi} \sin^2 \theta \cos \theta d\theta - 4 \sin \alpha \cos \alpha \int_{\theta=0}^{2\pi} \sin \theta \cos^2 \theta d\theta \\ & + 2 \sin^2 \alpha \int_{\theta=0}^{2\pi} \cos^3 \theta d\theta + \frac{\Gamma \cos \alpha}{\pi R V_\infty} \int_{\theta=0}^{2\pi} \sin \theta \cos \theta d\theta - \frac{\Gamma \sin \alpha}{\pi R V_\infty} \int_{\theta=0}^{2\pi} \cos^2 \theta d\theta\end{aligned}$$

Noting that

$$\begin{aligned}\int_{\theta=0}^{2\pi} \sin^2 \theta \cos \theta d\theta = 0, \quad \int_{\theta=0}^{2\pi} \sin \theta \cos^2 \theta d\theta = 0, \quad \int_{\theta=0}^{2\pi} \cos^3 \theta d\theta = 0, \\ \int_{\theta=0}^{2\pi} \sin \theta \cos \theta d\theta = 0, \quad \int_{\theta=0}^{2\pi} \cos^2 \theta d\theta = \pi\end{aligned}$$

we obtain the result

$$\boxed{\tilde{C}_A = -\frac{\Gamma}{R V_\infty} \sin \alpha} \quad (2.13.14)$$

Likewise, the section normal force coefficient as obtained from Eq. (2.13.11) is

$$\tilde{C}_N = - \int_{\theta=0}^{2\pi} \left[\frac{1}{2} - 2 \sin^2(\theta - \alpha) - \frac{\Gamma}{\pi R V_\infty} \sin(\theta - \alpha) - \frac{1}{8} \left(\frac{\Gamma}{\pi R V_\infty} \right)^2 \right] \sin \theta d\theta$$

and after performing this integration in a manner similar to that used to obtain the result given by Eq. (2.13.14), this yields

$$\boxed{\tilde{C}_N = \frac{\Gamma}{R V_\infty} \cos \alpha} \quad (2.13.15)$$

The reader should notice from Eqs. (2.13.14) and (2.13.15) that **the section aerodynamic force acting on this circular cylinder immersed in this steady, inviscid, irrotational flow field is proportional to the section circulation. With no circulation, the resultant aerodynamic force acting on this cylinder is zero.**

From Eqs. (2.13.5)–(2.13.7) combined with Eqs. (2.13.10) and (2.13.11), the nondimensional section aerodynamic force vector acting on this circular cross-section in this two-dimensional flow field can be written as

$$\tilde{\mathbf{F}}_a / (\frac{1}{2} \rho V_\infty^2 c) = \tilde{C}_A \mathbf{i}_x + \tilde{C}_N \mathbf{i}_y + 0 \mathbf{i}_z \quad (2.13.16)$$

Because the freestream for this flow field is normal to the z axis and makes the angle α with the x axis, the unit vector in the direction of the relative wind is

$$\mathbf{i}_\infty = \cos \alpha \mathbf{i}_x + \sin \alpha \mathbf{i}_y + 0 \mathbf{i}_z \quad (2.13.17)$$

Using Eqs. (2.13.16) and (2.13.17) with the definition in Eq. (2.13.8), the section drag coefficient for this inviscid flow is obtained from

$$\tilde{C}_D \equiv \tilde{\mathbf{F}}_a \cdot \mathbf{i}_\infty / (\frac{1}{2} \rho V_\infty^2 c) = (\tilde{C}_A \mathbf{i}_x + \tilde{C}_N \mathbf{i}_y + 0 \mathbf{i}_z) \cdot (\cos \alpha \mathbf{i}_x + \sin \alpha \mathbf{i}_y + 0 \mathbf{i}_z)$$

or

$$\tilde{C}_D = \tilde{C}_A \cos \alpha + \tilde{C}_N \sin \alpha \quad (2.13.18)$$

Likewise, from the definition in Eq. (2.13.9), the section lift coefficient can be written as

$$\tilde{C}_L \equiv \tilde{\mathbf{F}}_a \cdot (\mathbf{i}_z \times \mathbf{i}_\infty) / (\frac{1}{2} \rho V_\infty^2 c) \quad (2.13.19)$$

From Eq. (2.13.17) we obtain

$$\mathbf{i}_z \times \mathbf{i}_\infty = \begin{vmatrix} \mathbf{i}_x & \mathbf{i}_y & \mathbf{i}_z \\ 0 & 0 & 1 \\ \cos \alpha & \sin \alpha & 0 \end{vmatrix} = -\sin \alpha \mathbf{i}_x + \cos \alpha \mathbf{i}_y + 0 \mathbf{i}_z \quad (2.13.20)$$

Using Eqs. (2.13.16) and (2.13.20) in Eq. (2.13.19) yields

$$\tilde{C}_L = -\tilde{C}_A \sin \alpha + \tilde{C}_N \cos \alpha \quad (2.13.21)$$

After applying Eqs. (2.13.14) and (2.13.15) to Eq. (2.13.18), we find that the section drag acting on this circular cross-section is zero

$$\boxed{\tilde{C}_D = 0} \quad (2.13.22)$$

In a similar manner, after using Eqs. (2.13.14) and (2.13.15) in Eq. (2.13.21) and applying the trigonometric identity $\sin^2 \theta + \cos^2 \theta = 1$ we obtain

$$\boxed{\tilde{C}_L = \frac{\Gamma}{R V_\infty}} \quad (2.13.23)$$

Hence, for **steady, inviscid, irrotational flow over a long circular cylinder**, we obtain the remarkably simply result that the section drag is exactly zero and the section lift is linearly proportional to the section circulation.

2.14. d'Alembert's Paradox and the Kutta–Joukowski Law

The result of zero drag acting on a body moving with constant velocity relative to an inviscid fluid is more general than the special case of two-dimensional flow over a long circular cylinder, which was considered in Example 2.13.1. As we will see in Chapter 4, **for steady, two-dimensional, incompressible, potential flow over a body of arbitrary cross-section, with or without circulation, the section drag is always zero, provided that the flow is uniform far from the body and that mass is neither added to nor extracted from the freestream fluid as it flows about the cross-section.**

An analytically predicted result of zero drag acting on a body moving through a fluid is in direct conflict with common experience. Hence, it is easy to imagine the frustration of eighteenth- and nineteenth-century mathematicians and scientists when it was found that potential flow theory predicts zero drag. This contradiction between experimental observations and the zero-drag results predicted by potential flow theory is commonly known as *d'Alembert's paradox*, named after the famous French mathematician and scientist Jean-Baptiste le Rond d'Alembert (1717–1783), who studied aerodynamic drag extensively during the period from 1744 to 1768.

In a paper published in 1744, d'Alembert first obtained the result of zero drag for two-dimensional, incompressible, potential flow over a closed cross-section. In 1752, he published a similar result obtained by different means. He continued to work on this problem until 1768, but was unable to resolve the paradox. In his 1768 paper d'Alembert states, *“I do not see then, I admit, how one can explain the resistance of fluids by the theory in a satisfactory manner. It seems to me on the contrary that this theory, dealt with and studied with profound attention gives, at least in most cases, a resistance of absolutely zero: a singular paradox which I leave to future geometricians to explain.”*

Consequences of d'Alembert's paradox were devastating to the emerging field of theoretical fluid dynamics in the late eighteenth and early nineteenth centuries, because it foiled attempts by mathematicians like d'Alembert and Euler to describe the dynamics of slightly viscous fluids like air and water using potential flow theory, which is governed by Laplace's equation. The net effect of d'Alembert's paradox was to split the scientific community into two factions. On one hand, the mathematicians developing theoretical fluid mechanics were clinging to potential flow, which seemed to predict phenomena that were not observed. On the other hand, experimentalists in the field of applied hydraulics were observing phenomena that could not be explained.

Thanks to contributions made in the nineteenth and early twentieth centuries by mathematicians and scientists like Navier, Stokes, and Prandtl, today we know that the aerodynamic drag not predicted by potential flow theory is due to viscous effects, which generate shear stress at the body surface and cause the flow to separate from the surface on the downstream side of a bluff body. Even in slightly viscous fluids like air and water at high Reynolds numbers, viscous shear stresses remain important in a thin boundary layer next to the body surface. These viscous shear stresses cause *skin friction drag* on streamlined objects; and for bluff bodies they cause flow separation, which produces a low-pressure wake behind the object resulting in what is called *form drag*. Although potential flow theory does not apply in the boundary layer next to the body surface, as will be shown later in this textbook, potential flow theory can be used in combination with *Prandtl's boundary-layer theory* to accurately predict aerodynamic drag.

Although the prediction of zero drag obtained from Eq. (2.13.22) is erroneous and must be corrected using boundary-layer theory or empirical results, the prediction of lift from Eq. (2.13.23) is quite realistic. From the definition of the section lift coefficient discussed in Sec. 1.16, the section lift for incompressible flow can be written as

$$\tilde{L} = \frac{1}{2} \rho V_\infty^2 c \tilde{C}_L \quad (2.14.1)$$

The section chord length for the circular cross-section used to obtain the result given in Eq. (2.13.23) is $c=2R$. Thus, using Eq. (2.13.23) in Eq. (2.14.1), we obtain

$$\tilde{L} = \frac{1}{2} \rho V_\infty^2 2R \frac{\Gamma}{RV_\infty}$$

which yields

$$\boxed{\tilde{L} = \rho V_\infty \Gamma} \quad (2.14.2)$$

This remarkably simple result was developed here by computing the section aerodynamic lift for two-dimensional potential flow over a long circular cylinder. Equation (2.14.2) is a very important relation in theoretical aerodynamics, and it is referred to commonly as the *Kutta–Joukowski law* or the *Kutta–Joukowski theorem*. This relation between lift and the product of density, velocity, and circulation is named for work published by the well known German mathematician M. Wilhelm Kutta (1902) and the famed Russian physicist Nikolai E. Joukowski (1906), who each developed this relation independently.

Equation (2.14.2) is more universal than might be implied from its development for the special case of potential flow over a long circular cylinder, which was considered in Example 2.13.1. As will be shown in Chapter 4, **the Kutta–Joukowski theorem also applies to steady, two-dimensional, incompressible, potential flow over any body of arbitrary cross-section**. As was discussed in Sec. 1.15, for high-Reynolds-number flows about streamlined bodies at low angles of attack, the lift that is predicted from inviscid flow theory agrees very closely with experimental observations, because viscous shear stresses in the boundary layer have little effect on the production of lift in such cases. Even when boundary-layer separation occurs, the Kutta–Joukowski law provides reasonable results, because the reduction in lift caused by boundary-layer separation is accompanied by a corresponding reduction in circulation.

To better understand the important relation between lift and circulation, we shall return to our consideration of steady flow normal to a long circular cylinder, which was introduced in Example 1.14.1, discussed in more detail in Sec. 1.15, and considered again in Examples 2.7.1, 2.12.1, and 2.13.1. When two-dimensional flow about a fixed circular cylinder is observed in a low-speed wind tunnel, the streamlines will appear as shown in Fig. 1.15.5. At high Reynolds numbers, the streamlines about the front of the cylinder are similar to those predicted analytically for inviscid flow and shown in Fig. 1.14.3. However, viscous shear stresses in the boundary layer cause flow separation over the back of the cylinder, which produces a low-pressure wake behind the cylinder. This flow separation can be seen in Prandtl and Tietjens' historical flow-visualization photograph, which is reproduced here in Fig. 2.14.1 (a). Notice the similarity between Fig. 2.14.1 (a) and the streamlines that are shown in Fig. 1.15.5 for a Reynolds number of 1,000.

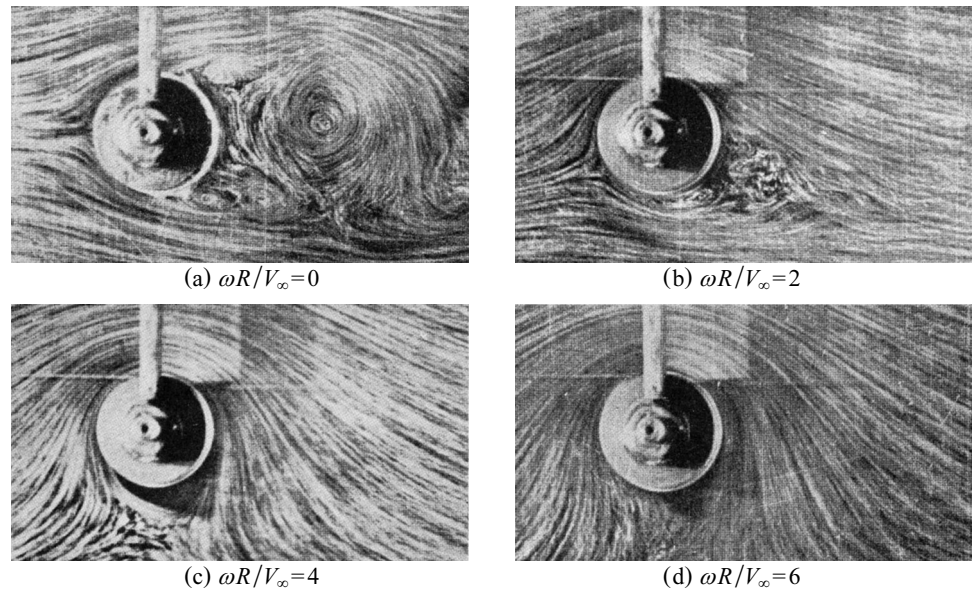


Figure 2.14.1. Early photographs presented by Prandtl and Tietjens (1934b) showing the results of flow-visualization experiments for flow about a spinning circular cylinder at different values of the ratio of the cylinder's peripheral velocity ωR to the freestream velocity V_∞ .

When two-dimensional flow about a circular cylinder spinning clockwise is observed in a low-speed wind tunnel with the freestream moving from left to right, the streamlines appear similar to those shown in Fig. 2.12.3. In the 1920s, similar results were observed by Ludwig Prandtl in a water tunnel at the Kaiser Wilhelm Institute for Flow Research in Göttingen, Germany. Many of Prandtl's flow-visualization photographs were published in Prandtl and Tietjens (1934b), and four of these photographs are reproduced here in Fig. 2.14.1. Notice the similarity between Fig. 2.14.1 (b)–(d) and the streamlines shown in Fig. 2.12.3. As the cylinder spins, shear stress in the boundary layer causes adjacent fluid to spin with the cylinder (i.e., the no-slip boundary condition is enforced in the boundary layer). When the cylinder spins clockwise and the freestream moves from left to right, the velocity of the fluid above the cylinder is increased and that below the cylinder is decreased. This moves the stagnation points to the lower part of the cylinder and creates lift on the cylinder through Bernoulli's effect, as was discussed in Sec. 2.7. Because the flow circulation is generated within the boundary layer and the potential flow equations do not apply in the boundary layer, the closed curve used to compute Γ for use in Eq. (2.14.2) must enclose the boundary layer as well as the cylinder.

The same principles that produce lift on a spinning cylinder in a wind tunnel will also produce lift on a spinning sphere. This is why a baseball pitcher is able to throw a *curve ball* by applying spin to the ball as it is released. Similarly, this is why a golf ball will *hook* or *slice* when hit so that the clubface imparts spin to the ball. In such cases, the spinning ball will experience an aerodynamic force that is perpendicular to both the ball's angular and translational velocity vectors.